

# Supplementary material for *couplr: Optimal Matching with Verifiable Certificates and Sparse Edge Generation*

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## 1 What this file holds

The article reports summaries of four measurements that are too large to print in full: the solver grid over problem regimes, the edge-generation loop over covariate counts, point clouds, calipers and sizes, the peak memory of every mode and comparator, and the raw runs behind the scaling table. This file carries them in full, and names the script that produced each one so any of them can be re-measured. It also carries the parts of the software design the article points at: the object model, the C++ organisation and the dependency footprint.

```
## run      20260828-170754
## host     Darwin Mac 25.5.0 Darwin Kernel Version 25.5.0: Mon Apr 27 20:41:15 PDT 2026; root:xnu-12
## commit   5a1693c 40 modified paths
## R version 4.5.3 (2026-03-11)
## aarch64-apple-darwin25.3.0
## couplr   1.6.2
## MatchIt  4.7.2
## optmatch 0.10.8
## microbenchmark 1.5.0
```

| Section                         | Script                        | Files                                                                              |
|---------------------------------|-------------------------------|------------------------------------------------------------------------------------|
| Solver selection across regimes | scripts/bench_regimes.R       | regime-runs.csv,<br>regime-cells.csv,<br>regime-results.csv,<br>regime-verdict.csv |
| The edge-generation loop        | scripts/bench_implicit_grid.R | implicit-grid-runs.csv,<br>implicit-grid-results.csv                               |
| Peak memory                     | scripts/bench_memory.R        | memory-results.csv                                                                 |
| Representation against solver   | scripts/bench_implicit.R      | implicit-results.csv                                                               |
| Scaling, every run              | scripts/bench_scaling.R       | scaling-runs.csv,<br>scaling-results.csv                                           |

Every script is resume-safe: it writes its rows as it goes and skips a cell it already holds, so a run that is interrupted continues where it stopped. Moving the CSV aside forces a full re-measurement. `paper/run_bench_suite.sh` runs all of them in order on one machine, which is how the numbers here were produced.

The scripts load couplr from a source checkout with `pkgload::load_all()` and write their CSVs under `paper/`, so they run from a clone of the package repository rather than from the submission archive:

```
git clone https://github.com/gcol33/couplr
cd couplr
sh paper/run_bench_suite.sh
```

The scaling, implicit, implicit-grid, regimes and memory scripts also take `--quick`, which runs a reduced grid.

## 2 Solver selection across problem regimes

`assignment(method = "auto")` chooses a solver from five rules read off one pass over the cost matrix: enumerate an at most 8 by 8 problem, use a cardinality algorithm where the finite costs are constant or binary, use the sparse Jonker-Volgenant variant where more than half the entries are forbidden, use the shortest-augmenting-path solver where there are at least three times as many columns as rows, and otherwise use Jonker-Volgenant. The article's solver figure varies problem size on square dense integer costs, which is the last rule's regime and no other. This grid varies the properties the rules are stated over.

A cell is a cost regime, a forbidden pattern and a shape. Each cell is generated as several independent instances, each instance is timed several times, and every solver in the panel sees the same instance.

Table 1: The factors the solver grid crosses. Shapes and sizes are in the next table.

| Factor            | Level            | Meaning                                       |
|-------------------|------------------|-----------------------------------------------|
| cost regime       | int_uniform      | integer, uniform on 1..10,000                 |
| cost regime       | dbl_uniform      | double, uniform on (0, 1)                     |
| cost regime       | binary           | binary, 0 or 1                                |
| cost regime       | constant         | constant                                      |
| cost regime       | tied             | integer, five distinct values                 |
| cost regime       | heavy_tailed     | double, lognormal                             |
| cost regime       | metric_uniform   | euclidean over uniform points in 8 dimensions |
| cost regime       | metric_clustered | euclidean over eight clusters in 8 dimensions |
| forbidden pattern | none             | complete: every pair admissible               |
| forbidden pattern | random_60        | 60 percent of pairs admissible, unstructured  |
| forbidden pattern | random_25        | 25 percent admissible, unstructured           |

| Factor            | Level   | Meaning                                               |
|-------------------|---------|-------------------------------------------------------|
| forbidden pattern | block_4 | 25 percent admissible in four disconnected components |

Table 2: Shapes. The base tier runs the whole panel at a size every solver in it reaches; the large tier repeats the crossing at a size only the scalable solvers reach.

| tier  | Shape      | aspect | Solvers in the panel |
|-------|------------|--------|----------------------|
| base  | 500x500    | 1:1    | 12                   |
| base  | 500x1500   | 1:3    | 12                   |
| base  | 500x5000   | 1:10   | 12                   |
| large | 1000x10000 | 1:10   | 5                    |
| large | 1500x1500  | 1:1    | 5                    |
| large | 1500x4500  | 1:3    | 5                    |

## 2.1 What the rules cost

For each cell, the solver "auto" selected is timed as a solver of its own, so its number carries the probing pass the rules cost, and it is compared against the fastest named solver in the same cell.

Table 3: Each dispatch rule, over the cells where it fired. Ratio is the time the dispatched solver took over the time the cell's fastest named solver took, so 1.00x means the rule picked the best available solver and 2.00x means it cost twice the best.

| Rule             | Cells | Picked the cell's fastest | Median ratio | Worst ratio |
|------------------|-------|---------------------------|--------------|-------------|
| sparse           | 48    | 2                         | 1.16x        | 12.44x      |
| very_rectangular | 32    | 0                         | 5.75x        | 13.38x      |
| no_cost_scale    | 30    | 24                        | 1.00x        | 1.02x       |
| default          | 16    | 10                        | 1.00x        | 10.28x      |

Table 4: The 25 cells where the dispatched solver was furthest off the cell's fastest.

| Regime           | Pattern  | Shape    | Rule             | Picked | Best            | Dispatched (ms) | Best (ms) | Ratio  |
|------------------|----------|----------|------------------|--------|-----------------|-----------------|-----------|--------|
| tied             | complete | 500x1500 | very_rectangular | sap    | network_simplex | 273             | 20.4      | 13.38x |
| tied             | 25%      | 500x500  | sparse           | lapmod | auction_scaled  | 73              | 5.9       | 12.44x |
| tied             | complete | 500x500  | default          | jv     | network_simplex | 57.1            | 5.6       | 10.28x |
| metric_uniform   | complete | 500x5000 | very_rectangular | sap    | jv              | 110             | 11        | 10.02x |
| metric_clustered | complete | 500x5000 | very_rectangular | sap    | jv              | 107             | 11.1      | 9.68x  |
| heavy_tailed     | complete | 500x5000 | very_rectangular | sap    | hungarian       | 100             | 11.2      | 9.00x  |
| dbl_uniform      | complete | 500x5000 | very_rectangular | sap    | jv              | 102             | 11.4      | 8.94x  |
| int_uniform      | complete | 500x5000 | very_rectangular | sap    | hungarian       | 101             | 11.9      | 8.50x  |
| tied             | 60%      | 500x500  | default          | jv     | network_simplex | 62.5            | 7.5       | 8.28x  |
| tied             | complete | 500x5000 | very_rectangular | sap    | network_simplex | 864             | 118       | 7.29x  |
| metric_uniform   | complete | 500x1500 | very_rectangular | sap    | jv              | 26.8            | 3.7       | 7.24x  |
| heavy_tailed     | complete | 500x1500 | very_rectangular | sap    | jv              | 26.7            | 3.7       | 7.13x  |
| dbl_uniform      | complete | 500x1500 | very_rectangular | sap    | hungarian       | 26.8            | 3.9       | 6.82x  |

| Regime           | Pattern  | Shape      | Rule             | Picked | Best            | Dispatched<br>(ms) | Best<br>(ms) | Ratio |
|------------------|----------|------------|------------------|--------|-----------------|--------------------|--------------|-------|
| int_uniform      | complete | 500x1500   | very_rectangular | sap    | hungarian       | 27.2               | 4.1          | 6.57x |
| metric_clustered | complete | 500x1500   | very_rectangular | sap    | jv              | 25.4               | 3.9          | 6.50x |
| tied             | 25%      | 500x1500   | sparse           | lapmod | network_simplex | 222                | 35.3         | 6.29x |
| dbl_uniform      | complete | 1000x10000 | very_rectangular | sap    | jv              | 475                | 77.1         | 6.16x |
| int_uniform      | complete | 1000x10000 | very_rectangular | sap    | jv              | 474                | 78.6         | 6.02x |
| metric_clustered | complete | 1000x10000 | very_rectangular | sap    | jv              | 464                | 77           | 6.02x |
| heavy_tailed     | complete | 1000x10000 | very_rectangular | sap    | jv              | 465                | 77.5         | 6.00x |
| tied             | 60%      | 500x1500   | very_rectangular | sap    | network_simplex | 195                | 35.5         | 5.49x |
| dbl_uniform      | complete | 1500x4500  | very_rectangular | sap    | jv              | 320                | 59           | 5.43x |
| metric_clustered | complete | 1500x4500  | very_rectangular | sap    | jv              | 316                | 59.2         | 5.34x |
| int_uniform      | complete | 1500x4500  | very_rectangular | sap    | jv              | 319                | 60.6         | 5.27x |
| heavy_tailed     | complete | 1500x4500  | very_rectangular | sap    | jv              | 319                | 62.8         | 5.08x |

## 2.2 Every cell

Table 5: Every cell of the solver grid: the rule that fired, the solver it picked, the fastest solver in the cell, and the ratio between them.

| Regime      | Pattern  | Shape    | Rule             | Picked | Best           | Ratio |
|-------------|----------|----------|------------------|--------|----------------|-------|
| binary      | 4 blocks | 500x1500 | no_cost_scale    | hk01   | hk01           | 1.00x |
| binary      | 4 blocks | 500x500  | no_cost_scale    | hk01   | hk01           | 1.01x |
| binary      | 4 blocks | 500x5000 | no_cost_scale    | hk01   | hk01           | 1.00x |
| binary      | complete | 500x1500 | no_cost_scale    | hk01   | hk01           | 1.00x |
| binary      | complete | 500x500  | no_cost_scale    | hk01   | hk01           | 1.01x |
| binary      | complete | 500x5000 | no_cost_scale    | hk01   | hk01           | 1.00x |
| binary      | 25%      | 500x1500 | no_cost_scale    | hk01   | hk01           | 1.00x |
| binary      | 25%      | 500x500  | no_cost_scale    | hk01   | hk01           | 1.00x |
| binary      | 25%      | 500x5000 | no_cost_scale    | hk01   | hk01           | 1.00x |
| binary      | 60%      | 500x1500 | no_cost_scale    | hk01   | hk01           | 1.01x |
| binary      | 60%      | 500x500  | no_cost_scale    | hk01   | hk01           | 1.02x |
| binary      | 60%      | 500x5000 | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | 4 blocks | 500x1500 | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | 4 blocks | 500x500  | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | 4 blocks | 500x5000 | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | complete | 500x1500 | no_cost_scale    | hk01   | hk01           | 1.01x |
| constant    | complete | 500x500  | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | complete | 500x5000 | no_cost_scale    | hk01   | hk01           | 1.01x |
| constant    | 25%      | 500x1500 | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | 25%      | 500x500  | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | 25%      | 500x5000 | no_cost_scale    | hk01   | hk01           | 0.99x |
| constant    | 60%      | 500x1500 | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | 60%      | 500x500  | no_cost_scale    | hk01   | hk01           | 1.00x |
| constant    | 60%      | 500x5000 | no_cost_scale    | hk01   | hk01           | 1.00x |
| dbl_uniform | 4 blocks | 500x1500 | sparse           | lapmod | hungarian      | 1.17x |
| dbl_uniform | 4 blocks | 500x500  | sparse           | lapmod | auction_scaled | 1.60x |
| dbl_uniform | 4 blocks | 500x5000 | sparse           | lapmod | hungarian      | 1.14x |
| dbl_uniform | complete | 500x1500 | very_rectangular | sap    | hungarian      | 6.82x |
| dbl_uniform | complete | 500x500  | default          | jv     | jv             | 1.01x |
| dbl_uniform | complete | 500x5000 | very_rectangular | sap    | jv             | 8.94x |
| dbl_uniform | 25%      | 500x1500 | sparse           | lapmod | jv             | 1.15x |

| Regime           | Pattern  | Shape    | Rule             | Picked | Best           | Ratio  |
|------------------|----------|----------|------------------|--------|----------------|--------|
| dbl_uniform      | 25%      | 500x500  | sparse           | lapmod | auktion_scaled | 2.42x  |
| dbl_uniform      | 25%      | 500x5000 | sparse           | lapmod | hungarian      | 1.18x  |
| dbl_uniform      | 60%      | 500x1500 | very_rectangular | sap    | jv             | 2.94x  |
| dbl_uniform      | 60%      | 500x500  | default          | jv     | auktion_scaled | 1.19x  |
| dbl_uniform      | 60%      | 500x5000 | very_rectangular | sap    | jv             | 3.06x  |
| heavy_tailed     | 4 blocks | 500x1500 | sparse           | lapmod | jv             | 1.16x  |
| heavy_tailed     | 4 blocks | 500x500  | sparse           | lapmod | hungarian      | 1.17x  |
| heavy_tailed     | 4 blocks | 500x5000 | sparse           | lapmod | jv             | 1.13x  |
| heavy_tailed     | complete | 500x1500 | very_rectangular | sap    | jv             | 7.13x  |
| heavy_tailed     | complete | 500x500  | default          | jv     | jv             | 1.00x  |
| heavy_tailed     | complete | 500x5000 | very_rectangular | sap    | hungarian      | 9.00x  |
| heavy_tailed     | 25%      | 500x1500 | sparse           | lapmod | jv             | 1.13x  |
| heavy_tailed     | 25%      | 500x500  | sparse           | lapmod | auktion_scaled | 1.64x  |
| heavy_tailed     | 25%      | 500x5000 | sparse           | lapmod | hungarian      | 1.18x  |
| heavy_tailed     | 60%      | 500x1500 | very_rectangular | sap    | jv             | 2.93x  |
| heavy_tailed     | 60%      | 500x500  | default          | jv     | jv             | 1.00x  |
| heavy_tailed     | 60%      | 500x5000 | very_rectangular | sap    | jv             | 3.08x  |
| int_uniform      | 4 blocks | 500x1500 | sparse           | lapmod | hungarian      | 1.16x  |
| int_uniform      | 4 blocks | 500x500  | sparse           | lapmod | auktion_scaled | 1.48x  |
| int_uniform      | 4 blocks | 500x5000 | sparse           | lapmod | hungarian      | 1.12x  |
| int_uniform      | complete | 500x1500 | very_rectangular | sap    | hungarian      | 6.57x  |
| int_uniform      | complete | 500x500  | default          | jv     | jv             | 1.02x  |
| int_uniform      | complete | 500x5000 | very_rectangular | sap    | hungarian      | 8.50x  |
| int_uniform      | 25%      | 500x1500 | sparse           | lapmod | hungarian      | 1.15x  |
| int_uniform      | 25%      | 500x500  | sparse           | lapmod | auktion_scaled | 1.73x  |
| int_uniform      | 25%      | 500x5000 | sparse           | lapmod | hungarian      | 1.19x  |
| int_uniform      | 60%      | 500x1500 | very_rectangular | sap    | hungarian      | 2.95x  |
| int_uniform      | 60%      | 500x500  | default          | jv     | auktion_scaled | 1.08x  |
| int_uniform      | 60%      | 500x5000 | very_rectangular | sap    | hungarian      | 3.08x  |
| metric_clustered | 4 blocks | 500x1500 | sparse           | lapmod | hungarian      | 1.16x  |
| metric_clustered | 4 blocks | 500x500  | sparse           | lapmod | auktion_scaled | 2.17x  |
| metric_clustered | 4 blocks | 500x5000 | sparse           | lapmod | jv             | 1.15x  |
| metric_clustered | complete | 500x1500 | very_rectangular | sap    | jv             | 6.50x  |
| metric_clustered | complete | 500x500  | default          | jv     | jv             | 1.00x  |
| metric_clustered | complete | 500x5000 | very_rectangular | sap    | jv             | 9.68x  |
| metric_clustered | 25%      | 500x1500 | sparse           | lapmod | jv             | 1.12x  |
| metric_clustered | 25%      | 500x500  | sparse           | lapmod | auktion_scaled | 3.56x  |
| metric_clustered | 25%      | 500x5000 | sparse           | lapmod | jv             | 0.87x  |
| metric_clustered | 60%      | 500x1500 | very_rectangular | sap    | jv             | 2.90x  |
| metric_clustered | 60%      | 500x500  | default          | jv     | auktion_scaled | 1.79x  |
| metric_clustered | 60%      | 500x5000 | very_rectangular | sap    | jv             | 3.01x  |
| metric_uniform   | 4 blocks | 500x1500 | sparse           | lapmod | jv             | 1.18x  |
| metric_uniform   | 4 blocks | 500x500  | sparse           | lapmod | auktion_scaled | 1.96x  |
| metric_uniform   | 4 blocks | 500x5000 | sparse           | lapmod | hungarian      | 0.74x  |
| metric_uniform   | complete | 500x1500 | very_rectangular | sap    | jv             | 7.24x  |
| metric_uniform   | complete | 500x500  | default          | jv     | jv             | 1.00x  |
| metric_uniform   | complete | 500x5000 | very_rectangular | sap    | jv             | 10.02x |
| metric_uniform   | 25%      | 500x1500 | sparse           | lapmod | jv             | 1.14x  |
| metric_uniform   | 25%      | 500x500  | sparse           | lapmod | auktion_scaled | 3.11x  |
| metric_uniform   | 25%      | 500x5000 | sparse           | lapmod | jv             | 1.16x  |
| metric_uniform   | 60%      | 500x1500 | very_rectangular | sap    | jv             | 2.97x  |
| metric_uniform   | 60%      | 500x500  | default          | jv     | auktion_scaled | 1.51x  |

| Regime           | Pattern  | Shape      | Rule             | Picked | Best            | Ratio  |
|------------------|----------|------------|------------------|--------|-----------------|--------|
| metric_uniform   | 60%      | 500x5000   | very_rectangular | sap    | hungarian       | 3.09x  |
| tied             | 4 blocks | 500x1500   | sparse           | lapmod | network_simplex | 1.91x  |
| tied             | 4 blocks | 500x500    | sparse           | lapmod | auction_scaled  | 4.61x  |
| tied             | 4 blocks | 500x5000   | sparse           | lapmod | sap             | 2.09x  |
| tied             | complete | 500x1500   | very_rectangular | sap    | network_simplex | 13.38x |
| tied             | complete | 500x500    | default          | jv     | network_simplex | 10.28x |
| tied             | complete | 500x5000   | very_rectangular | sap    | network_simplex | 7.29x  |
| tied             | 25%      | 500x1500   | sparse           | lapmod | network_simplex | 6.29x  |
| tied             | 25%      | 500x500    | sparse           | lapmod | auction_scaled  | 12.44x |
| tied             | 25%      | 500x5000   | sparse           | lapmod | network_simplex | 3.42x  |
| tied             | 60%      | 500x1500   | very_rectangular | sap    | network_simplex | 5.49x  |
| tied             | 60%      | 500x500    | default          | jv     | network_simplex | 8.28x  |
| tied             | 60%      | 500x5000   | very_rectangular | sap    | network_simplex | 3.35x  |
| binary           | complete | 1000x10000 | no_cost_scale    | hk01   | jv              | 0.03x  |
| binary           | complete | 1500x1500  | no_cost_scale    | hk01   | jv              | 0.02x  |
| binary           | complete | 1500x4500  | no_cost_scale    | hk01   | jv              | 0.02x  |
| binary           | 25%      | 1000x10000 | no_cost_scale    | hk01   | sap             | 0.07x  |
| binary           | 25%      | 1500x1500  | no_cost_scale    | hk01   | sap             | 0.03x  |
| binary           | 25%      | 1500x4500  | no_cost_scale    | hk01   | sap             | 0.05x  |
| dbl_uniform      | complete | 1000x10000 | very_rectangular | sap    | jv              | 6.16x  |
| dbl_uniform      | complete | 1500x1500  | default          | jv     | jv              | 0.99x  |
| dbl_uniform      | complete | 1500x4500  | very_rectangular | sap    | jv              | 5.43x  |
| dbl_uniform      | 25%      | 1000x10000 | sparse           | lapmod | jv              | 1.14x  |
| dbl_uniform      | 25%      | 1500x1500  | sparse           | lapmod | jv              | 1.21x  |
| dbl_uniform      | 25%      | 1500x4500  | sparse           | lapmod | lapmod          | 0.95x  |
| heavy_tailed     | complete | 1000x10000 | very_rectangular | sap    | jv              | 6.00x  |
| heavy_tailed     | complete | 1500x1500  | default          | jv     | jv              | 1.00x  |
| heavy_tailed     | complete | 1500x4500  | very_rectangular | sap    | jv              | 5.08x  |
| heavy_tailed     | 25%      | 1000x10000 | sparse           | lapmod | jv              | 1.11x  |
| heavy_tailed     | 25%      | 1500x1500  | sparse           | lapmod | jv              | 1.16x  |
| heavy_tailed     | 25%      | 1500x4500  | sparse           | lapmod | lapmod          | 1.00x  |
| int_uniform      | complete | 1000x10000 | very_rectangular | sap    | jv              | 6.02x  |
| int_uniform      | complete | 1500x1500  | default          | jv     | jv              | 1.00x  |
| int_uniform      | complete | 1500x4500  | very_rectangular | sap    | jv              | 5.27x  |
| int_uniform      | 25%      | 1000x10000 | sparse           | lapmod | jv              | 1.12x  |
| int_uniform      | 25%      | 1500x1500  | sparse           | lapmod | jv              | 1.24x  |
| int_uniform      | 25%      | 1500x4500  | sparse           | lapmod | jv              | 1.12x  |
| metric_clustered | complete | 1000x10000 | very_rectangular | sap    | jv              | 6.02x  |
| metric_clustered | complete | 1500x1500  | default          | jv     | jv              | 1.00x  |
| metric_clustered | complete | 1500x4500  | very_rectangular | sap    | jv              | 5.34x  |
| metric_clustered | 25%      | 1000x10000 | sparse           | lapmod | jv              | 1.13x  |
| metric_clustered | 25%      | 1500x1500  | sparse           | lapmod | sap             | 1.23x  |
| metric_clustered | 25%      | 1500x4500  | sparse           | lapmod | jv              | 1.04x  |

## 2.3 Every solver, every cell

Table 6: Milliseconds per solver and cell: the median across instances, with the quartiles beside it. An instance contributes the median of its own repetitions, so the quartiles are between problems rather than between repetitions of one problem. A solver that errored or timed out on a cell has no row there.

| Regime | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|--------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| binary | 4 blocks | 500x500  | hk01            | 3         | 1.66           | 1.65        | 1.67        |
| binary | 4 blocks | 500x500  | auto            | 3         | 1.68           | 1.68        | 1.69        |
| binary | 4 blocks | 500x500  | network_simplex | 3         | 4.31           | 4.14        | 4.65        |
| binary | 4 blocks | 500x500  | auction_scaled  | 3         | 4.64           | 4.63        | 4.66        |
| binary | 4 blocks | 500x500  | csa             | 3         | 4.8            | 4.77        | 4.8         |
| binary | 4 blocks | 500x500  | sap             | 3         | 11.2           | 11.2        | 11.3        |
| binary | 4 blocks | 500x500  | csflow          | 3         | 11.3           | 11.2        | 11.3        |
| binary | 4 blocks | 500x500  | ramshaw_tarjan  | 3         | 13.7           | 13.7        | 13.7        |
| binary | 4 blocks | 500x500  | hungarian       | 3         | 17.2           | 17.2        | 17.2        |
| binary | 4 blocks | 500x500  | jv              | 3         | 17.2           | 17.2        | 17.3        |
| binary | 4 blocks | 500x500  | lapmod          | 3         | 18.5           | 18.5        | 18.5        |
| binary | 4 blocks | 500x500  | sap_dense       | 3         | 25             | 24.9        | 25          |
| binary | 4 blocks | 500x1500 | hk01            | 3         | 4.68           | 4.68        | 4.7         |
| binary | 4 blocks | 500x1500 | auto            | 3         | 4.68           | 4.67        | 4.7         |
| binary | 4 blocks | 500x1500 | sap             | 3         | 26.1           | 26.1        | 26.2        |
| binary | 4 blocks | 500x1500 | csflow          | 3         | 26.2           | 26.1        | 26.2        |
| binary | 4 blocks | 500x1500 | network_simplex | 3         | 30.1           | 28.3        | 30.7        |
| binary | 4 blocks | 500x1500 | ramshaw_tarjan  | 3         | 39.3           | 39.2        | 39.4        |
| binary | 4 blocks | 500x1500 | hungarian       | 3         | 52.2           | 52.2        | 52.3        |
| binary | 4 blocks | 500x1500 | jv              | 3         | 52.3           | 52.3        | 52.3        |
| binary | 4 blocks | 500x1500 | lapmod          | 3         | 56.6           | 56.6        | 56.6        |
| binary | 4 blocks | 500x1500 | sap_dense       | 3         | 74.9           | 74.9        | 75.3        |
| binary | 4 blocks | 500x1500 | csa             | 3         | 102            | 97          | 102         |
| binary | 4 blocks | 500x1500 | auction_scaled  | 3         | 160            | 158         | 163         |
| binary | 4 blocks | 500x5000 | auto            | 3         | 15.9           | 15.9        | 15.9        |
| binary | 4 blocks | 500x5000 | hk01            | 3         | 16             | 15.9        | 16          |
| binary | 4 blocks | 500x5000 | sap             | 3         | 77.5           | 77.5        | 77.6        |
| binary | 4 blocks | 500x5000 | csflow          | 3         | 77.6           | 77.6        | 77.7        |
| binary | 4 blocks | 500x5000 | ramshaw_tarjan  | 3         | 127            | 127         | 127         |
| binary | 4 blocks | 500x5000 | network_simplex | 3         | 161            | 161         | 169         |
| binary | 4 blocks | 500x5000 | hungarian       | 3         | 175            | 175         | 175         |
| binary | 4 blocks | 500x5000 | jv              | 3         | 175            | 175         | 175         |
| binary | 4 blocks | 500x5000 | lapmod          | 3         | 189            | 189         | 189         |
| binary | 4 blocks | 500x5000 | sap_dense       | 3         | 250            | 250         | 250         |
| binary | 4 blocks | 500x5000 | csa             | 3         | 1.37e+03       | 1.36e+03    | 1.39e+03    |
| binary | 4 blocks | 500x5000 | auction_scaled  | 3         | 2.08e+03       | 2.08e+03    | 2.16e+03    |
| binary | complete | 500x500  | hk01            | 3         | 3.45           | 3.45        | 3.47        |
| binary | complete | 500x500  | auto            | 3         | 3.49           | 3.46        | 3.5         |
| binary | complete | 500x500  | network_simplex | 3         | 3.84           | 3.04        | 3.87        |
| binary | complete | 500x500  | csa             | 3         | 14.1           | 14          | 14.1        |
| binary | complete | 500x500  | auction_scaled  | 3         | 14.3           | 14.2        | 14.3        |
| binary | complete | 500x500  | jv              | 3         | 58.4           | 57.2        | 58.6        |
| binary | complete | 500x500  | hungarian       | 3         | 58.7           | 58.2        | 59.2        |
| binary | complete | 500x500  | ramshaw_tarjan  | 3         | 63.1           | 63          | 63.2        |
| binary | complete | 500x500  | sap_dense       | 3         | 87.2           | 86.9        | 88.4        |
| binary | complete | 500x500  | lapmod          | 3         | 95.1           | 95          | 95.4        |

| Regime | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|--------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| binary | complete | 500x500  | sap             | 3         | 106            | 106         | 106         |
| binary | complete | 500x500  | csflow          | 3         | 107            | 107         | 107         |
| binary | complete | 500x1500 | hk01            | 3         | 10             | 9.92        | 10          |
| binary | complete | 500x1500 | auto            | 3         | 10             | 9.97        | 10          |
| binary | complete | 500x1500 | network_simplex | 3         | 21.8           | 20.9        | 26.8        |
| binary | complete | 500x1500 | csa             | 3         | 87.5           | 82.6        | 87.6        |
| binary | complete | 500x1500 | auction_scaled  | 3         | 128            | 127         | 128         |
| binary | complete | 500x1500 | jv              | 3         | 193            | 193         | 194         |
| binary | complete | 500x1500 | hungarian       | 3         | 194            | 194         | 195         |
| binary | complete | 500x1500 | ramshaw_tarjan  | 3         | 211            | 211         | 211         |
| binary | complete | 500x1500 | sap             | 3         | 254            | 254         | 255         |
| binary | complete | 500x1500 | csflow          | 3         | 257            | 257         | 257         |
| binary | complete | 500x1500 | sap_dense       | 3         | 282            | 282         | 282         |
| binary | complete | 500x1500 | lapmod          | 3         | 294            | 293         | 294         |
| binary | complete | 500x5000 | auto            | 3         | 33.7           | 33.7        | 33.8        |
| binary | complete | 500x5000 | hk01            | 3         | 33.8           | 33.7        | 33.8        |
| binary | complete | 500x5000 | network_simplex | 3         | 96.4           | 88.9        | 101         |
| binary | complete | 500x5000 | jv              | 3         | 665            | 665         | 665         |
| binary | complete | 500x5000 | hungarian       | 3         | 666            | 665         | 666         |
| binary | complete | 500x5000 | ramshaw_tarjan  | 3         | 764            | 764         | 775         |
| binary | complete | 500x5000 | sap             | 3         | 803            | 803         | 803         |
| binary | complete | 500x5000 | csflow          | 3         | 812            | 812         | 812         |
| binary | complete | 500x5000 | sap_dense       | 3         | 959            | 959         | 962         |
| binary | complete | 500x5000 | lapmod          | 3         | 1.01e+03       | 1.01e+03    | 1.01e+03    |
| binary | complete | 500x5000 | csa             | 3         | 1.1e+03        | 1.07e+03    | 1.24e+03    |
| binary | complete | 500x5000 | auction_scaled  | 3         | 1.69e+03       | 1.69e+03    | 1.74e+03    |
| binary | 25%      | 500x500  | auto            | 3         | 2.74           | 2.74        | 2.75        |
| binary | 25%      | 500x500  | hk01            | 3         | 2.75           | 2.75        | 2.78        |
| binary | 25%      | 500x500  | auction_scaled  | 3         | 6.13           | 6.12        | 6.16        |
| binary | 25%      | 500x500  | network_simplex | 3         | 6.35           | 6.19        | 6.38        |
| binary | 25%      | 500x500  | csa             | 3         | 6.41           | 6.4         | 6.43        |
| binary | 25%      | 500x500  | sap             | 3         | 45.1           | 45          | 45.1        |
| binary | 25%      | 500x500  | csflow          | 3         | 45.2           | 45.1        | 45.2        |
| binary | 25%      | 500x500  | jv              | 3         | 61.4           | 61.1        | 61.8        |
| binary | 25%      | 500x500  | hungarian       | 3         | 61.6           | 61.5        | 61.8        |
| binary | 25%      | 500x500  | lapmod          | 3         | 71.4           | 71.4        | 71.5        |
| binary | 25%      | 500x500  | ramshaw_tarjan  | 3         | 99.4           | 99.4        | 99.7        |
| binary | 25%      | 500x500  | sap_dense       | 3         | 127            | 127         | 127         |
| binary | 25%      | 500x1500 | hk01            | 3         | 7.95           | 7.94        | 7.96        |
| binary | 25%      | 500x1500 | auto            | 3         | 7.96           | 7.93        | 7.96        |
| binary | 25%      | 500x1500 | network_simplex | 3         | 30.1           | 29.6        | 30.3        |
| binary | 25%      | 500x1500 | csa             | 3         | 84.4           | 79.6        | 84.4        |
| binary | 25%      | 500x1500 | sap             | 3         | 100            | 99.9        | 100         |
| binary | 25%      | 500x1500 | csflow          | 3         | 101            | 100         | 101         |
| binary | 25%      | 500x1500 | auction_scaled  | 3         | 109            | 106         | 115         |
| binary | 25%      | 500x1500 | jv              | 3         | 199            | 199         | 199         |
| binary | 25%      | 500x1500 | hungarian       | 3         | 200            | 199         | 200         |
| binary | 25%      | 500x1500 | lapmod          | 3         | 220            | 220         | 221         |
| binary | 25%      | 500x1500 | ramshaw_tarjan  | 3         | 385            | 385         | 385         |
| binary | 25%      | 500x1500 | sap_dense       | 3         | 478            | 478         | 479         |
| binary | 25%      | 500x5000 | hk01            | 3         | 26.6           | 26.5        | 26.6        |



| Regime   | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|----------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| binary   | 25%      | 500x5000 | auto            | 3         | 26.6           | 26.5        | 26.6        |
| binary   | 25%      | 500x5000 | network_simplex | 3         | 169            | 166         | 174         |
| binary   | 25%      | 500x5000 | sap             | 3         | 296            | 295         | 296         |
| binary   | 25%      | 500x5000 | csflow          | 3         | 299            | 298         | 300         |
| binary   | 25%      | 500x5000 | hungarian       | 3         | 683            | 683         | 684         |
| binary   | 25%      | 500x5000 | jv              | 3         | 683            | 683         | 684         |
| binary   | 25%      | 500x5000 | lapmod          | 3         | 740            | 740         | 741         |
| binary   | 25%      | 500x5000 | csa             | 3         | 1.11e+03       | 1.11e+03    | 1.23e+03    |
| binary   | 25%      | 500x5000 | ramshaw_tarjan  | 3         | 1.37e+03       | 1.37e+03    | 1.37e+03    |
| binary   | 25%      | 500x5000 | auction_scaled  | 3         | 1.67e+03       | 1.63e+03    | 1.69e+03    |
| binary   | 25%      | 500x5000 | sap_dense       | 3         | 1.7e+03        | 1.7e+03     | 1.7e+03     |
| binary   | 60%      | 500x500  | hk01            | 3         | 4.25           | 4.24        | 4.26        |
| binary   | 60%      | 500x500  | auto            | 3         | 4.32           | 4.27        | 4.34        |
| binary   | 60%      | 500x500  | network_simplex | 3         | 6.15           | 5.92        | 6.2         |
| binary   | 60%      | 500x500  | auction_scaled  | 3         | 11.3           | 11.2        | 11.3        |
| binary   | 60%      | 500x500  | csa             | 3         | 11.4           | 11.4        | 11.5        |
| binary   | 60%      | 500x500  | hungarian       | 3         | 61.3           | 61          | 61.7        |
| binary   | 60%      | 500x500  | jv              | 3         | 61.4           | 61.3        | 62.4        |
| binary   | 60%      | 500x500  | sap             | 3         | 75.1           | 74.9        | 75.1        |
| binary   | 60%      | 500x500  | csflow          | 3         | 75.5           | 75.4        | 75.6        |
| binary   | 60%      | 500x500  | lapmod          | 3         | 85.4           | 85.4        | 85.6        |
| binary   | 60%      | 500x500  | ramshaw_tarjan  | 3         | 140            | 140         | 140         |
| binary   | 60%      | 500x500  | sap_dense       | 3         | 142            | 142         | 142         |
| binary   | 60%      | 500x1500 | hk01            | 3         | 12.2           | 12.2        | 12.4        |
| binary   | 60%      | 500x1500 | auto            | 3         | 12.3           | 12.3        | 12.3        |
| binary   | 60%      | 500x1500 | network_simplex | 3         | 35.6           | 33.9        | 37.1        |
| binary   | 60%      | 500x1500 | csa             | 3         | 95.3           | 92.9        | 95.5        |
| binary   | 60%      | 500x1500 | auction_scaled  | 3         | 126            | 124         | 137         |
| binary   | 60%      | 500x1500 | sap             | 3         | 177            | 176         | 177         |
| binary   | 60%      | 500x1500 | csflow          | 3         | 178            | 178         | 178         |
| binary   | 60%      | 500x1500 | hungarian       | 3         | 200            | 200         | 201         |
| binary   | 60%      | 500x1500 | jv              | 3         | 201            | 201         | 201         |
| binary   | 60%      | 500x1500 | lapmod          | 3         | 260            | 260         | 261         |
| binary   | 60%      | 500x1500 | ramshaw_tarjan  | 3         | 542            | 541         | 543         |
| binary   | 60%      | 500x1500 | sap_dense       | 3         | 592            | 591         | 592         |
| binary   | 60%      | 500x5000 | auto            | 3         | 41.6           | 41.5        | 41.6        |
| binary   | 60%      | 500x5000 | hk01            | 3         | 41.6           | 41.5        | 41.7        |
| binary   | 60%      | 500x5000 | network_simplex | 3         | 167            | 163         | 170         |
| binary   | 60%      | 500x5000 | sap             | 3         | 552            | 552         | 553         |
| binary   | 60%      | 500x5000 | csflow          | 3         | 558            | 558         | 558         |
| binary   | 60%      | 500x5000 | jv              | 3         | 686            | 686         | 687         |
| binary   | 60%      | 500x5000 | hungarian       | 3         | 687            | 686         | 687         |
| binary   | 60%      | 500x5000 | lapmod          | 3         | 874            | 874         | 875         |
| binary   | 60%      | 500x5000 | csa             | 3         | 1.3e+03        | 1.21e+03    | 1.33e+03    |
| binary   | 60%      | 500x5000 | auction_scaled  | 3         | 1.65e+03       | 1.59e+03    | 1.66e+03    |
| binary   | 60%      | 500x5000 | ramshaw_tarjan  | 3         | 1.96e+03       | 1.95e+03    | 1.96e+03    |
| binary   | 60%      | 500x5000 | sap_dense       | 3         | 2.13e+03       | 2.13e+03    | 2.13e+03    |
| constant | 4 blocks | 500x500  | hk01            | 3         | 1.25           | 1.25        | 1.27        |
| constant | 4 blocks | 500x500  | auto            | 3         | 1.25           | 1.25        | 1.26        |
| constant | 4 blocks | 500x500  | network_simplex | 3         | 2.48           | 2.47        | 2.53        |
| constant | 4 blocks | 500x500  | auction_scaled  | 3         | 3.1            | 3.1         | 3.12        |

| Regime   | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|----------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| constant | 4 blocks | 500x500  | csa             | 3         | 3.16           | 3.16        | 3.2         |
| constant | 4 blocks | 500x500  | sap             | 3         | 8.78           | 8.77        | 8.8         |
| constant | 4 blocks | 500x500  | csflow          | 3         | 8.81           | 8.79        | 8.85        |
| constant | 4 blocks | 500x500  | ramshaw_tarjan  | 3         | 12.5           | 12.5        | 12.5        |
| constant | 4 blocks | 500x500  | hungarian       | 3         | 16.8           | 16.8        | 16.9        |
| constant | 4 blocks | 500x500  | jv              | 3         | 16.9           | 16.9        | 16.9        |
| constant | 4 blocks | 500x500  | lapmod          | 3         | 17.7           | 17.7        | 17.7        |
| constant | 4 blocks | 500x500  | sap_dense       | 3         | 24.7           | 24.6        | 24.7        |
| constant | 4 blocks | 500x1500 | auto            | 3         | 3.55           | 3.54        | 3.55        |
| constant | 4 blocks | 500x1500 | hk01            | 3         | 3.55           | 3.54        | 3.55        |
| constant | 4 blocks | 500x1500 | network_simplex | 3         | 16.4           | 16.4        | 16.4        |
| constant | 4 blocks | 500x1500 | csflow          | 3         | 21.9           | 21.8        | 21.9        |
| constant | 4 blocks | 500x1500 | sap             | 3         | 21.9           | 21.8        | 21.9        |
| constant | 4 blocks | 500x1500 | ramshaw_tarjan  | 3         | 36.3           | 36.3        | 36.4        |
| constant | 4 blocks | 500x1500 | hungarian       | 3         | 51.3           | 51.3        | 51.3        |
| constant | 4 blocks | 500x1500 | jv              | 3         | 51.3           | 51.3        | 51.3        |
| constant | 4 blocks | 500x1500 | lapmod          | 3         | 55.7           | 55.7        | 55.7        |
| constant | 4 blocks | 500x1500 | sap_dense       | 3         | 74             | 74          | 74.1        |
| constant | 4 blocks | 500x1500 | csa             | 3         | 101            | 101         | 101         |
| constant | 4 blocks | 500x1500 | auction_scaled  | 3         | 149            | 149         | 149         |
| constant | 4 blocks | 500x5000 | auto            | 3         | 12             | 12          | 12.1        |
| constant | 4 blocks | 500x5000 | hk01            | 3         | 12.1           | 12          | 12.1        |
| constant | 4 blocks | 500x5000 | sap             | 3         | 66.5           | 66.5        | 66.5        |
| constant | 4 blocks | 500x5000 | csflow          | 3         | 67             | 66.9        | 67          |
| constant | 4 blocks | 500x5000 | network_simplex | 3         | 93.2           | 93.1        | 93.3        |
| constant | 4 blocks | 500x5000 | ramshaw_tarjan  | 3         | 119            | 119         | 119         |
| constant | 4 blocks | 500x5000 | hungarian       | 3         | 173            | 172         | 173         |
| constant | 4 blocks | 500x5000 | jv              | 3         | 173            | 173         | 173         |
| constant | 4 blocks | 500x5000 | lapmod          | 3         | 186            | 186         | 186         |
| constant | 4 blocks | 500x5000 | sap_dense       | 3         | 247            | 247         | 247         |
| constant | 4 blocks | 500x5000 | csa             | 3         | 1.3e+03        | 1.3e+03     | 1.3e+03     |
| constant | 4 blocks | 500x5000 | auction_scaled  | 3         | 2.47e+03       | 2.47e+03    | 2.47e+03    |
| constant | complete | 500x500  | hk01            | 3         | 1.74           | 1.74        | 1.74        |
| constant | complete | 500x500  | auto            | 3         | 1.74           | 1.74        | 1.75        |
| constant | complete | 500x500  | network_simplex | 3         | 2.3            | 2.29        | 2.37        |
| constant | complete | 500x500  | csa             | 3         | 8.87           | 8.86        | 8.87        |
| constant | complete | 500x500  | auction_scaled  | 3         | 9.19           | 9.19        | 9.2         |
| constant | complete | 500x500  | jv              | 3         | 56.2           | 56.2        | 57          |
| constant | complete | 500x500  | hungarian       | 3         | 56.7           | 56.3        | 57.4        |
| constant | complete | 500x500  | ramshaw_tarjan  | 3         | 58.9           | 58.7        | 59          |
| constant | complete | 500x500  | sap_dense       | 3         | 85.5           | 85.3        | 86.1        |
| constant | complete | 500x500  | lapmod          | 3         | 91.5           | 91.1        | 91.5        |
| constant | complete | 500x500  | sap             | 3         | 97.5           | 97.4        | 97.5        |
| constant | complete | 500x500  | csflow          | 3         | 98.3           | 98.3        | 98.5        |
| constant | complete | 500x1500 | hk01            | 3         | 4.96           | 4.93        | 4.99        |
| constant | complete | 500x1500 | auto            | 3         | 5              | 4.97        | 5.05        |
| constant | complete | 500x1500 | network_simplex | 3         | 7.85           | 7.8         | 7.9         |
| constant | complete | 500x1500 | csa             | 3         | 77.3           | 76.5        | 77.3        |
| constant | complete | 500x1500 | auction_scaled  | 3         | 184            | 184         | 184         |
| constant | complete | 500x1500 | hungarian       | 3         | 190            | 190         | 191         |
| constant | complete | 500x1500 | jv              | 3         | 190            | 190         | 191         |

| Regime   | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|----------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| constant | complete | 500x1500 | ramshaw_tarjan  | 3         | 202            | 202         | 203         |
| constant | complete | 500x1500 | sap             | 3         | 237            | 237         | 238         |
| constant | complete | 500x1500 | csflow          | 3         | 240            | 240         | 240         |
| constant | complete | 500x1500 | sap_dense       | 3         | 278            | 278         | 278         |
| constant | complete | 500x1500 | lapmod          | 3         | 287            | 287         | 288         |
| constant | complete | 500x5000 | hk01            | 3         | 16.5           | 16.5        | 16.6        |
| constant | complete | 500x5000 | auto            | 3         | 16.6           | 16.6        | 16.7        |
| constant | complete | 500x5000 | network_simplex | 3         | 27.6           | 27.6        | 27.8        |
| constant | complete | 500x5000 | hungarian       | 3         | 653            | 653         | 654         |
| constant | complete | 500x5000 | jv              | 3         | 654            | 653         | 654         |
| constant | complete | 500x5000 | ramshaw_tarjan  | 3         | 738            | 736         | 746         |
| constant | complete | 500x5000 | sap             | 3         | 757            | 757         | 777         |
| constant | complete | 500x5000 | csflow          | 3         | 765            | 765         | 766         |
| constant | complete | 500x5000 | csa             | 3         | 926            | 926         | 926         |
| constant | complete | 500x5000 | sap_dense       | 3         | 948            | 947         | 951         |
| constant | complete | 500x5000 | lapmod          | 3         | 980            | 980         | 981         |
| constant | complete | 500x5000 | auction_scaled  | 3         | 2.31e+03       | 2.31e+03    | 2.32e+03    |
| constant | 25%      | 500x500  | hk01            | 3         | 2.35           | 2.34        | 2.36        |
| constant | 25%      | 500x500  | auto            | 3         | 2.36           | 2.36        | 2.38        |
| constant | 25%      | 500x500  | network_simplex | 3         | 3.66           | 3.57        | 3.66        |
| constant | 25%      | 500x500  | auction_scaled  | 3         | 5.59           | 5.33        | 5.6         |
| constant | 25%      | 500x500  | csa             | 3         | 5.75           | 5.52        | 5.77        |
| constant | 25%      | 500x500  | sap             | 3         | 37             | 36.9        | 37          |
| constant | 25%      | 500x500  | csflow          | 3         | 37.1           | 37.1        | 37.1        |
| constant | 25%      | 500x500  | hungarian       | 3         | 60.6           | 59.9        | 61.1        |
| constant | 25%      | 500x500  | jv              | 3         | 60.8           | 60.8        | 61.3        |
| constant | 25%      | 500x500  | lapmod          | 3         | 69.9           | 69.8        | 69.9        |
| constant | 25%      | 500x500  | ramshaw_tarjan  | 3         | 93.6           | 93.2        | 93.8        |
| constant | 25%      | 500x500  | sap_dense       | 3         | 127            | 126         | 127         |
| constant | 25%      | 500x1500 | auto            | 3         | 6.66           | 6.63        | 6.73        |
| constant | 25%      | 500x1500 | hk01            | 3         | 6.67           | 6.65        | 6.7         |
| constant | 25%      | 500x1500 | network_simplex | 3         | 23.6           | 23.4        | 24.6        |
| constant | 25%      | 500x1500 | csa             | 3         | 65.7           | 64.4        | 65.9        |
| constant | 25%      | 500x1500 | sap             | 3         | 82.4           | 82.4        | 82.8        |
| constant | 25%      | 500x1500 | csflow          | 3         | 83.1           | 83          | 83.5        |
| constant | 25%      | 500x1500 | auction_scaled  | 3         | 128            | 126         | 133         |
| constant | 25%      | 500x1500 | hungarian       | 3         | 198            | 198         | 198         |
| constant | 25%      | 500x1500 | jv              | 3         | 199            | 198         | 199         |
| constant | 25%      | 500x1500 | lapmod          | 3         | 218            | 218         | 218         |
| constant | 25%      | 500x1500 | ramshaw_tarjan  | 3         | 373            | 372         | 373         |
| constant | 25%      | 500x1500 | sap_dense       | 3         | 477            | 476         | 477         |
| constant | 25%      | 500x5000 | auto            | 3         | 22             | 22          | 22.2        |
| constant | 25%      | 500x5000 | hk01            | 3         | 22.1           | 22.1        | 22.2        |
| constant | 25%      | 500x5000 | network_simplex | 3         | 116            | 115         | 116         |
| constant | 25%      | 500x5000 | csflow          | 3         | 250            | 249         | 250         |
| constant | 25%      | 500x5000 | sap             | 3         | 250            | 248         | 250         |
| constant | 25%      | 500x5000 | hungarian       | 3         | 676            | 676         | 677         |
| constant | 25%      | 500x5000 | jv              | 3         | 677            | 677         | 677         |
| constant | 25%      | 500x5000 | lapmod          | 3         | 732            | 732         | 732         |
| constant | 25%      | 500x5000 | csa             | 3         | 968            | 887         | 1e+03       |
| constant | 25%      | 500x5000 | ramshaw_tarjan  | 3         | 1.34e+03       | 1.33e+03    | 1.34e+03    |

| Regime      | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|-------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| constant    | 25%      | 500x5000 | sap_dense       | 3         | 1.69e+03       | 1.69e+03    | 1.69e+03    |
| constant    | 25%      | 500x5000 | auction_scaled  | 3         | 1.88e+03       | 1.87e+03    | 2.03e+03    |
| constant    | 60%      | 500x500  | auto            | 3         | 3.22           | 3.22        | 3.27        |
| constant    | 60%      | 500x500  | hk01            | 3         | 3.23           | 3.22        | 3.24        |
| constant    | 60%      | 500x500  | network_simplex | 3         | 4.18           | 4.13        | 4.21        |
| constant    | 60%      | 500x500  | auction_scaled  | 3         | 8.94           | 8.5         | 9.39        |
| constant    | 60%      | 500x500  | csa             | 3         | 9.14           | 8.71        | 9.58        |
| constant    | 60%      | 500x500  | hungarian       | 3         | 61.4           | 61          | 61.4        |
| constant    | 60%      | 500x500  | jv              | 3         | 62             | 61.3        | 62.1        |
| constant    | 60%      | 500x500  | sap             | 3         | 67.6           | 67.6        | 67.7        |
| constant    | 60%      | 500x500  | csflow          | 3         | 68             | 68          | 68.1        |
| constant    | 60%      | 500x500  | lapmod          | 3         | 82.2           | 81.7        | 82.3        |
| constant    | 60%      | 500x500  | ramshaw_tarjan  | 3         | 134            | 134         | 134         |
| constant    | 60%      | 500x500  | sap_dense       | 3         | 140            | 140         | 140         |
| constant    | 60%      | 500x1500 | hk01            | 3         | 9.12           | 9.12        | 9.25        |
| constant    | 60%      | 500x1500 | auto            | 3         | 9.16           | 9.1         | 9.19        |
| constant    | 60%      | 500x1500 | network_simplex | 3         | 27.3           | 24.8        | 27.5        |
| constant    | 60%      | 500x1500 | csa             | 3         | 83.6           | 78.7        | 83.6        |
| constant    | 60%      | 500x1500 | sap             | 3         | 160            | 160         | 160         |
| constant    | 60%      | 500x1500 | csflow          | 3         | 162            | 162         | 162         |
| constant    | 60%      | 500x1500 | auction_scaled  | 3         | 164            | 138         | 171         |
| constant    | 60%      | 500x1500 | hungarian       | 3         | 197            | 196         | 198         |
| constant    | 60%      | 500x1500 | jv              | 3         | 198            | 198         | 199         |
| constant    | 60%      | 500x1500 | lapmod          | 3         | 256            | 256         | 256         |
| constant    | 60%      | 500x1500 | ramshaw_tarjan  | 3         | 531            | 530         | 531         |
| constant    | 60%      | 500x1500 | sap_dense       | 3         | 591            | 590         | 591         |
| constant    | 60%      | 500x5000 | auto            | 3         | 30.5           | 30.5        | 30.9        |
| constant    | 60%      | 500x5000 | hk01            | 3         | 30.5           | 30.4        | 30.8        |
| constant    | 60%      | 500x5000 | network_simplex | 3         | 102            | 98.7        | 108         |
| constant    | 60%      | 500x5000 | csflow          | 3         | 510            | 510         | 525         |
| constant    | 60%      | 500x5000 | sap             | 3         | 537            | 521         | 537         |
| constant    | 60%      | 500x5000 | hungarian       | 3         | 676            | 676         | 676         |
| constant    | 60%      | 500x5000 | jv              | 3         | 677            | 677         | 677         |
| constant    | 60%      | 500x5000 | lapmod          | 3         | 861            | 861         | 861         |
| constant    | 60%      | 500x5000 | csa             | 3         | 1.04e+03       | 995         | 1.09e+03    |
| constant    | 60%      | 500x5000 | ramshaw_tarjan  | 3         | 1.9e+03        | 1.9e+03     | 1.91e+03    |
| constant    | 60%      | 500x5000 | sap_dense       | 3         | 2.12e+03       | 2.12e+03    | 2.12e+03    |
| constant    | 60%      | 500x5000 | auction_scaled  | 3         | 2.16e+03       | 2.05e+03    | 2.25e+03    |
| dbl_uniform | 4 blocks | 500x500  | auction_scaled  | 3         | 2.64           | 2.62        | 2.64        |
| dbl_uniform | 4 blocks | 500x500  | csa             | 3         | 3.37           | 3.36        | 3.4         |
| dbl_uniform | 4 blocks | 500x500  | hungarian       | 3         | 3.63           | 3.58        | 3.67        |
| dbl_uniform | 4 blocks | 500x500  | jv              | 3         | 3.68           | 3.6         | 3.7         |
| dbl_uniform | 4 blocks | 500x500  | lapmod          | 3         | 4.21           | 4.14        | 4.25        |
| dbl_uniform | 4 blocks | 500x500  | auto            | 3         | 4.22           | 4.14        | 4.25        |
| dbl_uniform | 4 blocks | 500x500  | ramshaw_tarjan  | 3         | 4.94           | 4.88        | 4.96        |
| dbl_uniform | 4 blocks | 500x500  | csflow          | 3         | 4.97           | 4.93        | 5           |
| dbl_uniform | 4 blocks | 500x500  | sap             | 3         | 5.03           | 4.99        | 5.1         |
| dbl_uniform | 4 blocks | 500x500  | sap_dense       | 3         | 5.48           | 5.36        | 5.5         |
| dbl_uniform | 4 blocks | 500x500  | network_simplex | 3         | 27.9           | 27.3        | 28.2        |
| dbl_uniform | 4 blocks | 500x1500 | hungarian       | 3         | 3.67           | 3.66        | 3.72        |
| dbl_uniform | 4 blocks | 500x1500 | jv              | 3         | 3.67           | 3.67        | 3.73        |

| Regime      | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|-------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| dbl_uniform | 4 blocks | 500x1500 | auto            | 3         | 4.28           | 4.26        | 4.35        |
| dbl_uniform | 4 blocks | 500x1500 | lapmod          | 3         | 4.39           | 4.36        | 4.4         |
| dbl_uniform | 4 blocks | 500x1500 | sap_dense       | 3         | 6.2            | 6.16        | 6.26        |
| dbl_uniform | 4 blocks | 500x1500 | ramshaw_tarjan  | 3         | 7.61           | 7.52        | 7.66        |
| dbl_uniform | 4 blocks | 500x1500 | csflow          | 3         | 9.45           | 9.34        | 9.46        |
| dbl_uniform | 4 blocks | 500x1500 | sap             | 3         | 9.69           | 9.57        | 9.76        |
| dbl_uniform | 4 blocks | 500x1500 | network_simplex | 3         | 92.8           | 91.2        | 95.3        |
| dbl_uniform | 4 blocks | 500x1500 | auction_scaled  | 3         | 281            | 268         | 286         |
| dbl_uniform | 4 blocks | 500x1500 | csa             | 3         | 506            | 498         | 564         |
| dbl_uniform | 4 blocks | 500x5000 | hungarian       | 3         | 11.9           | 11.9        | 11.9        |
| dbl_uniform | 4 blocks | 500x5000 | jv              | 3         | 12             | 11.9        | 12          |
| dbl_uniform | 4 blocks | 500x5000 | auto            | 3         | 13.5           | 13.4        | 13.6        |
| dbl_uniform | 4 blocks | 500x5000 | lapmod          | 3         | 13.5           | 13.5        | 13.5        |
| dbl_uniform | 4 blocks | 500x5000 | sap_dense       | 3         | 19             | 19          | 19.1        |
| dbl_uniform | 4 blocks | 500x5000 | ramshaw_tarjan  | 3         | 23.8           | 23.7        | 23.8        |
| dbl_uniform | 4 blocks | 500x5000 | csflow          | 3         | 30.3           | 30.3        | 30.6        |
| dbl_uniform | 4 blocks | 500x5000 | sap             | 3         | 31.1           | 30.9        | 31.1        |
| dbl_uniform | 4 blocks | 500x5000 | network_simplex | 3         | 527            | 518         | 534         |
| dbl_uniform | 4 blocks | 500x5000 | auction_scaled  | 3         | 3.75e+03       | 3.69e+03    | 3.78e+03    |
| dbl_uniform | 4 blocks | 500x5000 | csa             | 3         | 1.08e+04       | 1.07e+04    | 1.11e+04    |
| dbl_uniform | complete | 500x500  | jv              | 3         | 3.19           | 3.17        | 3.38        |
| dbl_uniform | complete | 500x500  | auto            | 3         | 3.23           | 3.18        | 3.4         |
| dbl_uniform | complete | 500x500  | hungarian       | 3         | 7.7            | 7.63        | 7.96        |
| dbl_uniform | complete | 500x500  | auction_scaled  | 3         | 8.14           | 8.07        | 8.25        |
| dbl_uniform | complete | 500x500  | sap_dense       | 3         | 10.3           | 10.2        | 10.7        |
| dbl_uniform | complete | 500x500  | csa             | 3         | 10.3           | 10.2        | 10.4        |
| dbl_uniform | complete | 500x500  | lapmod          | 3         | 10.7           | 10.6        | 11.1        |
| dbl_uniform | complete | 500x500  | ramshaw_tarjan  | 3         | 13.4           | 13.3        | 13.7        |
| dbl_uniform | complete | 500x500  | sap             | 3         | 19.4           | 19.3        | 19.8        |
| dbl_uniform | complete | 500x500  | csflow          | 3         | 19.4           | 19.3        | 19.7        |
| dbl_uniform | complete | 500x500  | network_simplex | 3         | 91.5           | 85.5        | 98          |
| dbl_uniform | complete | 500x1500 | hungarian       | 3         | 3.93           | 3.86        | 4.01        |
| dbl_uniform | complete | 500x1500 | jv              | 3         | 3.94           | 3.84        | 4.04        |
| dbl_uniform | complete | 500x1500 | lapmod          | 3         | 5.41           | 5.28        | 5.45        |
| dbl_uniform | complete | 500x1500 | sap_dense       | 3         | 6.17           | 6           | 6.21        |
| dbl_uniform | complete | 500x1500 | ramshaw_tarjan  | 3         | 14.2           | 14          | 14.4        |
| dbl_uniform | complete | 500x1500 | auto            | 3         | 26.8           | 26.7        | 27          |
| dbl_uniform | complete | 500x1500 | csflow          | 3         | 26.9           | 26.9        | 27.2        |
| dbl_uniform | complete | 500x1500 | sap             | 3         | 27.1           | 27          | 27.1        |
| dbl_uniform | complete | 500x1500 | network_simplex | 3         | 95.3           | 94.4        | 120         |
| dbl_uniform | complete | 500x1500 | auction_scaled  | 3         | 266            | 251         | 273         |
| dbl_uniform | complete | 500x1500 | csa             | 3         | 507            | 493         | 544         |
| dbl_uniform | complete | 500x5000 | jv              | 3         | 11.4           | 11.3        | 11.5        |
| dbl_uniform | complete | 500x5000 | hungarian       | 3         | 11.5           | 11.3        | 11.6        |
| dbl_uniform | complete | 500x5000 | lapmod          | 3         | 16.4           | 16.3        | 16.4        |
| dbl_uniform | complete | 500x5000 | sap_dense       | 3         | 18.1           | 18          | 18.2        |
| dbl_uniform | complete | 500x5000 | ramshaw_tarjan  | 3         | 46.4           | 46.1        | 46.4        |
| dbl_uniform | complete | 500x5000 | sap             | 3         | 100            | 100         | 101         |
| dbl_uniform | complete | 500x5000 | csflow          | 3         | 100            | 100         | 101         |
| dbl_uniform | complete | 500x5000 | auto            | 3         | 102            | 101         | 102         |
| dbl_uniform | complete | 500x5000 | network_simplex | 3         | 544            | 543         | 568         |

| Regime      | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|-------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| dbl_uniform | complete | 500x5000 | auction_scaled  | 3         | 3.23e+03       | 3.22e+03    | 3.27e+03    |
| dbl_uniform | complete | 500x5000 | csa             | 3         | 1.01e+04       | 9.89e+03    | 1.06e+04    |
| dbl_uniform | 25%      | 500x500  | auction_scaled  | 3         | 4.2            | 4.16        | 4.27        |
| dbl_uniform | 25%      | 500x500  | csa             | 3         | 5.41           | 5.38        | 5.41        |
| dbl_uniform | 25%      | 500x500  | jv              | 3         | 8.55           | 8.22        | 9.05        |
| dbl_uniform | 25%      | 500x500  | hungarian       | 3         | 8.56           | 8.22        | 9.05        |
| dbl_uniform | 25%      | 500x500  | csflow          | 3         | 9.91           | 9.77        | 10.4        |
| dbl_uniform | 25%      | 500x500  | sap             | 3         | 9.93           | 9.73        | 10.4        |
| dbl_uniform | 25%      | 500x500  | lapmod          | 3         | 10.1           | 9.63        | 10.8        |
| dbl_uniform | 25%      | 500x500  | auto            | 3         | 10.2           | 9.66        | 10.8        |
| dbl_uniform | 25%      | 500x500  | sap_dense       | 3         | 17.4           | 16.5        | 18.5        |
| dbl_uniform | 25%      | 500x500  | ramshaw_tarjan  | 3         | 17.6           | 16.8        | 18.8        |
| dbl_uniform | 25%      | 500x500  | network_simplex | 3         | 85.8           | 81.8        | 91.2        |
| dbl_uniform | 25%      | 500x1500 | jv              | 3         | 6.18           | 6.18        | 6.22        |
| dbl_uniform | 25%      | 500x1500 | hungarian       | 3         | 6.2            | 6.18        | 6.21        |
| dbl_uniform | 25%      | 500x1500 | lapmod          | 3         | 7.09           | 7.09        | 7.11        |
| dbl_uniform | 25%      | 500x1500 | auto            | 3         | 7.14           | 7.1         | 7.15        |
| dbl_uniform | 25%      | 500x1500 | sap_dense       | 3         | 10.1           | 10.1        | 10.2        |
| dbl_uniform | 25%      | 500x1500 | ramshaw_tarjan  | 3         | 11.3           | 11.3        | 11.4        |
| dbl_uniform | 25%      | 500x1500 | csflow          | 3         | 12.5           | 12.5        | 12.6        |
| dbl_uniform | 25%      | 500x1500 | sap             | 3         | 12.6           | 12.6        | 12.6        |
| dbl_uniform | 25%      | 500x1500 | network_simplex | 3         | 109            | 104         | 154         |
| dbl_uniform | 25%      | 500x1500 | auction_scaled  | 3         | 239            | 238         | 242         |
| dbl_uniform | 25%      | 500x1500 | csa             | 3         | 577            | 528         | 606         |
| dbl_uniform | 25%      | 500x5000 | hungarian       | 3         | 19.6           | 19.5        | 19.6        |
| dbl_uniform | 25%      | 500x5000 | jv              | 3         | 19.8           | 19.7        | 20.2        |
| dbl_uniform | 25%      | 500x5000 | auto            | 3         | 23.1           | 23.1        | 23.2        |
| dbl_uniform | 25%      | 500x5000 | lapmod          | 3         | 23.2           | 23.1        | 23.2        |
| dbl_uniform | 25%      | 500x5000 | sap_dense       | 3         | 32             | 31.9        | 32.4        |
| dbl_uniform | 25%      | 500x5000 | ramshaw_tarjan  | 3         | 36             | 35.9        | 36.1        |
| dbl_uniform | 25%      | 500x5000 | csflow          | 3         | 41             | 41          | 41.1        |
| dbl_uniform | 25%      | 500x5000 | sap             | 3         | 41.6           | 41.6        | 41.8        |
| dbl_uniform | 25%      | 500x5000 | network_simplex | 3         | 575            | 574         | 595         |
| dbl_uniform | 25%      | 500x5000 | auction_scaled  | 3         | 3.4e+03        | 3.4e+03     | 3.44e+03    |
| dbl_uniform | 25%      | 500x5000 | csa             | 3         | 1.03e+04       | 1e+04       | 1.09e+04    |
| dbl_uniform | 60%      | 500x500  | auction_scaled  | 3         | 7.48           | 7.33        | 7.57        |
| dbl_uniform | 60%      | 500x500  | auto            | 3         | 8.94           | 8.58        | 9.64        |
| dbl_uniform | 60%      | 500x500  | hungarian       | 3         | 8.96           | 8.58        | 9.64        |
| dbl_uniform | 60%      | 500x500  | jv              | 3         | 8.97           | 8.58        | 9.64        |
| dbl_uniform | 60%      | 500x500  | csa             | 3         | 9.73           | 9.6         | 9.81        |
| dbl_uniform | 60%      | 500x500  | lapmod          | 3         | 12.3           | 11.8        | 13.8        |
| dbl_uniform | 60%      | 500x500  | csflow          | 3         | 16.3           | 15.8        | 17          |
| dbl_uniform | 60%      | 500x500  | sap             | 3         | 16.4           | 15.9        | 17          |
| dbl_uniform | 60%      | 500x500  | sap_dense       | 3         | 19.2           | 18.3        | 21.1        |
| dbl_uniform | 60%      | 500x500  | ramshaw_tarjan  | 3         | 22.1           | 21.1        | 23.9        |
| dbl_uniform | 60%      | 500x500  | network_simplex | 3         | 98.8           | 87          | 99.4        |
| dbl_uniform | 60%      | 500x1500 | jv              | 3         | 7.92           | 7.89        | 8.02        |
| dbl_uniform | 60%      | 500x1500 | hungarian       | 3         | 7.98           | 7.93        | 8.01        |
| dbl_uniform | 60%      | 500x1500 | lapmod          | 3         | 9.58           | 9.56        | 9.64        |
| dbl_uniform | 60%      | 500x1500 | sap_dense       | 3         | 13.1           | 13          | 13.1        |
| dbl_uniform | 60%      | 500x1500 | ramshaw_tarjan  | 3         | 17.5           | 17.3        | 17.7        |

| Regime       | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|--------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| dbl_uniform  | 60%      | 500x1500 | csflow          | 3         | 22.9           | 22.9        | 23.2        |
| dbl_uniform  | 60%      | 500x1500 | auto            | 3         | 23.3           | 23.1        | 23.5        |
| dbl_uniform  | 60%      | 500x1500 | sap             | 3         | 23.5           | 23.2        | 23.6        |
| dbl_uniform  | 60%      | 500x1500 | network_simplex | 3         | 117            | 116         | 129         |
| dbl_uniform  | 60%      | 500x1500 | auction_scaled  | 3         | 271            | 267         | 285         |
| dbl_uniform  | 60%      | 500x1500 | csa             | 3         | 579            | 550         | 779         |
| dbl_uniform  | 60%      | 500x5000 | jv              | 3         | 24.6           | 24.5        | 24.8        |
| dbl_uniform  | 60%      | 500x5000 | hungarian       | 3         | 24.6           | 24.5        | 24.8        |
| dbl_uniform  | 60%      | 500x5000 | lapmod          | 3         | 30.8           | 30.7        | 30.8        |
| dbl_uniform  | 60%      | 500x5000 | sap_dense       | 3         | 41.3           | 41.2        | 41.4        |
| dbl_uniform  | 60%      | 500x5000 | ramshaw_tarjan  | 3         | 54.7           | 54.6        | 55          |
| dbl_uniform  | 60%      | 500x5000 | csflow          | 3         | 74.8           | 74.7        | 74.8        |
| dbl_uniform  | 60%      | 500x5000 | sap             | 3         | 75.2           | 74.7        | 75.2        |
| dbl_uniform  | 60%      | 500x5000 | auto            | 3         | 75.2           | 74.7        | 75.3        |
| dbl_uniform  | 60%      | 500x5000 | network_simplex | 3         | 684            | 645         | 697         |
| dbl_uniform  | 60%      | 500x5000 | auction_scaled  | 3         | 3.19e+03       | 3e+03       | 3.33e+03    |
| dbl_uniform  | 60%      | 500x5000 | csa             | 3         | 1e+04          | 1e+04       | 1.02e+04    |
| heavy_tailed | 4 blocks | 500x500  | hungarian       | 3         | 4.18           | 3.91        | 4.29        |
| heavy_tailed | 4 blocks | 500x500  | jv              | 3         | 4.18           | 3.92        | 4.31        |
| heavy_tailed | 4 blocks | 500x500  | auto            | 3         | 4.87           | 4.56        | 5           |
| heavy_tailed | 4 blocks | 500x500  | lapmod          | 3         | 4.87           | 4.56        | 5           |
| heavy_tailed | 4 blocks | 500x500  | csflow          | 3         | 5.3            | 5.12        | 5.37        |
| heavy_tailed | 4 blocks | 500x500  | sap             | 3         | 5.33           | 5.16        | 5.46        |
| heavy_tailed | 4 blocks | 500x500  | ramshaw_tarjan  | 3         | 5.42           | 5.23        | 5.51        |
| heavy_tailed | 4 blocks | 500x500  | auction_scaled  | 3         | 5.82           | 5.78        | 5.87        |
| heavy_tailed | 4 blocks | 500x500  | sap_dense       | 3         | 6.19           | 5.84        | 6.35        |
| heavy_tailed | 4 blocks | 500x500  | csa             | 3         | 7              | 6.83        | 7.06        |
| heavy_tailed | 4 blocks | 500x500  | network_simplex | 3         | 28.9           | 27.3        | 29.9        |
| heavy_tailed | 4 blocks | 500x1500 | jv              | 3         | 3.66           | 3.66        | 3.66        |
| heavy_tailed | 4 blocks | 500x1500 | hungarian       | 3         | 3.69           | 3.67        | 3.76        |
| heavy_tailed | 4 blocks | 500x1500 | lapmod          | 3         | 4.21           | 4.21        | 4.26        |
| heavy_tailed | 4 blocks | 500x1500 | auto            | 3         | 4.24           | 4.22        | 4.3         |
| heavy_tailed | 4 blocks | 500x1500 | sap_dense       | 3         | 6.06           | 6.06        | 6.14        |
| heavy_tailed | 4 blocks | 500x1500 | ramshaw_tarjan  | 3         | 7.44           | 7.41        | 7.45        |
| heavy_tailed | 4 blocks | 500x1500 | csflow          | 3         | 9.4            | 9.32        | 9.42        |
| heavy_tailed | 4 blocks | 500x1500 | sap             | 3         | 9.59           | 9.49        | 9.59        |
| heavy_tailed | 4 blocks | 500x1500 | network_simplex | 3         | 109            | 103         | 110         |
| heavy_tailed | 4 blocks | 500x1500 | csa             | 3         | 184            | 182         | 203         |
| heavy_tailed | 4 blocks | 500x1500 | auction_scaled  | 1         | 264            | 264         | 264         |
| heavy_tailed | 4 blocks | 500x5000 | jv              | 3         | 12             | 12          | 12          |
| heavy_tailed | 4 blocks | 500x5000 | lapmod          | 3         | 13.5           | 13.5        | 13.6        |
| heavy_tailed | 4 blocks | 500x5000 | auto            | 3         | 13.6           | 13.6        | 18          |
| heavy_tailed | 4 blocks | 500x5000 | hungarian       | 3         | 18.8           | 15.4        | 18.8        |
| heavy_tailed | 4 blocks | 500x5000 | ramshaw_tarjan  | 3         | 23.8           | 23.7        | 27.2        |
| heavy_tailed | 4 blocks | 500x5000 | csflow          | 3         | 30.5           | 30.5        | 31          |
| heavy_tailed | 4 blocks | 500x5000 | sap_dense       | 3         | 30.7           | 30.1        | 30.8        |
| heavy_tailed | 4 blocks | 500x5000 | sap             | 3         | 39.1           | 38.4        | 39.5        |
| heavy_tailed | 4 blocks | 500x5000 | network_simplex | 3         | 669            | 655         | 686         |
| heavy_tailed | 4 blocks | 500x5000 | csa             | 3         | 7.43e+03       | 7.38e+03    | 7.43e+03    |
| heavy_tailed | complete | 500x500  | auto            | 3         | 3.62           | 3.61        | 3.81        |
| heavy_tailed | complete | 500x500  | jv              | 3         | 3.64           | 3.63        | 3.84        |

| Regime       | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|--------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| heavy_tailed | complete | 500x500  | hungarian       | 3         | 8.11           | 7.62        | 8.67        |
| heavy_tailed | complete | 500x500  | sap_dense       | 3         | 10.9           | 10.2        | 11.7        |
| heavy_tailed | complete | 500x500  | lapmod          | 3         | 11.4           | 10.7        | 12.3        |
| heavy_tailed | complete | 500x500  | ramshaw_tarjan  | 3         | 13.7           | 13.2        | 14.4        |
| heavy_tailed | complete | 500x500  | csa             | 3         | 18.4           | 17.8        | 18.7        |
| heavy_tailed | complete | 500x500  | csflow          | 3         | 19.6           | 19          | 20.5        |
| heavy_tailed | complete | 500x500  | sap             | 3         | 19.7           | 19.1        | 20.5        |
| heavy_tailed | complete | 500x500  | auction_scaled  | 3         | 20.6           | 20.2        | 21          |
| heavy_tailed | complete | 500x500  | network_simplex | 3         | 88.1           | 87          | 92.6        |
| heavy_tailed | complete | 500x1500 | jv              | 3         | 3.74           | 3.72        | 3.78        |
| heavy_tailed | complete | 500x1500 | hungarian       | 3         | 3.82           | 3.76        | 3.86        |
| heavy_tailed | complete | 500x1500 | lapmod          | 3         | 5.28           | 5.26        | 5.47        |
| heavy_tailed | complete | 500x1500 | sap_dense       | 3         | 5.9            | 5.87        | 5.96        |
| heavy_tailed | complete | 500x1500 | ramshaw_tarjan  | 3         | 13.9           | 13.8        | 14          |
| heavy_tailed | complete | 500x1500 | sap             | 3         | 26.5           | 26.4        | 26.6        |
| heavy_tailed | complete | 500x1500 | auto            | 3         | 26.7           | 26.6        | 26.8        |
| heavy_tailed | complete | 500x1500 | csflow          | 3         | 26.7           | 26.5        | 26.7        |
| heavy_tailed | complete | 500x1500 | network_simplex | 3         | 112            | 107         | 130         |
| heavy_tailed | complete | 500x1500 | csa             | 3         | 148            | 147         | 150         |
| heavy_tailed | complete | 500x1500 | auction_scaled  | 1         | 257            | 257         | 257         |
| heavy_tailed | complete | 500x5000 | hungarian       | 3         | 11.2           | 11.1        | 14.9        |
| heavy_tailed | complete | 500x5000 | jv              | 3         | 11.2           | 11.2        | 11.3        |
| heavy_tailed | complete | 500x5000 | lapmod          | 3         | 16.6           | 16.4        | 21.4        |
| heavy_tailed | complete | 500x5000 | sap_dense       | 3         | 22.4           | 20.2        | 25.7        |
| heavy_tailed | complete | 500x5000 | ramshaw_tarjan  | 3         | 52.7           | 49.2        | 54.1        |
| heavy_tailed | complete | 500x5000 | auto            | 3         | 100            | 100         | 105         |
| heavy_tailed | complete | 500x5000 | csflow          | 3         | 101            | 100         | 102         |
| heavy_tailed | complete | 500x5000 | sap             | 3         | 108            | 104         | 110         |
| heavy_tailed | complete | 500x5000 | network_simplex | 3         | 787            | 740         | 808         |
| heavy_tailed | complete | 500x5000 | csa             | 3         | 7.08e+03       | 6.95e+03    | 7.09e+03    |
| heavy_tailed | 25%      | 500x500  | auction_scaled  | 3         | 7.39           | 7.22        | 7.55        |
| heavy_tailed | 25%      | 500x500  | csa             | 3         | 8.69           | 8.53        | 9.04        |
| heavy_tailed | 25%      | 500x500  | jv              | 3         | 10.1           | 8.74        | 10.1        |
| heavy_tailed | 25%      | 500x500  | hungarian       | 3         | 10.1           | 8.76        | 10.2        |
| heavy_tailed | 25%      | 500x500  | csflow          | 3         | 10.8           | 9.99        | 11          |
| heavy_tailed | 25%      | 500x500  | sap             | 3         | 10.8           | 10          | 11          |
| heavy_tailed | 25%      | 500x500  | lapmod          | 3         | 12.1           | 10.3        | 12.2        |
| heavy_tailed | 25%      | 500x500  | auto            | 3         | 12.1           | 10.3        | 12.2        |
| heavy_tailed | 25%      | 500x500  | sap_dense       | 3         | 20.9           | 17.7        | 21          |
| heavy_tailed | 25%      | 500x500  | ramshaw_tarjan  | 3         | 21             | 17.9        | 21          |
| heavy_tailed | 25%      | 500x500  | network_simplex | 3         | 93.8           | 89.2        | 95.2        |
| heavy_tailed | 25%      | 500x1500 | jv              | 3         | 6.3            | 6.29        | 6.34        |
| heavy_tailed | 25%      | 500x1500 | hungarian       | 3         | 6.31           | 6.25        | 6.33        |
| heavy_tailed | 25%      | 500x1500 | lapmod          | 3         | 7.09           | 7.09        | 7.12        |
| heavy_tailed | 25%      | 500x1500 | auto            | 3         | 7.13           | 7.1         | 7.17        |
| heavy_tailed | 25%      | 500x1500 | sap_dense       | 3         | 10.2           | 10.2        | 10.2        |
| heavy_tailed | 25%      | 500x1500 | ramshaw_tarjan  | 3         | 11.5           | 11.4        | 11.5        |
| heavy_tailed | 25%      | 500x1500 | csflow          | 3         | 12.6           | 12.5        | 12.6        |
| heavy_tailed | 25%      | 500x1500 | sap             | 3         | 12.7           | 12.7        | 12.8        |
| heavy_tailed | 25%      | 500x1500 | csa             | 3         | 127            | 119         | 129         |
| heavy_tailed | 25%      | 500x1500 | network_simplex | 3         | 133            | 128         | 147         |



| Regime       | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|--------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| heavy_tailed | 25%      | 500x1500 | auction_scaled  | 3         | 231            | 227         | 248         |
| heavy_tailed | 25%      | 500x5000 | hungarian       | 3         | 19.8           | 19.7        | 23.1        |
| heavy_tailed | 25%      | 500x5000 | lapmod          | 3         | 23.2           | 23.1        | 23.2        |
| heavy_tailed | 25%      | 500x5000 | auto            | 3         | 23.2           | 23.2        | 27.5        |
| heavy_tailed | 25%      | 500x5000 | jv              | 3         | 26.5           | 23.1        | 27.5        |
| heavy_tailed | 25%      | 500x5000 | ramshaw_tarjan  | 3         | 36             | 36          | 39.5        |
| heavy_tailed | 25%      | 500x5000 | sap_dense       | 3         | 37.1           | 37.1        | 40.5        |
| heavy_tailed | 25%      | 500x5000 | csflow          | 3         | 41.4           | 41.3        | 46.2        |
| heavy_tailed | 25%      | 500x5000 | sap             | 3         | 42.3           | 41.9        | 46.1        |
| heavy_tailed | 25%      | 500x5000 | network_simplex | 3         | 698            | 684         | 752         |
| heavy_tailed | 25%      | 500x5000 | csa             | 3         | 6.77e+03       | 6.67e+03    | 6.79e+03    |
| heavy_tailed | 60%      | 500x500  | jv              | 3         | 8.58           | 8.53        | 9.18        |
| heavy_tailed | 60%      | 500x500  | hungarian       | 3         | 8.61           | 8.56        | 9.18        |
| heavy_tailed | 60%      | 500x500  | auto            | 3         | 8.61           | 8.55        | 9.19        |
| heavy_tailed | 60%      | 500x500  | lapmod          | 3         | 12.2           | 12.1        | 13.3        |
| heavy_tailed | 60%      | 500x500  | auction_scaled  | 3         | 14.2           | 14.2        | 14.3        |
| heavy_tailed | 60%      | 500x500  | csa             | 3         | 14.3           | 14.1        | 14.5        |
| heavy_tailed | 60%      | 500x500  | sap             | 3         | 15.8           | 15.8        | 16.5        |
| heavy_tailed | 60%      | 500x500  | csflow          | 3         | 15.8           | 15.8        | 16.5        |
| heavy_tailed | 60%      | 500x500  | sap_dense       | 3         | 18.4           | 18.2        | 20          |
| heavy_tailed | 60%      | 500x500  | ramshaw_tarjan  | 3         | 20.9           | 20.8        | 22.6        |
| heavy_tailed | 60%      | 500x500  | network_simplex | 3         | 85.6           | 84.9        | 87.6        |
| heavy_tailed | 60%      | 500x1500 | jv              | 3         | 7.9            | 7.83        | 7.9         |
| heavy_tailed | 60%      | 500x1500 | hungarian       | 3         | 7.92           | 7.78        | 7.95        |
| heavy_tailed | 60%      | 500x1500 | lapmod          | 3         | 9.52           | 9.48        | 9.55        |
| heavy_tailed | 60%      | 500x1500 | sap_dense       | 3         | 12.8           | 12.8        | 13          |
| heavy_tailed | 60%      | 500x1500 | ramshaw_tarjan  | 3         | 17.2           | 17.2        | 17.4        |
| heavy_tailed | 60%      | 500x1500 | csflow          | 3         | 22.8           | 22.8        | 23          |
| heavy_tailed | 60%      | 500x1500 | sap             | 3         | 23             | 23          | 23.1        |
| heavy_tailed | 60%      | 500x1500 | auto            | 3         | 23.2           | 23          | 23.2        |
| heavy_tailed | 60%      | 500x1500 | csa             | 3         | 132            | 127         | 138         |
| heavy_tailed | 60%      | 500x1500 | network_simplex | 3         | 168            | 159         | 185         |
| heavy_tailed | 60%      | 500x5000 | jv              | 3         | 27.2           | 26.3        | 29.5        |
| heavy_tailed | 60%      | 500x5000 | hungarian       | 3         | 27.5           | 27.2        | 29.6        |
| heavy_tailed | 60%      | 500x5000 | lapmod          | 3         | 30.9           | 30.8        | 36.7        |
| heavy_tailed | 60%      | 500x5000 | sap_dense       | 3         | 45.6           | 45.5        | 49.7        |
| heavy_tailed | 60%      | 500x5000 | ramshaw_tarjan  | 3         | 55.2           | 55.1        | 58.9        |
| heavy_tailed | 60%      | 500x5000 | csflow          | 3         | 77             | 76.1        | 77.1        |
| heavy_tailed | 60%      | 500x5000 | sap             | 3         | 77.5           | 76.6        | 81.4        |
| heavy_tailed | 60%      | 500x5000 | auto            | 3         | 83.8           | 79.3        | 84.8        |
| heavy_tailed | 60%      | 500x5000 | network_simplex | 3         | 740            | 726         | 751         |
| heavy_tailed | 60%      | 500x5000 | csa             | 3         | 6.95e+03       | 6.92e+03    | 6.95e+03    |
| int_uniform  | 4 blocks | 500x500  | auction_scaled  | 3         | 2.91           | 2.89        | 2.96        |
| int_uniform  | 4 blocks | 500x500  | csa             | 3         | 3.07           | 3.04        | 3.12        |
| int_uniform  | 4 blocks | 500x500  | jv              | 3         | 3.72           | 3.71        | 3.87        |
| int_uniform  | 4 blocks | 500x500  | hungarian       | 3         | 3.76           | 3.74        | 3.91        |
| int_uniform  | 4 blocks | 500x500  | lapmod          | 3         | 4.31           | 4.3         | 4.5         |
| int_uniform  | 4 blocks | 500x500  | auto            | 3         | 4.32           | 4.31        | 4.51        |
| int_uniform  | 4 blocks | 500x500  | ramshaw_tarjan  | 3         | 5.08           | 5.06        | 5.18        |
| int_uniform  | 4 blocks | 500x500  | csflow          | 3         | 5.16           | 5.11        | 5.21        |
| int_uniform  | 4 blocks | 500x500  | sap             | 3         | 5.29           | 5.2         | 5.3         |

| Regime      | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|-------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| int_uniform | 4 blocks | 500x500  | sap_dense       | 3         | 5.59           | 5.56        | 5.83        |
| int_uniform | 4 blocks | 500x500  | network_simplex | 3         | 28.4           | 27.4        | 29.5        |
| int_uniform | 4 blocks | 500x1500 | hungarian       | 3         | 3.74           | 3.68        | 3.74        |
| int_uniform | 4 blocks | 500x1500 | jv              | 3         | 3.76           | 3.67        | 3.8         |
| int_uniform | 4 blocks | 500x1500 | lapmod          | 3         | 4.3            | 4.28        | 4.36        |
| int_uniform | 4 blocks | 500x1500 | auto            | 3         | 4.34           | 4.25        | 4.37        |
| int_uniform | 4 blocks | 500x1500 | sap_dense       | 3         | 6.31           | 6.2         | 6.33        |
| int_uniform | 4 blocks | 500x1500 | ramshaw_tarjan  | 3         | 7.52           | 7.39        | 7.54        |
| int_uniform | 4 blocks | 500x1500 | csflow          | 3         | 9.39           | 9.36        | 9.42        |
| int_uniform | 4 blocks | 500x1500 | sap             | 3         | 9.54           | 9.39        | 9.68        |
| int_uniform | 4 blocks | 500x1500 | network_simplex | 3         | 89.1           | 88.9        | 89.3        |
| int_uniform | 4 blocks | 500x1500 | csa             | 3         | 480            | 399         | 493         |
| int_uniform | 4 blocks | 500x1500 | auction_scaled  | 3         | 527            | 458         | 538         |
| int_uniform | 4 blocks | 500x5000 | hungarian       | 3         | 11.9           | 11.8        | 11.9        |
| int_uniform | 4 blocks | 500x5000 | jv              | 3         | 12             | 11.9        | 12          |
| int_uniform | 4 blocks | 500x5000 | auto            | 3         | 13.4           | 13.4        | 13.5        |
| int_uniform | 4 blocks | 500x5000 | lapmod          | 3         | 13.5           | 13.4        | 13.5        |
| int_uniform | 4 blocks | 500x5000 | sap_dense       | 3         | 19.8           | 19.8        | 19.9        |
| int_uniform | 4 blocks | 500x5000 | ramshaw_tarjan  | 3         | 23.7           | 23.7        | 23.8        |
| int_uniform | 4 blocks | 500x5000 | csflow          | 3         | 30.2           | 30.1        | 30.2        |
| int_uniform | 4 blocks | 500x5000 | sap             | 3         | 30.7           | 30.6        | 30.8        |
| int_uniform | 4 blocks | 500x5000 | network_simplex | 3         | 507            | 507         | 521         |
| int_uniform | 4 blocks | 500x5000 | csa             | 3         | 4.88e+03       | 4.6e+03     | 4.93e+03    |
| int_uniform | 4 blocks | 500x5000 | auction_scaled  | 3         | 6.11e+03       | 6.07e+03    | 6.96e+03    |
| int_uniform | complete | 500x500  | jv              | 3         | 4.45           | 4.05        | 4.49        |
| int_uniform | complete | 500x500  | auto            | 3         | 4.54           | 4.11        | 4.55        |
| int_uniform | complete | 500x500  | hungarian       | 3         | 8.46           | 8.39        | 8.66        |
| int_uniform | complete | 500x500  | csa             | 3         | 9.81           | 9.7         | 9.94        |
| int_uniform | complete | 500x500  | auction_scaled  | 3         | 9.93           | 9.82        | 10.1        |
| int_uniform | complete | 500x500  | sap_dense       | 3         | 11.3           | 11.1        | 11.6        |
| int_uniform | complete | 500x500  | lapmod          | 3         | 12             | 11.8        | 12.3        |
| int_uniform | complete | 500x500  | ramshaw_tarjan  | 3         | 14.3           | 14.3        | 14.5        |
| int_uniform | complete | 500x500  | sap             | 3         | 20.3           | 20.2        | 20.6        |
| int_uniform | complete | 500x500  | csflow          | 3         | 20.4           | 20.3        | 20.7        |
| int_uniform | complete | 500x500  | network_simplex | 3         | 95.7           | 89.7        | 96.6        |
| int_uniform | complete | 500x1500 | hungarian       | 3         | 4.14           | 4.11        | 4.2         |
| int_uniform | complete | 500x1500 | jv              | 3         | 4.16           | 4.14        | 4.2         |
| int_uniform | complete | 500x1500 | lapmod          | 3         | 5.57           | 5.53        | 5.64        |
| int_uniform | complete | 500x1500 | sap_dense       | 3         | 6.06           | 5.98        | 6.09        |
| int_uniform | complete | 500x1500 | ramshaw_tarjan  | 3         | 14.5           | 14.5        | 14.5        |
| int_uniform | complete | 500x1500 | csflow          | 3         | 26.9           | 26.7        | 27.1        |
| int_uniform | complete | 500x1500 | sap             | 3         | 27.2           | 27.1        | 27.2        |
| int_uniform | complete | 500x1500 | auto            | 3         | 27.2           | 27.1        | 27.3        |
| int_uniform | complete | 500x1500 | network_simplex | 3         | 87.3           | 86.4        | 92.9        |
| int_uniform | complete | 500x1500 | csa             | 3         | 389            | 383         | 425         |
| int_uniform | complete | 500x1500 | auction_scaled  | 3         | 534            | 514         | 596         |
| int_uniform | complete | 500x5000 | hungarian       | 3         | 11.9           | 11.8        | 11.9        |
| int_uniform | complete | 500x5000 | jv              | 3         | 12.1           | 12.1        | 12.2        |
| int_uniform | complete | 500x5000 | lapmod          | 3         | 17             | 17          | 17.1        |
| int_uniform | complete | 500x5000 | sap_dense       | 3         | 18.4           | 18.4        | 18.5        |
| int_uniform | complete | 500x5000 | ramshaw_tarjan  | 3         | 46.6           | 46.6        | 47          |

| Regime      | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|-------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| int_uniform | complete | 500x5000 | sap             | 3         | 99.6           | 99.5        | 100         |
| int_uniform | complete | 500x5000 | csflow          | 3         | 101            | 100         | 101         |
| int_uniform | complete | 500x5000 | auto            | 3         | 101            | 101         | 101         |
| int_uniform | complete | 500x5000 | network_simplex | 3         | 541            | 538         | 628         |
| int_uniform | complete | 500x5000 | csa             | 3         | 4.71e+03       | 4.7e+03     | 4.84e+03    |
| int_uniform | complete | 500x5000 | auction_scaled  | 3         | 4.76e+03       | 4.48e+03    | 5.26e+03    |
| int_uniform | 25%      | 500x500  | auction_scaled  | 3         | 4.43           | 4.43        | 4.54        |
| int_uniform | 25%      | 500x500  | csa             | 3         | 4.71           | 4.68        | 4.82        |
| int_uniform | 25%      | 500x500  | jv              | 3         | 6.75           | 6.74        | 7.75        |
| int_uniform | 25%      | 500x500  | hungarian       | 3         | 6.77           | 6.75        | 7.75        |
| int_uniform | 25%      | 500x500  | lapmod          | 3         | 7.68           | 7.6         | 9.04        |
| int_uniform | 25%      | 500x500  | auto            | 3         | 7.69           | 7.64        | 9.02        |
| int_uniform | 25%      | 500x500  | csflow          | 3         | 8.95           | 8.9         | 9.57        |
| int_uniform | 25%      | 500x500  | sap             | 3         | 9.02           | 8.9         | 9.65        |
| int_uniform | 25%      | 500x500  | sap_dense       | 3         | 13.1           | 13          | 15.5        |
| int_uniform | 25%      | 500x500  | ramshaw_tarjan  | 3         | 13.6           | 13.6        | 15.9        |
| int_uniform | 25%      | 500x500  | network_simplex | 3         | 102            | 97.1        | 103         |
| int_uniform | 25%      | 500x1500 | hungarian       | 3         | 6.22           | 6.21        | 6.29        |
| int_uniform | 25%      | 500x1500 | jv              | 3         | 6.32           | 6.25        | 6.34        |
| int_uniform | 25%      | 500x1500 | lapmod          | 3         | 7.13           | 7.1         | 7.15        |
| int_uniform | 25%      | 500x1500 | auto            | 3         | 7.17           | 7.17        | 7.19        |
| int_uniform | 25%      | 500x1500 | sap_dense       | 3         | 10.2           | 10.1        | 10.2        |
| int_uniform | 25%      | 500x1500 | ramshaw_tarjan  | 3         | 11.4           | 11.4        | 11.5        |
| int_uniform | 25%      | 500x1500 | csflow          | 3         | 12.6           | 12.5        | 12.7        |
| int_uniform | 25%      | 500x1500 | sap             | 3         | 12.7           | 12.7        | 12.8        |
| int_uniform | 25%      | 500x1500 | network_simplex | 3         | 102            | 101         | 129         |
| int_uniform | 25%      | 500x1500 | auction_scaled  | 3         | 365            | 354         | 455         |
| int_uniform | 25%      | 500x1500 | csa             | 3         | 463            | 387         | 495         |
| int_uniform | 25%      | 500x5000 | hungarian       | 3         | 19.5           | 19.5        | 19.6        |
| int_uniform | 25%      | 500x5000 | jv              | 3         | 19.7           | 19.7        | 19.7        |
| int_uniform | 25%      | 500x5000 | lapmod          | 3         | 23.1           | 23          | 23.2        |
| int_uniform | 25%      | 500x5000 | auto            | 3         | 23.2           | 23.1        | 23.2        |
| int_uniform | 25%      | 500x5000 | sap_dense       | 3         | 32.7           | 32.7        | 32.8        |
| int_uniform | 25%      | 500x5000 | ramshaw_tarjan  | 3         | 35.9           | 35.9        | 36          |
| int_uniform | 25%      | 500x5000 | csflow          | 3         | 41.2           | 41.1        | 41.2        |
| int_uniform | 25%      | 500x5000 | sap             | 3         | 41.7           | 41.6        | 41.8        |
| int_uniform | 25%      | 500x5000 | network_simplex | 3         | 587            | 578         | 602         |
| int_uniform | 25%      | 500x5000 | auction_scaled  | 3         | 5.5e+03        | 5.42e+03    | 6.25e+03    |
| int_uniform | 25%      | 500x5000 | csa             | 3         | 6.17e+03       | 5.56e+03    | 6.31e+03    |
| int_uniform | 60%      | 500x500  | auction_scaled  | 3         | 8.17           | 8.16        | 8.19        |
| int_uniform | 60%      | 500x500  | csa             | 3         | 8.39           | 8.38        | 8.46        |
| int_uniform | 60%      | 500x500  | jv              | 3         | 8.8            | 8.75        | 8.93        |
| int_uniform | 60%      | 500x500  | hungarian       | 3         | 8.8            | 8.75        | 8.93        |
| int_uniform | 60%      | 500x500  | auto            | 3         | 8.82           | 8.77        | 8.93        |
| int_uniform | 60%      | 500x500  | lapmod          | 3         | 12.6           | 12.1        | 12.7        |
| int_uniform | 60%      | 500x500  | sap             | 3         | 16.2           | 16.1        | 16.4        |
| int_uniform | 60%      | 500x500  | csflow          | 3         | 16.2           | 16.1        | 16.4        |
| int_uniform | 60%      | 500x500  | sap_dense       | 3         | 19.2           | 18.9        | 19.4        |
| int_uniform | 60%      | 500x500  | ramshaw_tarjan  | 3         | 21.6           | 21.4        | 22.1        |
| int_uniform | 60%      | 500x500  | network_simplex | 3         | 85.8           | 85.1        | 94.8        |
| int_uniform | 60%      | 500x1500 | hungarian       | 3         | 7.96           | 7.91        | 8.16        |

| Regime           | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|------------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| int_uniform      | 60%      | 500x1500 | jv              | 3         | 8.06           | 8.05        | 8.12        |
| int_uniform      | 60%      | 500x1500 | lapmod          | 3         | 9.68           | 9.58        | 9.77        |
| int_uniform      | 60%      | 500x1500 | sap_dense       | 3         | 13.2           | 13.1        | 13.4        |
| int_uniform      | 60%      | 500x1500 | ramshaw_tarjan  | 3         | 17.8           | 17.6        | 18          |
| int_uniform      | 60%      | 500x1500 | csflow          | 3         | 23.2           | 23.1        | 23.5        |
| int_uniform      | 60%      | 500x1500 | auto            | 3         | 23.5           | 23.5        | 23.7        |
| int_uniform      | 60%      | 500x1500 | sap             | 3         | 23.6           | 23.4        | 23.8        |
| int_uniform      | 60%      | 500x1500 | network_simplex | 3         | 130            | 124         | 145         |
| int_uniform      | 60%      | 500x1500 | csa             | 3         | 369            | 358         | 385         |
| int_uniform      | 60%      | 500x1500 | auction_scaled  | 3         | 445            | 416         | 452         |
| int_uniform      | 60%      | 500x5000 | hungarian       | 3         | 24.7           | 24.7        | 24.8        |
| int_uniform      | 60%      | 500x5000 | jv              | 3         | 25.2           | 25.1        | 25.3        |
| int_uniform      | 60%      | 500x5000 | lapmod          | 3         | 30.9           | 30.8        | 31          |
| int_uniform      | 60%      | 500x5000 | sap_dense       | 3         | 41.4           | 41.4        | 41.4        |
| int_uniform      | 60%      | 500x5000 | ramshaw_tarjan  | 3         | 55.5           | 55.4        | 55.6        |
| int_uniform      | 60%      | 500x5000 | csflow          | 3         | 74.8           | 74.7        | 75.4        |
| int_uniform      | 60%      | 500x5000 | sap             | 3         | 75             | 75          | 75.2        |
| int_uniform      | 60%      | 500x5000 | auto            | 3         | 75.9           | 75.8        | 76          |
| int_uniform      | 60%      | 500x5000 | network_simplex | 3         | 623            | 621         | 679         |
| int_uniform      | 60%      | 500x5000 | csa             | 3         | 4.51e+03       | 4.46e+03    | 4.7e+03     |
| int_uniform      | 60%      | 500x5000 | auction_scaled  | 3         | 5.39e+03       | 5.17e+03    | 5.61e+03    |
| metric_clustered | 4 blocks | 500x500  | auction_scaled  | 3         | 2.73           | 2.7         | 2.75        |
| metric_clustered | 4 blocks | 500x500  | csa             | 3         | 3.35           | 3.35        | 3.4         |
| metric_clustered | 4 blocks | 500x500  | jv              | 3         | 5.05           | 4.75        | 5.06        |
| metric_clustered | 4 blocks | 500x500  | hungarian       | 3         | 5.05           | 4.77        | 5.05        |
| metric_clustered | 4 blocks | 500x500  | lapmod          | 3         | 5.87           | 5.54        | 5.9         |
| metric_clustered | 4 blocks | 500x500  | auto            | 3         | 5.91           | 5.56        | 5.92        |
| metric_clustered | 4 blocks | 500x500  | csflow          | 3         | 6.06           | 5.93        | 6.16        |
| metric_clustered | 4 blocks | 500x500  | sap             | 3         | 6.1            | 6.01        | 6.26        |
| metric_clustered | 4 blocks | 500x500  | ramshaw_tarjan  | 3         | 6.41           | 6.17        | 6.5         |
| metric_clustered | 4 blocks | 500x500  | sap_dense       | 3         | 7.43           | 7.05        | 7.45        |
| metric_clustered | 4 blocks | 500x500  | network_simplex | 3         | 19.3           | 19          | 20.5        |
| metric_clustered | 4 blocks | 500x1500 | hungarian       | 3         | 3.84           | 3.81        | 3.87        |
| metric_clustered | 4 blocks | 500x1500 | jv              | 3         | 3.85           | 3.83        | 3.87        |
| metric_clustered | 4 blocks | 500x1500 | lapmod          | 3         | 4.44           | 4.41        | 4.52        |
| metric_clustered | 4 blocks | 500x1500 | auto            | 3         | 4.47           | 4.45        | 4.55        |
| metric_clustered | 4 blocks | 500x1500 | sap_dense       | 3         | 6.34           | 6.32        | 6.35        |
| metric_clustered | 4 blocks | 500x1500 | ramshaw_tarjan  | 3         | 7.28           | 7.17        | 7.36        |
| metric_clustered | 4 blocks | 500x1500 | sap             | 3         | 8.95           | 8.91        | 9.01        |
| metric_clustered | 4 blocks | 500x1500 | csflow          | 3         | 8.98           | 8.95        | 9.12        |
| metric_clustered | 4 blocks | 500x1500 | network_simplex | 3         | 83.4           | 81.5        | 88.7        |
| metric_clustered | 4 blocks | 500x1500 | auction_scaled  | 3         | 354            | 350         | 370         |
| metric_clustered | 4 blocks | 500x1500 | csa             | 3         | 654            | 594         | 661         |
| metric_clustered | 4 blocks | 500x5000 | jv              | 3         | 12.1           | 12          | 15.5        |
| metric_clustered | 4 blocks | 500x5000 | lapmod          | 3         | 13.6           | 13.5        | 18          |
| metric_clustered | 4 blocks | 500x5000 | auto            | 3         | 13.9           | 13.7        | 18.1        |
| metric_clustered | 4 blocks | 500x5000 | hungarian       | 3         | 18.1           | 15.8        | 18.5        |
| metric_clustered | 4 blocks | 500x5000 | sap_dense       | 3         | 28.2           | 25.8        | 29.2        |
| metric_clustered | 4 blocks | 500x5000 | csflow          | 3         | 29.3           | 29.2        | 29.5        |
| metric_clustered | 4 blocks | 500x5000 | sap             | 3         | 29.8           | 29.6        | 29.9        |
| metric_clustered | 4 blocks | 500x5000 | ramshaw_tarjan  | 3         | 30.1           | 26.7        | 30.9        |

| Regime           | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|------------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| metric_clustered | 4 blocks | 500x5000 | network_simplex | 3         | 598            | 596         | 600         |
| metric_clustered | 4 blocks | 500x5000 | auction_scaled  | 3         | 4.61e+03       | 4.49e+03    | 4.64e+03    |
| metric_clustered | 4 blocks | 500x5000 | csa             | 3         | 1.34e+04       | 1.21e+04    | 1.4e+04     |
| metric_clustered | complete | 500x500  | jv              | 3         | 6.25           | 6.05        | 7.08        |
| metric_clustered | complete | 500x500  | auto            | 3         | 6.25           | 6.04        | 7.08        |
| metric_clustered | complete | 500x500  | auction_scaled  | 3         | 7.74           | 7.54        | 8.33        |
| metric_clustered | complete | 500x500  | csa             | 3         | 9.53           | 9.36        | 10.3        |
| metric_clustered | complete | 500x500  | hungarian       | 3         | 9.88           | 9.47        | 11.4        |
| metric_clustered | complete | 500x500  | sap_dense       | 3         | 13.2           | 12.7        | 15.4        |
| metric_clustered | complete | 500x500  | lapmod          | 3         | 14.1           | 13.5        | 16.6        |
| metric_clustered | complete | 500x500  | ramshaw_tarjan  | 3         | 17.6           | 16.8        | 19.2        |
| metric_clustered | complete | 500x500  | sap             | 3         | 23.6           | 22.8        | 25.7        |
| metric_clustered | complete | 500x500  | csflow          | 3         | 23.7           | 22.8        | 25.8        |
| metric_clustered | complete | 500x500  | network_simplex | 3         | 35             | 34          | 36.5        |
| metric_clustered | complete | 500x1500 | jv              | 3         | 3.91           | 3.85        | 3.92        |
| metric_clustered | complete | 500x1500 | hungarian       | 3         | 3.91           | 3.85        | 3.91        |
| metric_clustered | complete | 500x1500 | lapmod          | 3         | 5.46           | 5.37        | 5.47        |
| metric_clustered | complete | 500x1500 | sap_dense       | 3         | 6.07           | 6           | 6.07        |
| metric_clustered | complete | 500x1500 | ramshaw_tarjan  | 3         | 13.3           | 13.2        | 13.3        |
| metric_clustered | complete | 500x1500 | csflow          | 3         | 24.9           | 24.8        | 25.1        |
| metric_clustered | complete | 500x1500 | auto            | 3         | 25.4           | 25.2        | 25.5        |
| metric_clustered | complete | 500x1500 | sap             | 3         | 25.5           | 25.2        | 25.8        |
| metric_clustered | complete | 500x1500 | network_simplex | 3         | 98             | 97.7        | 103         |
| metric_clustered | complete | 500x1500 | auction_scaled  | 3         | 350            | 341         | 379         |
| metric_clustered | complete | 500x1500 | csa             | 3         | 512            | 498         | 611         |
| metric_clustered | complete | 500x5000 | jv              | 3         | 11.1           | 11          | 14.8        |
| metric_clustered | complete | 500x5000 | hungarian       | 3         | 11.1           | 11          | 11.2        |
| metric_clustered | complete | 500x5000 | lapmod          | 3         | 16.7           | 16.6        | 16.7        |
| metric_clustered | complete | 500x5000 | sap_dense       | 3         | 21.4           | 21.3        | 21.7        |
| metric_clustered | complete | 500x5000 | ramshaw_tarjan  | 3         | 45.8           | 45.7        | 49.7        |
| metric_clustered | complete | 500x5000 | csflow          | 3         | 97.9           | 97.1        | 98.3        |
| metric_clustered | complete | 500x5000 | sap             | 3         | 98.9           | 98          | 104         |
| metric_clustered | complete | 500x5000 | auto            | 3         | 107            | 107         | 109         |
| metric_clustered | complete | 500x5000 | network_simplex | 3         | 614            | 603         | 648         |
| metric_clustered | complete | 500x5000 | auction_scaled  | 3         | 4.1e+03        | 4.09e+03    | 4.54e+03    |
| metric_clustered | complete | 500x5000 | csa             | 3         | 1.05e+04       | 1.04e+04    | 1.21e+04    |
| metric_clustered | 25%      | 500x500  | auction_scaled  | 3         | 4.53           | 4.52        | 4.55        |
| metric_clustered | 25%      | 500x500  | csa             | 3         | 5.65           | 5.63        | 5.77        |
| metric_clustered | 25%      | 500x500  | hungarian       | 3         | 13.6           | 12.9        | 14.1        |
| metric_clustered | 25%      | 500x500  | jv              | 3         | 13.6           | 12.9        | 14.1        |
| metric_clustered | 25%      | 500x500  | sap             | 3         | 14.4           | 13.9        | 14.7        |
| metric_clustered | 25%      | 500x500  | csflow          | 3         | 14.4           | 13.9        | 14.7        |
| metric_clustered | 25%      | 500x500  | auto            | 3         | 16.1           | 15.3        | 16.8        |
| metric_clustered | 25%      | 500x500  | lapmod          | 3         | 16.2           | 15.3        | 16.8        |
| metric_clustered | 25%      | 500x500  | sap_dense       | 3         | 28.1           | 26.7        | 29.5        |
| metric_clustered | 25%      | 500x500  | ramshaw_tarjan  | 3         | 28.8           | 27.4        | 30.2        |
| metric_clustered | 25%      | 500x500  | network_simplex | 3         | 52.6           | 51.5        | 52.8        |
| metric_clustered | 25%      | 500x1500 | jv              | 3         | 6.6            | 6.59        | 6.66        |
| metric_clustered | 25%      | 500x1500 | hungarian       | 3         | 6.67           | 6.66        | 6.7         |
| metric_clustered | 25%      | 500x1500 | lapmod          | 3         | 7.36           | 7.33        | 7.39        |
| metric_clustered | 25%      | 500x1500 | auto            | 3         | 7.39           | 7.38        | 7.44        |

| Regime           | Pattern | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|------------------|---------|----------|-----------------|-----------|----------------|-------------|-------------|
| metric_clustered | 25%     | 500x1500 | sap_dense       | 3         | 10.6           | 10.6        | 10.6        |
| metric_clustered | 25%     | 500x1500 | ramshaw_tarjan  | 3         | 12             | 12          | 12.1        |
| metric_clustered | 25%     | 500x1500 | csflow          | 3         | 13.2           | 13.2        | 13.3        |
| metric_clustered | 25%     | 500x1500 | sap             | 3         | 13.4           | 13.3        | 13.5        |
| metric_clustered | 25%     | 500x1500 | network_simplex | 3         | 112            | 108         | 112         |
| metric_clustered | 25%     | 500x1500 | auction_scaled  | 3         | 340            | 328         | 358         |
| metric_clustered | 25%     | 500x1500 | csa             | 3         | 587            | 534         | 599         |
| metric_clustered | 25%     | 500x5000 | auto            | 3         | 23.2           | 23.2        | 23.3        |
| metric_clustered | 25%     | 500x5000 | jv              | 3         | 26.6           | 23.2        | 26.7        |
| metric_clustered | 25%     | 500x5000 | hungarian       | 3         | 28.8           | 24.3        | 29.5        |
| metric_clustered | 25%     | 500x5000 | lapmod          | 3         | 32             | 31.9        | 32          |
| metric_clustered | 25%     | 500x5000 | ramshaw_tarjan  | 3         | 36.8           | 36.5        | 42.7        |
| metric_clustered | 25%     | 500x5000 | csflow          | 3         | 41.5           | 41.2        | 41.6        |
| metric_clustered | 25%     | 500x5000 | sap_dense       | 3         | 42.3           | 40.9        | 43.2        |
| metric_clustered | 25%     | 500x5000 | sap             | 3         | 45.2           | 44          | 45.8        |
| metric_clustered | 25%     | 500x5000 | network_simplex | 3         | 664            | 657         | 672         |
| metric_clustered | 25%     | 500x5000 | auction_scaled  | 3         | 4.14e+03       | 4.07e+03    | 4.96e+03    |
| metric_clustered | 25%     | 500x5000 | csa             | 3         | 1.02e+04       | 1.02e+04    | 1.1e+04     |
| metric_clustered | 60%     | 500x500  | auction_scaled  | 3         | 7.61           | 7.58        | 7.65        |
| metric_clustered | 60%     | 500x500  | csa             | 3         | 9.6            | 9.49        | 9.64        |
| metric_clustered | 60%     | 500x500  | hungarian       | 3         | 13.6           | 12.1        | 14          |
| metric_clustered | 60%     | 500x500  | jv              | 3         | 13.6           | 12.1        | 14          |
| metric_clustered | 60%     | 500x500  | auto            | 3         | 13.7           | 12.1        | 14.1        |
| metric_clustered | 60%     | 500x500  | lapmod          | 3         | 20.8           | 17.8        | 21.1        |
| metric_clustered | 60%     | 500x500  | sap             | 3         | 22.1           | 20.4        | 22.6        |
| metric_clustered | 60%     | 500x500  | csflow          | 3         | 22.2           | 20.4        | 22.6        |
| metric_clustered | 60%     | 500x500  | sap_dense       | 3         | 30.5           | 26.4        | 31.3        |
| metric_clustered | 60%     | 500x500  | ramshaw_tarjan  | 3         | 34.4           | 30.1        | 35.5        |
| metric_clustered | 60%     | 500x500  | network_simplex | 3         | 37.4           | 35.9        | 37.6        |
| metric_clustered | 60%     | 500x1500 | jv              | 3         | 8.4            | 8.35        | 8.46        |
| metric_clustered | 60%     | 500x1500 | hungarian       | 3         | 8.55           | 8.4         | 8.56        |
| metric_clustered | 60%     | 500x1500 | lapmod          | 3         | 10.1           | 10          | 10.1        |
| metric_clustered | 60%     | 500x1500 | sap_dense       | 3         | 13.6           | 13.6        | 13.7        |
| metric_clustered | 60%     | 500x1500 | ramshaw_tarjan  | 3         | 18.6           | 18.5        | 18.8        |
| metric_clustered | 60%     | 500x1500 | sap             | 3         | 24.3           | 24.2        | 24.5        |
| metric_clustered | 60%     | 500x1500 | csflow          | 3         | 24.3           | 24.2        | 24.4        |
| metric_clustered | 60%     | 500x1500 | auto            | 3         | 24.3           | 24.3        | 24.5        |
| metric_clustered | 60%     | 500x1500 | network_simplex | 3         | 118            | 116         | 118         |
| metric_clustered | 60%     | 500x1500 | auction_scaled  | 3         | 349            | 345         | 377         |
| metric_clustered | 60%     | 500x1500 | csa             | 3         | 631            | 563         | 644         |
| metric_clustered | 60%     | 500x5000 | jv              | 3         | 25.1           | 25.1        | 25.2        |
| metric_clustered | 60%     | 500x5000 | hungarian       | 3         | 26.9           | 26          | 28.4        |
| metric_clustered | 60%     | 500x5000 | lapmod          | 3         | 30.7           | 30.7        | 35.2        |
| metric_clustered | 60%     | 500x5000 | sap_dense       | 3         | 50.6           | 48          | 51.1        |
| metric_clustered | 60%     | 500x5000 | ramshaw_tarjan  | 3         | 55.5           | 55.5        | 58.8        |
| metric_clustered | 60%     | 500x5000 | auto            | 3         | 75.7           | 75.6        | 82.2        |
| metric_clustered | 60%     | 500x5000 | csflow          | 3         | 76.4           | 76.1        | 76.6        |
| metric_clustered | 60%     | 500x5000 | sap             | 3         | 78.9           | 77.6        | 83.8        |
| metric_clustered | 60%     | 500x5000 | network_simplex | 3         | 665            | 664         | 665         |
| metric_clustered | 60%     | 500x5000 | auction_scaled  | 3         | 4.25e+03       | 4.12e+03    | 4.3e+03     |
| metric_clustered | 60%     | 500x5000 | csa             | 3         | 1.04e+04       | 1.02e+04    | 1.04e+04    |

| Regime         | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|----------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| metric_uniform | 4 blocks | 500x500  | auction_scaled  | 3         | 2.48           | 2.44        | 2.48        |
| metric_uniform | 4 blocks | 500x500  | csa             | 3         | 3.12           | 3.12        | 3.17        |
| metric_uniform | 4 blocks | 500x500  | jv              | 3         | 4.15           | 4.14        | 4.3         |
| metric_uniform | 4 blocks | 500x500  | hungarian       | 3         | 4.19           | 4.16        | 4.32        |
| metric_uniform | 4 blocks | 500x500  | lapmod          | 3         | 4.84           | 4.83        | 5           |
| metric_uniform | 4 blocks | 500x500  | auto            | 3         | 4.85           | 4.83        | 5.01        |
| metric_uniform | 4 blocks | 500x500  | sap             | 3         | 5.48           | 5.46        | 5.58        |
| metric_uniform | 4 blocks | 500x500  | csflow          | 3         | 5.51           | 5.48        | 5.58        |
| metric_uniform | 4 blocks | 500x500  | ramshaw_tarjan  | 3         | 5.66           | 5.58        | 5.74        |
| metric_uniform | 4 blocks | 500x500  | sap_dense       | 3         | 6.12           | 6.12        | 6.37        |
| metric_uniform | 4 blocks | 500x500  | network_simplex | 3         | 25.3           | 24.2        | 25.8        |
| metric_uniform | 4 blocks | 500x1500 | jv              | 3         | 3.64           | 3.62        | 3.66        |
| metric_uniform | 4 blocks | 500x1500 | hungarian       | 3         | 3.66           | 3.63        | 3.67        |
| metric_uniform | 4 blocks | 500x1500 | auto            | 3         | 4.28           | 4.25        | 4.29        |
| metric_uniform | 4 blocks | 500x1500 | lapmod          | 3         | 4.29           | 4.26        | 4.32        |
| metric_uniform | 4 blocks | 500x1500 | sap_dense       | 3         | 6.03           | 6.02        | 6.05        |
| metric_uniform | 4 blocks | 500x1500 | ramshaw_tarjan  | 3         | 7.41           | 7.4         | 7.48        |
| metric_uniform | 4 blocks | 500x1500 | csflow          | 3         | 9.31           | 9.3         | 9.35        |
| metric_uniform | 4 blocks | 500x1500 | sap             | 3         | 9.42           | 9.38        | 9.45        |
| metric_uniform | 4 blocks | 500x1500 | network_simplex | 3         | 89.1           | 85.6        | 90.1        |
| metric_uniform | 4 blocks | 500x1500 | auction_scaled  | 3         | 322            | 304         | 333         |
| metric_uniform | 4 blocks | 500x1500 | csa             | 3         | 595            | 510         | 600         |
| metric_uniform | 4 blocks | 500x5000 | auto            | 3         | 13.5           | 13.4        | 17.9        |
| metric_uniform | 4 blocks | 500x5000 | hungarian       | 3         | 18.2           | 18.1        | 18.5        |
| metric_uniform | 4 blocks | 500x5000 | jv              | 3         | 18.7           | 15.3        | 18.9        |
| metric_uniform | 4 blocks | 500x5000 | lapmod          | 3         | 20.2           | 18.3        | 21.2        |
| metric_uniform | 4 blocks | 500x5000 | ramshaw_tarjan  | 3         | 23.9           | 23.8        | 28.1        |
| metric_uniform | 4 blocks | 500x5000 | csflow          | 3         | 30.2           | 30.2        | 30.5        |
| metric_uniform | 4 blocks | 500x5000 | sap             | 3         | 31             | 30.8        | 34.9        |
| metric_uniform | 4 blocks | 500x5000 | sap_dense       | 3         | 31.6           | 27.6        | 34.4        |
| metric_uniform | 4 blocks | 500x5000 | network_simplex | 3         | 509            | 507         | 516         |
| metric_uniform | 4 blocks | 500x5000 | auction_scaled  | 3         | 4.27e+03       | 4.15e+03    | 4.74e+03    |
| metric_uniform | 4 blocks | 500x5000 | csa             | 3         | 1.17e+04       | 1.11e+04    | 1.17e+04    |
| metric_uniform | complete | 500x500  | auto            | 3         | 5.7            | 5.01        | 5.95        |
| metric_uniform | complete | 500x500  | jv              | 3         | 5.7            | 5.01        | 5.95        |
| metric_uniform | complete | 500x500  | auction_scaled  | 3         | 6.98           | 6.97        | 7.07        |
| metric_uniform | complete | 500x500  | csa             | 3         | 9.11           | 8.98        | 9.24        |
| metric_uniform | complete | 500x500  | hungarian       | 3         | 9.43           | 8.49        | 9.85        |
| metric_uniform | complete | 500x500  | sap_dense       | 3         | 12.5           | 11.3        | 13.1        |
| metric_uniform | complete | 500x500  | lapmod          | 3         | 13.3           | 11.8        | 14          |
| metric_uniform | complete | 500x500  | ramshaw_tarjan  | 3         | 15.9           | 14.9        | 16.4        |
| metric_uniform | complete | 500x500  | csflow          | 3         | 22.4           | 21          | 23          |
| metric_uniform | complete | 500x500  | sap             | 3         | 22.4           | 21.1        | 23          |
| metric_uniform | complete | 500x500  | network_simplex | 3         | 67.5           | 65.1        | 79.5        |
| metric_uniform | complete | 500x1500 | jv              | 3         | 3.7            | 3.68        | 3.84        |
| metric_uniform | complete | 500x1500 | hungarian       | 3         | 3.72           | 3.69        | 3.83        |
| metric_uniform | complete | 500x1500 | lapmod          | 3         | 5.18           | 5.16        | 5.32        |
| metric_uniform | complete | 500x1500 | sap_dense       | 3         | 6.05           | 5.95        | 6.05        |
| metric_uniform | complete | 500x1500 | ramshaw_tarjan  | 3         | 14.1           | 14.1        | 14.4        |
| metric_uniform | complete | 500x1500 | csflow          | 3         | 26.7           | 26.5        | 27          |
| metric_uniform | complete | 500x1500 | sap             | 3         | 26.7           | 26.7        | 27.1        |

| Regime         | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|----------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| metric_uniform | complete | 500x1500 | auto            | 3         | 26.8           | 26.7        | 27.3        |
| metric_uniform | complete | 500x1500 | network_simplex | 3         | 112            | 102         | 177         |
| metric_uniform | complete | 500x1500 | auction_scaled  | 3         | 326            | 304         | 350         |
| metric_uniform | complete | 500x1500 | csa             | 3         | 416            | 416         | 544         |
| metric_uniform | complete | 500x5000 | jv              | 3         | 11             | 10.9        | 11.1        |
| metric_uniform | complete | 500x5000 | hungarian       | 3         | 11.3           | 11.2        | 14.7        |
| metric_uniform | complete | 500x5000 | lapmod          | 3         | 16.4           | 16.3        | 16.5        |
| metric_uniform | complete | 500x5000 | sap_dense       | 3         | 21.2           | 21.2        | 21.4        |
| metric_uniform | complete | 500x5000 | ramshaw_tarjan  | 3         | 46.1           | 45.9        | 50.4        |
| metric_uniform | complete | 500x5000 | sap             | 3         | 100            | 99          | 104         |
| metric_uniform | complete | 500x5000 | csflow          | 3         | 100            | 99.4        | 100         |
| metric_uniform | complete | 500x5000 | auto            | 3         | 110            | 109         | 111         |
| metric_uniform | complete | 500x5000 | network_simplex | 3         | 580            | 552         | 624         |
| metric_uniform | complete | 500x5000 | auction_scaled  | 3         | 3.97e+03       | 3.96e+03    | 4.1e+03     |
| metric_uniform | complete | 500x5000 | csa             | 3         | 1.18e+04       | 1.15e+04    | 1.19e+04    |
| metric_uniform | 25%      | 500x500  | auction_scaled  | 3         | 4.03           | 4           | 4.08        |
| metric_uniform | 25%      | 500x500  | csa             | 3         | 5.15           | 5.13        | 5.18        |
| metric_uniform | 25%      | 500x500  | hungarian       | 3         | 10.6           | 9.65        | 10.9        |
| metric_uniform | 25%      | 500x500  | jv              | 3         | 10.6           | 9.66        | 10.9        |
| metric_uniform | 25%      | 500x500  | csflow          | 3         | 11.9           | 11.2        | 12          |
| metric_uniform | 25%      | 500x500  | sap             | 3         | 12             | 11.2        | 12          |
| metric_uniform | 25%      | 500x500  | lapmod          | 3         | 12.5           | 11.2        | 13          |
| metric_uniform | 25%      | 500x500  | auto            | 3         | 12.5           | 11.2        | 13          |
| metric_uniform | 25%      | 500x500  | sap_dense       | 3         | 21.7           | 19.5        | 22.5        |
| metric_uniform | 25%      | 500x500  | ramshaw_tarjan  | 3         | 22.2           | 19.9        | 22.7        |
| metric_uniform | 25%      | 500x500  | network_simplex | 3         | 85.4           | 84.6        | 90.3        |
| metric_uniform | 25%      | 500x1500 | jv              | 3         | 6.3            | 6.29        | 6.32        |
| metric_uniform | 25%      | 500x1500 | hungarian       | 3         | 6.38           | 6.35        | 6.42        |
| metric_uniform | 25%      | 500x1500 | lapmod          | 3         | 7.05           | 7.01        | 7.11        |
| metric_uniform | 25%      | 500x1500 | auto            | 3         | 7.18           | 7.17        | 7.2         |
| metric_uniform | 25%      | 500x1500 | sap_dense       | 3         | 10.1           | 10.1        | 10.2        |
| metric_uniform | 25%      | 500x1500 | ramshaw_tarjan  | 3         | 11.5           | 11.5        | 11.5        |
| metric_uniform | 25%      | 500x1500 | csflow          | 3         | 12.7           | 12.6        | 12.7        |
| metric_uniform | 25%      | 500x1500 | sap             | 3         | 12.8           | 12.7        | 12.8        |
| metric_uniform | 25%      | 500x1500 | network_simplex | 3         | 135            | 125         | 165         |
| metric_uniform | 25%      | 500x1500 | auction_scaled  | 3         | 296            | 296         | 301         |
| metric_uniform | 25%      | 500x1500 | csa             | 3         | 671            | 607         | 695         |
| metric_uniform | 25%      | 500x5000 | jv              | 3         | 20.1           | 20          | 23.4        |
| metric_uniform | 25%      | 500x5000 | auto            | 3         | 23.3           | 23.3        | 26.7        |
| metric_uniform | 25%      | 500x5000 | hungarian       | 3         | 23.6           | 22.2        | 26.5        |
| metric_uniform | 25%      | 500x5000 | lapmod          | 3         | 31.9           | 31          | 31.9        |
| metric_uniform | 25%      | 500x5000 | csflow          | 3         | 41.7           | 41.5        | 42.4        |
| metric_uniform | 25%      | 500x5000 | ramshaw_tarjan  | 3         | 43.5           | 39.9        | 44.3        |
| metric_uniform | 25%      | 500x5000 | sap             | 3         | 43.6           | 43          | 48.8        |
| metric_uniform | 25%      | 500x5000 | sap_dense       | 3         | 50.5           | 49.2        | 50.6        |
| metric_uniform | 25%      | 500x5000 | network_simplex | 3         | 648            | 638         | 688         |
| metric_uniform | 25%      | 500x5000 | auction_scaled  | 3         | 4e+03          | 3.89e+03    | 4.19e+03    |
| metric_uniform | 25%      | 500x5000 | csa             | 3         | 1.2e+04        | 1.14e+04    | 1.21e+04    |
| metric_uniform | 60%      | 500x500  | auction_scaled  | 3         | 6.61           | 6.59        | 6.78        |
| metric_uniform | 60%      | 500x500  | csa             | 3         | 8.72           | 8.67        | 8.94        |
| metric_uniform | 60%      | 500x500  | auto            | 3         | 10             | 9.38        | 10.4        |



| Regime         | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|----------------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| metric_uniform | 60%      | 500x500  | hungarian       | 3         | 10             | 9.37        | 10.4        |
| metric_uniform | 60%      | 500x500  | jv              | 3         | 10             | 9.4         | 10.4        |
| metric_uniform | 60%      | 500x500  | lapmod          | 3         | 14.4           | 13.2        | 15          |
| metric_uniform | 60%      | 500x500  | csflow          | 3         | 17.5           | 16.8        | 17.8        |
| metric_uniform | 60%      | 500x500  | sap             | 3         | 17.6           | 16.8        | 17.9        |
| metric_uniform | 60%      | 500x500  | sap_dense       | 3         | 21.7           | 20.1        | 22.7        |
| metric_uniform | 60%      | 500x500  | ramshaw_tarjan  | 3         | 24.5           | 22.9        | 25.6        |
| metric_uniform | 60%      | 500x500  | network_simplex | 3         | 77.8           | 76.7        | 78.8        |
| metric_uniform | 60%      | 500x1500 | jv              | 3         | 7.88           | 7.84        | 7.95        |
| metric_uniform | 60%      | 500x1500 | hungarian       | 3         | 7.93           | 7.84        | 7.98        |
| metric_uniform | 60%      | 500x1500 | lapmod          | 3         | 9.55           | 9.53        | 9.59        |
| metric_uniform | 60%      | 500x1500 | sap_dense       | 3         | 13             | 12.9        | 13          |
| metric_uniform | 60%      | 500x1500 | ramshaw_tarjan  | 3         | 17.6           | 17.4        | 17.7        |
| metric_uniform | 60%      | 500x1500 | csflow          | 3         | 23.1           | 23          | 23.4        |
| metric_uniform | 60%      | 500x1500 | sap             | 3         | 23.2           | 23.2        | 23.5        |
| metric_uniform | 60%      | 500x1500 | auto            | 3         | 23.4           | 23.2        | 23.5        |
| metric_uniform | 60%      | 500x1500 | network_simplex | 3         | 215            | 179         | 247         |
| metric_uniform | 60%      | 500x1500 | auction_scaled  | 3         | 279            | 273         | 301         |
| metric_uniform | 60%      | 500x1500 | csa             | 3         | 528            | 511         | 540         |
| metric_uniform | 60%      | 500x5000 | hungarian       | 3         | 24.9           | 24.8        | 25.8        |
| metric_uniform | 60%      | 500x5000 | lapmod          | 3         | 30.5           | 30.5        | 35.3        |
| metric_uniform | 60%      | 500x5000 | jv              | 3         | 31.8           | 28.5        | 31.8        |
| metric_uniform | 60%      | 500x5000 | sap_dense       | 3         | 46.8           | 45.8        | 49.2        |
| metric_uniform | 60%      | 500x5000 | ramshaw_tarjan  | 3         | 62.5           | 62.2        | 63.1        |
| metric_uniform | 60%      | 500x5000 | csflow          | 3         | 75.3           | 75.1        | 75.7        |
| metric_uniform | 60%      | 500x5000 | auto            | 3         | 76.9           | 76.2        | 77.9        |
| metric_uniform | 60%      | 500x5000 | sap             | 3         | 88.4           | 82.1        | 89.5        |
| metric_uniform | 60%      | 500x5000 | network_simplex | 3         | 629            | 616         | 664         |
| metric_uniform | 60%      | 500x5000 | auction_scaled  | 3         | 3.53e+03       | 3.51e+03    | 3.79e+03    |
| metric_uniform | 60%      | 500x5000 | csa             | 3         | 1.04e+04       | 1.02e+04    | 1.08e+04    |
| tied           | 4 blocks | 500x500  | auction_scaled  | 3         | 4.05           | 4.04        | 4.12        |
| tied           | 4 blocks | 500x500  | csa             | 3         | 4.23           | 4.21        | 4.29        |
| tied           | 4 blocks | 500x500  | network_simplex | 3         | 5.24           | 5.2         | 5.42        |
| tied           | 4 blocks | 500x500  | sap             | 3         | 13.2           | 13.1        | 13.2        |
| tied           | 4 blocks | 500x500  | csflow          | 3         | 13.2           | 13.2        | 13.2        |
| tied           | 4 blocks | 500x500  | ramshaw_tarjan  | 3         | 15.2           | 15.1        | 15.2        |
| tied           | 4 blocks | 500x500  | jv              | 3         | 17.4           | 17.4        | 17.4        |
| tied           | 4 blocks | 500x500  | hungarian       | 3         | 17.5           | 17.4        | 17.5        |
| tied           | 4 blocks | 500x500  | auto            | 3         | 18.7           | 18.7        | 18.7        |
| tied           | 4 blocks | 500x500  | lapmod          | 3         | 18.7           | 18.6        | 18.8        |
| tied           | 4 blocks | 500x500  | sap_dense       | 3         | 25.1           | 25.1        | 25.1        |
| tied           | 4 blocks | 500x1500 | network_simplex | 3         | 30             | 29.3        | 30.1        |
| tied           | 4 blocks | 500x1500 | csflow          | 3         | 30.5           | 30.5        | 30.7        |
| tied           | 4 blocks | 500x1500 | sap             | 3         | 30.6           | 30.6        | 30.6        |
| tied           | 4 blocks | 500x1500 | ramshaw_tarjan  | 3         | 42.8           | 42.7        | 42.9        |
| tied           | 4 blocks | 500x1500 | hungarian       | 3         | 52.8           | 52.8        | 52.8        |
| tied           | 4 blocks | 500x1500 | jv              | 3         | 52.8           | 52.8        | 52.9        |
| tied           | 4 blocks | 500x1500 | auto            | 3         | 57.2           | 57.1        | 57.3        |
| tied           | 4 blocks | 500x1500 | lapmod          | 3         | 57.2           | 57.2        | 57.3        |
| tied           | 4 blocks | 500x1500 | sap_dense       | 3         | 75.3           | 75.3        | 75.7        |
| tied           | 4 blocks | 500x1500 | csa             | 3         | 103            | 103         | 106         |

| Regime | Pattern  | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|--------|----------|----------|-----------------|-----------|----------------|-------------|-------------|
| tied   | 4 blocks | 500x1500 | auction_scaled  | 3         | 168            | 166         | 170         |
| tied   | 4 blocks | 500x5000 | sap             | 3         | 91.2           | 91          | 91.5        |
| tied   | 4 blocks | 500x5000 | csflow          | 3         | 91.5           | 91.3        | 91.7        |
| tied   | 4 blocks | 500x5000 | ramshaw_tarjan  | 3         | 138            | 138         | 139         |
| tied   | 4 blocks | 500x5000 | jv              | 3         | 177            | 176         | 177         |
| tied   | 4 blocks | 500x5000 | hungarian       | 3         | 177            | 176         | 177         |
| tied   | 4 blocks | 500x5000 | auto            | 3         | 191            | 190         | 191         |
| tied   | 4 blocks | 500x5000 | lapmod          | 3         | 191            | 190         | 191         |
| tied   | 4 blocks | 500x5000 | network_simplex | 3         | 195            | 192         | 198         |
| tied   | 4 blocks | 500x5000 | sap_dense       | 3         | 254            | 252         | 255         |
| tied   | 4 blocks | 500x5000 | csa             | 3         | 1.69e+03       | 1.68e+03    | 1.69e+03    |
| tied   | 4 blocks | 500x5000 | auction_scaled  | 3         | 2.67e+03       | 2.6e+03     | 2.74e+03    |
| tied   | complete | 500x500  | network_simplex | 3         | 5.56           | 5.11        | 5.62        |
| tied   | complete | 500x500  | csa             | 3         | 13.5           | 13.3        | 14.2        |
| tied   | complete | 500x500  | auction_scaled  | 3         | 13.7           | 13.5        | 14.4        |
| tied   | complete | 500x500  | jv              | 3         | 56.1           | 55.9        | 56.1        |
| tied   | complete | 500x500  | auto            | 3         | 57.1           | 56.6        | 57.5        |
| tied   | complete | 500x500  | hungarian       | 3         | 61.4           | 61          | 61.5        |
| tied   | complete | 500x500  | ramshaw_tarjan  | 3         | 69             | 68.9        | 69          |
| tied   | complete | 500x500  | sap_dense       | 3         | 90.1           | 89.9        | 90.2        |
| tied   | complete | 500x500  | lapmod          | 3         | 96.5           | 96.3        | 96.9        |
| tied   | complete | 500x500  | sap             | 3         | 113            | 113         | 114         |
| tied   | complete | 500x500  | csflow          | 3         | 114            | 114         | 115         |
| tied   | complete | 500x1500 | network_simplex | 3         | 20.4           | 20.2        | 20.8        |
| tied   | complete | 500x1500 | csa             | 3         | 92.9           | 92.3        | 105         |
| tied   | complete | 500x1500 | auction_scaled  | 3         | 149            | 146         | 151         |
| tied   | complete | 500x1500 | hungarian       | 3         | 199            | 199         | 199         |
| tied   | complete | 500x1500 | jv              | 3         | 199            | 199         | 199         |
| tied   | complete | 500x1500 | ramshaw_tarjan  | 3         | 226            | 226         | 226         |
| tied   | complete | 500x1500 | sap             | 3         | 273            | 273         | 273         |
| tied   | complete | 500x1500 | auto            | 3         | 273            | 273         | 273         |
| tied   | complete | 500x1500 | csflow          | 3         | 276            | 276         | 276         |
| tied   | complete | 500x1500 | sap_dense       | 3         | 287            | 287         | 287         |
| tied   | complete | 500x1500 | lapmod          | 3         | 297            | 296         | 297         |
| tied   | complete | 500x5000 | network_simplex | 3         | 119            | 117         | 121         |
| tied   | complete | 500x5000 | jv              | 3         | 681            | 681         | 682         |
| tied   | complete | 500x5000 | hungarian       | 3         | 681            | 680         | 683         |
| tied   | complete | 500x5000 | ramshaw_tarjan  | 3         | 810            | 807         | 822         |
| tied   | complete | 500x5000 | auto            | 3         | 864            | 863         | 864         |
| tied   | complete | 500x5000 | csflow          | 3         | 873            | 873         | 873         |
| tied   | complete | 500x5000 | sap             | 3         | 921            | 921         | 921         |
| tied   | complete | 500x5000 | sap_dense       | 3         | 979            | 978         | 979         |
| tied   | complete | 500x5000 | lapmod          | 3         | 1.01e+03       | 1.01e+03    | 1.01e+03    |
| tied   | complete | 500x5000 | csa             | 3         | 1.41e+03       | 1.34e+03    | 1.42e+03    |
| tied   | complete | 500x5000 | auction_scaled  | 3         | 1.82e+03       | 1.81e+03    | 1.88e+03    |
| tied   | 25%      | 500x500  | auction_scaled  | 3         | 5.87           | 5.86        | 5.91        |
| tied   | 25%      | 500x500  | csa             | 3         | 6.2            | 6.18        | 6.24        |
| tied   | 25%      | 500x500  | network_simplex | 3         | 8.6            | 8.4         | 8.62        |
| tied   | 25%      | 500x500  | sap             | 3         | 52             | 51.8        | 52          |
| tied   | 25%      | 500x500  | csflow          | 3         | 52.1           | 52          | 52.2        |
| tied   | 25%      | 500x500  | hungarian       | 3         | 62             | 61.5        | 62.1        |

| Regime | Pattern | Shape    | Solver          | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|--------|---------|----------|-----------------|-----------|----------------|-------------|-------------|
| tied   | 25%     | 500x500  | jv              | 3         | 62             | 61.6        | 62.1        |
| tied   | 25%     | 500x500  | auto            | 3         | 73             | 72.9        | 73.1        |
| tied   | 25%     | 500x500  | lapmod          | 3         | 73             | 72.9        | 73.2        |
| tied   | 25%     | 500x500  | ramshaw_tarjan  | 3         | 108            | 108         | 108         |
| tied   | 25%     | 500x500  | sap_dense       | 3         | 129            | 129         | 129         |
| tied   | 25%     | 500x1500 | network_simplex | 3         | 35.3           | 34.5        | 35.4        |
| tied   | 25%     | 500x1500 | csa             | 3         | 98.3           | 97.9        | 100         |
| tied   | 25%     | 500x1500 | sap             | 3         | 115            | 115         | 115         |
| tied   | 25%     | 500x1500 | csflow          | 3         | 115            | 115         | 115         |
| tied   | 25%     | 500x1500 | auction_scaled  | 3         | 139            | 129         | 141         |
| tied   | 25%     | 500x1500 | hungarian       | 3         | 198            | 198         | 198         |
| tied   | 25%     | 500x1500 | jv              | 3         | 198            | 198         | 198         |
| tied   | 25%     | 500x1500 | lapmod          | 3         | 222            | 222         | 222         |
| tied   | 25%     | 500x1500 | auto            | 3         | 222            | 222         | 222         |
| tied   | 25%     | 500x1500 | ramshaw_tarjan  | 3         | 395            | 395         | 396         |
| tied   | 25%     | 500x1500 | sap_dense       | 3         | 476            | 476         | 477         |
| tied   | 25%     | 500x5000 | network_simplex | 3         | 218            | 215         | 219         |
| tied   | 25%     | 500x5000 | sap             | 3         | 344            | 343         | 345         |
| tied   | 25%     | 500x5000 | csflow          | 3         | 349            | 347         | 350         |
| tied   | 25%     | 500x5000 | jv              | 3         | 684            | 683         | 685         |
| tied   | 25%     | 500x5000 | hungarian       | 3         | 688            | 687         | 689         |
| tied   | 25%     | 500x5000 | auto            | 3         | 745            | 744         | 746         |
| tied   | 25%     | 500x5000 | lapmod          | 3         | 745            | 744         | 747         |
| tied   | 25%     | 500x5000 | csa             | 3         | 1.35e+03       | 1.27e+03    | 1.36e+03    |
| tied   | 25%     | 500x5000 | ramshaw_tarjan  | 3         | 1.39e+03       | 1.39e+03    | 1.4e+03     |
| tied   | 25%     | 500x5000 | sap_dense       | 3         | 1.69e+03       | 1.69e+03    | 1.69e+03    |
| tied   | 25%     | 500x5000 | auction_scaled  | 3         | 1.74e+03       | 1.71e+03    | 1.81e+03    |
| tied   | 60%     | 500x500  | network_simplex | 3         | 7.55           | 7.38        | 7.67        |
| tied   | 60%     | 500x500  | auction_scaled  | 3         | 11.4           | 10.9        | 11.5        |
| tied   | 60%     | 500x500  | csa             | 3         | 11.7           | 11.1        | 11.8        |
| tied   | 60%     | 500x500  | auto            | 3         | 62.5           | 62.4        | 62.7        |
| tied   | 60%     | 500x500  | jv              | 3         | 63.1           | 63          | 63.2        |
| tied   | 60%     | 500x500  | hungarian       | 3         | 63.3           | 63.2        | 63.5        |
| tied   | 60%     | 500x500  | sap             | 3         | 82.8           | 82.8        | 82.9        |
| tied   | 60%     | 500x500  | csflow          | 3         | 83.4           | 83.3        | 83.4        |
| tied   | 60%     | 500x500  | lapmod          | 3         | 84.9           | 84.9        | 85.1        |
| tied   | 60%     | 500x500  | sap_dense       | 3         | 145            | 145         | 145         |
| tied   | 60%     | 500x500  | ramshaw_tarjan  | 3         | 149            | 149         | 149         |
| tied   | 60%     | 500x1500 | network_simplex | 3         | 35.5           | 34.3        | 36.5        |
| tied   | 60%     | 500x1500 | csa             | 3         | 105            | 102         | 107         |
| tied   | 60%     | 500x1500 | auction_scaled  | 3         | 142            | 137         | 145         |
| tied   | 60%     | 500x1500 | auto            | 3         | 195            | 195         | 195         |
| tied   | 60%     | 500x1500 | sap             | 3         | 195            | 195         | 195         |
| tied   | 60%     | 500x1500 | csflow          | 3         | 197            | 196         | 197         |
| tied   | 60%     | 500x1500 | hungarian       | 3         | 203            | 203         | 204         |
| tied   | 60%     | 500x1500 | jv              | 3         | 205            | 204         | 205         |
| tied   | 60%     | 500x1500 | lapmod          | 3         | 262            | 262         | 262         |
| tied   | 60%     | 500x1500 | ramshaw_tarjan  | 3         | 558            | 558         | 559         |
| tied   | 60%     | 500x1500 | sap_dense       | 3         | 595            | 595         | 597         |
| tied   | 60%     | 500x5000 | network_simplex | 3         | 181            | 179         | 181         |
| tied   | 60%     | 500x5000 | auto            | 3         | 609            | 608         | 609         |

| Regime      | Pattern  | Shape      | Solver         | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|-------------|----------|------------|----------------|-----------|----------------|-------------|-------------|
| tied        | 60%      | 500x5000   | sap            | 3         | 609            | 609         | 610         |
| tied        | 60%      | 500x5000   | csflow         | 3         | 644            | 629         | 645         |
| tied        | 60%      | 500x5000   | hungarian      | 3         | 697            | 697         | 697         |
| tied        | 60%      | 500x5000   | jv             | 3         | 700            | 699         | 701         |
| tied        | 60%      | 500x5000   | lapmod         | 3         | 884            | 884         | 886         |
| tied        | 60%      | 500x5000   | csa            | 3         | 1.38e+03       | 1.28e+03    | 1.46e+03    |
| tied        | 60%      | 500x5000   | auction_scaled | 3         | 1.52e+03       | 1.51e+03    | 1.7e+03     |
| tied        | 60%      | 500x5000   | ramshaw_tarjan | 3         | 1.98e+03       | 1.98e+03    | 1.99e+03    |
| tied        | 60%      | 500x5000   | sap_dense      | 3         | 2.14e+03       | 2.14e+03    | 2.14e+03    |
| binary      | complete | 1000x10000 | auto           | 2         | 168            | 167         | 169         |
| binary      | complete | 1000x10000 | jv             | 2         | 5.31e+03       | 5.3e+03     | 5.32e+03    |
| binary      | complete | 1000x10000 | ramshaw_tarjan | 2         | 5.86e+03       | 5.86e+03    | 5.86e+03    |
| binary      | complete | 1000x10000 | sap            | 2         | 5.96e+03       | 5.96e+03    | 5.96e+03    |
| binary      | complete | 1000x10000 | lapmod         | 2         | 7.95e+03       | 7.95e+03    | 7.95e+03    |
| binary      | complete | 1500x1500  | auto           | 2         | 36.8           | 36.2        | 37.4        |
| binary      | complete | 1500x1500  | jv             | 2         | 1.49e+03       | 1.49e+03    | 1.49e+03    |
| binary      | complete | 1500x1500  | ramshaw_tarjan | 2         | 1.51e+03       | 1.51e+03    | 1.51e+03    |
| binary      | complete | 1500x1500  | lapmod         | 2         | 2.52e+03       | 2.51e+03    | 2.52e+03    |
| binary      | complete | 1500x1500  | sap            | 2         | 2.52e+03       | 2.51e+03    | 2.52e+03    |
| binary      | complete | 1500x4500  | auto           | 2         | 116            | 115         | 116         |
| binary      | complete | 1500x4500  | jv             | 2         | 5.11e+03       | 5.11e+03    | 5.12e+03    |
| binary      | complete | 1500x4500  | ramshaw_tarjan | 2         | 5.45e+03       | 5.44e+03    | 5.45e+03    |
| binary      | complete | 1500x4500  | sap            | 2         | 6.39e+03       | 6.35e+03    | 6.43e+03    |
| binary      | complete | 1500x4500  | lapmod         | 2         | 7.88e+03       | 7.87e+03    | 7.88e+03    |
| binary      | 25%      | 1000x10000 | auto           | 2         | 136            | 136         | 137         |
| binary      | 25%      | 1000x10000 | sap            | 2         | 2.05e+03       | 2.05e+03    | 2.05e+03    |
| binary      | 25%      | 1000x10000 | jv             | 2         | 5.37e+03       | 5.37e+03    | 5.38e+03    |
| binary      | 25%      | 1000x10000 | lapmod         | 2         | 5.95e+03       | 5.95e+03    | 5.95e+03    |
| binary      | 25%      | 1000x10000 | ramshaw_tarjan | 2         | 1.04e+04       | 1.04e+04    | 1.04e+04    |
| binary      | 25%      | 1500x1500  | auto           | 2         | 26.8           | 26.8        | 26.8        |
| binary      | 25%      | 1500x1500  | sap            | 2         | 810            | 810         | 810         |
| binary      | 25%      | 1500x1500  | jv             | 2         | 1.52e+03       | 1.52e+03    | 1.52e+03    |
| binary      | 25%      | 1500x1500  | lapmod         | 2         | 1.89e+03       | 1.89e+03    | 1.89e+03    |
| binary      | 25%      | 1500x1500  | ramshaw_tarjan | 2         | 2.66e+03       | 2.66e+03    | 2.66e+03    |
| binary      | 25%      | 1500x4500  | auto           | 2         | 93.9           | 93.9        | 93.9        |
| binary      | 25%      | 1500x4500  | sap            | 2         | 1.98e+03       | 1.98e+03    | 1.98e+03    |
| binary      | 25%      | 1500x4500  | jv             | 2         | 5.17e+03       | 5.17e+03    | 5.18e+03    |
| binary      | 25%      | 1500x4500  | lapmod         | 2         | 5.78e+03       | 5.78e+03    | 5.78e+03    |
| binary      | 25%      | 1500x4500  | ramshaw_tarjan | 2         | 9.96e+03       | 9.96e+03    | 9.96e+03    |
| dbl_uniform | complete | 1000x10000 | jv             | 2         | 77.1           | 76.4        | 77.7        |
| dbl_uniform | complete | 1000x10000 | lapmod         | 2         | 99.2           | 98.7        | 99.7        |
| dbl_uniform | complete | 1000x10000 | ramshaw_tarjan | 2         | 225            | 224         | 227         |
| dbl_uniform | complete | 1000x10000 | sap            | 2         | 473            | 473         | 474         |
| dbl_uniform | complete | 1000x10000 | auto           | 2         | 475            | 475         | 475         |
| dbl_uniform | complete | 1500x1500  | auto           | 2         | 50.5           | 49.8        | 51.2        |
| dbl_uniform | complete | 1500x1500  | jv             | 2         | 50.8           | 49.7        | 52          |
| dbl_uniform | complete | 1500x1500  | ramshaw_tarjan | 2         | 158            | 156         | 159         |
| dbl_uniform | complete | 1500x1500  | lapmod         | 2         | 161            | 160         | 163         |
| dbl_uniform | complete | 1500x1500  | sap            | 2         | 259            | 257         | 261         |
| dbl_uniform | complete | 1500x4500  | jv             | 2         | 59             | 58.9        | 59.1        |
| dbl_uniform | complete | 1500x4500  | lapmod         | 2         | 73.1           | 72.8        | 73.4        |

| Regime       | Pattern  | Shape      | Solver         | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|--------------|----------|------------|----------------|-----------|----------------|-------------|-------------|
| dbl_uniform  | complete | 1500x4500  | ramshaw_tarjan | 2         | 165            | 165         | 166         |
| dbl_uniform  | complete | 1500x4500  | auto           | 2         | 320            | 319         | 321         |
| dbl_uniform  | complete | 1500x4500  | sap            | 2         | 320            | 319         | 321         |
| dbl_uniform  | 25%      | 1000x10000 | jv             | 2         | 112            | 111         | 112         |
| dbl_uniform  | 25%      | 1000x10000 | auto           | 2         | 127            | 125         | 129         |
| dbl_uniform  | 25%      | 1000x10000 | lapmod         | 2         | 128            | 126         | 129         |
| dbl_uniform  | 25%      | 1000x10000 | ramshaw_tarjan | 2         | 188            | 188         | 189         |
| dbl_uniform  | 25%      | 1000x10000 | sap            | 2         | 218            | 217         | 219         |
| dbl_uniform  | 25%      | 1500x1500  | jv             | 2         | 97.3           | 91.8        | 103         |
| dbl_uniform  | 25%      | 1500x1500  | sap            | 2         | 107            | 105         | 109         |
| dbl_uniform  | 25%      | 1500x1500  | auto           | 2         | 118            | 110         | 126         |
| dbl_uniform  | 25%      | 1500x1500  | lapmod         | 2         | 118            | 110         | 126         |
| dbl_uniform  | 25%      | 1500x1500  | ramshaw_tarjan | 2         | 213            | 199         | 226         |
| dbl_uniform  | 25%      | 1500x4500  | auto           | 2         | 88.3           | 86.9        | 89.8        |
| dbl_uniform  | 25%      | 1500x4500  | lapmod         | 2         | 92.9           | 89.3        | 96.4        |
| dbl_uniform  | 25%      | 1500x4500  | jv             | 2         | 94.6           | 94.5        | 94.7        |
| dbl_uniform  | 25%      | 1500x4500  | ramshaw_tarjan | 2         | 134            | 132         | 136         |
| dbl_uniform  | 25%      | 1500x4500  | sap            | 2         | 151            | 150         | 152         |
| heavy_tailed | complete | 1000x10000 | jv             | 2         | 77.5           | 77.4        | 77.6        |
| heavy_tailed | complete | 1000x10000 | lapmod         | 2         | 98.8           | 98.7        | 99          |
| heavy_tailed | complete | 1000x10000 | ramshaw_tarjan | 2         | 224            | 223         | 224         |
| heavy_tailed | complete | 1000x10000 | auto           | 2         | 465            | 465         | 466         |
| heavy_tailed | complete | 1000x10000 | sap            | 2         | 466            | 464         | 467         |
| heavy_tailed | complete | 1500x1500  | auto           | 2         | 50.3           | 47.5        | 53          |
| heavy_tailed | complete | 1500x1500  | jv             | 2         | 50.4           | 49.3        | 51.6        |
| heavy_tailed | complete | 1500x1500  | lapmod         | 2         | 142            | 140         | 143         |
| heavy_tailed | complete | 1500x1500  | ramshaw_tarjan | 2         | 145            | 145         | 146         |
| heavy_tailed | complete | 1500x1500  | sap            | 2         | 238            | 238         | 238         |
| heavy_tailed | complete | 1500x4500  | jv             | 2         | 62.8           | 60.8        | 64.8        |
| heavy_tailed | complete | 1500x4500  | lapmod         | 2         | 72.5           | 72.4        | 72.7        |
| heavy_tailed | complete | 1500x4500  | ramshaw_tarjan | 2         | 167            | 166         | 167         |
| heavy_tailed | complete | 1500x4500  | auto           | 2         | 319            | 318         | 320         |
| heavy_tailed | complete | 1500x4500  | sap            | 2         | 324            | 322         | 326         |
| heavy_tailed | 25%      | 1000x10000 | jv             | 2         | 111            | 111         | 111         |
| heavy_tailed | 25%      | 1000x10000 | lapmod         | 2         | 123            | 123         | 123         |
| heavy_tailed | 25%      | 1000x10000 | auto           | 2         | 123            | 123         | 123         |
| heavy_tailed | 25%      | 1000x10000 | ramshaw_tarjan | 2         | 187            | 186         | 189         |
| heavy_tailed | 25%      | 1000x10000 | sap            | 2         | 214            | 213         | 214         |
| heavy_tailed | 25%      | 1500x1500  | jv             | 2         | 101            | 99          | 102         |
| heavy_tailed | 25%      | 1500x1500  | sap            | 2         | 108            | 108         | 108         |
| heavy_tailed | 25%      | 1500x1500  | auto           | 2         | 117            | 117         | 118         |
| heavy_tailed | 25%      | 1500x1500  | lapmod         | 2         | 119            | 119         | 120         |
| heavy_tailed | 25%      | 1500x1500  | ramshaw_tarjan | 2         | 211            | 210         | 212         |
| heavy_tailed | 25%      | 1500x4500  | lapmod         | 2         | 85.1           | 85          | 85.2        |
| heavy_tailed | 25%      | 1500x4500  | auto           | 2         | 85.2           | 85.2        | 85.2        |
| heavy_tailed | 25%      | 1500x4500  | jv             | 2         | 86.5           | 85.9        | 87.1        |
| heavy_tailed | 25%      | 1500x4500  | ramshaw_tarjan | 2         | 130            | 129         | 131         |
| heavy_tailed | 25%      | 1500x4500  | sap            | 2         | 153            | 152         | 154         |
| int_uniform  | complete | 1000x10000 | jv             | 2         | 78.6           | 78.4        | 78.8        |
| int_uniform  | complete | 1000x10000 | lapmod         | 2         | 99.7           | 99.6        | 99.9        |
| int_uniform  | complete | 1000x10000 | ramshaw_tarjan | 2         | 225            | 225         | 225         |

| Regime           | Pattern  | Shape      | Solver         | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|------------------|----------|------------|----------------|-----------|----------------|-------------|-------------|
| int_uniform      | complete | 1000x10000 | auto           | 2         | 474            | 472         | 475         |
| int_uniform      | complete | 1000x10000 | sap            | 2         | 476            | 475         | 477         |
| int_uniform      | complete | 1500x1500  | jv             | 2         | 43.8           | 43.4        | 44.3        |
| int_uniform      | complete | 1500x1500  | auto           | 2         | 43.9           | 43.5        | 44.4        |
| int_uniform      | complete | 1500x1500  | lapmod         | 2         | 129            | 128         | 131         |
| int_uniform      | complete | 1500x1500  | ramshaw_tarjan | 2         | 140            | 138         | 141         |
| int_uniform      | complete | 1500x1500  | sap            | 2         | 225            | 223         | 227         |
| int_uniform      | complete | 1500x4500  | jv             | 2         | 60.6           | 60.3        | 60.9        |
| int_uniform      | complete | 1500x4500  | lapmod         | 2         | 75.1           | 74.7        | 75.5        |
| int_uniform      | complete | 1500x4500  | ramshaw_tarjan | 2         | 162            | 162         | 162         |
| int_uniform      | complete | 1500x4500  | auto           | 2         | 319            | 318         | 321         |
| int_uniform      | complete | 1500x4500  | sap            | 2         | 321            | 320         | 321         |
| int_uniform      | 25%      | 1000x10000 | jv             | 2         | 108            | 108         | 109         |
| int_uniform      | 25%      | 1000x10000 | lapmod         | 2         | 121            | 121         | 121         |
| int_uniform      | 25%      | 1000x10000 | auto           | 2         | 122            | 122         | 122         |
| int_uniform      | 25%      | 1000x10000 | ramshaw_tarjan | 2         | 181            | 180         | 181         |
| int_uniform      | 25%      | 1000x10000 | sap            | 2         | 213            | 212         | 213         |
| int_uniform      | 25%      | 1500x1500  | jv             | 2         | 98.4           | 98.2        | 98.6        |
| int_uniform      | 25%      | 1500x1500  | sap            | 2         | 106            | 105         | 108         |
| int_uniform      | 25%      | 1500x1500  | lapmod         | 2         | 122            | 121         | 123         |
| int_uniform      | 25%      | 1500x1500  | auto           | 2         | 122            | 121         | 122         |
| int_uniform      | 25%      | 1500x1500  | ramshaw_tarjan | 2         | 220            | 217         | 223         |
| int_uniform      | 25%      | 1500x4500  | jv             | 2         | 78             | 77.5        | 78.5        |
| int_uniform      | 25%      | 1500x4500  | auto           | 2         | 87.3           | 85.8        | 88.9        |
| int_uniform      | 25%      | 1500x4500  | lapmod         | 2         | 88.5           | 87          | 90          |
| int_uniform      | 25%      | 1500x4500  | ramshaw_tarjan | 2         | 127            | 127         | 127         |
| int_uniform      | 25%      | 1500x4500  | sap            | 2         | 161            | 156         | 166         |
| metric_clustered | complete | 1000x10000 | jv             | 2         | 77             | 76.8        | 77.1        |
| metric_clustered | complete | 1000x10000 | lapmod         | 2         | 98.4           | 98.3        | 98.5        |
| metric_clustered | complete | 1000x10000 | ramshaw_tarjan | 2         | 221            | 220         | 221         |
| metric_clustered | complete | 1000x10000 | auto           | 2         | 464            | 463         | 465         |
| metric_clustered | complete | 1000x10000 | sap            | 2         | 468            | 465         | 471         |
| metric_clustered | complete | 1500x1500  | auto           | 2         | 153            | 140         | 166         |
| metric_clustered | complete | 1500x1500  | jv             | 2         | 153            | 140         | 166         |
| metric_clustered | complete | 1500x1500  | ramshaw_tarjan | 2         | 275            | 260         | 290         |
| metric_clustered | complete | 1500x1500  | lapmod         | 2         | 305            | 284         | 325         |
| metric_clustered | complete | 1500x1500  | sap            | 2         | 412            | 391         | 433         |
| metric_clustered | complete | 1500x4500  | jv             | 2         | 59.2           | 58.7        | 59.7        |
| metric_clustered | complete | 1500x4500  | lapmod         | 2         | 77.8           | 76.6        | 79          |
| metric_clustered | complete | 1500x4500  | ramshaw_tarjan | 2         | 165            | 161         | 168         |
| metric_clustered | complete | 1500x4500  | auto           | 2         | 316            | 314         | 318         |
| metric_clustered | complete | 1500x4500  | sap            | 2         | 317            | 316         | 318         |
| metric_clustered | 25%      | 1000x10000 | jv             | 2         | 113            | 113         | 113         |
| metric_clustered | 25%      | 1000x10000 | lapmod         | 2         | 124            | 124         | 124         |
| metric_clustered | 25%      | 1000x10000 | auto           | 2         | 127            | 125         | 129         |
| metric_clustered | 25%      | 1000x10000 | ramshaw_tarjan | 2         | 189            | 187         | 190         |
| metric_clustered | 25%      | 1000x10000 | sap            | 2         | 218            | 218         | 218         |
| metric_clustered | 25%      | 1500x1500  | sap            | 2         | 148            | 144         | 152         |
| metric_clustered | 25%      | 1500x1500  | jv             | 2         | 156            | 150         | 161         |
| metric_clustered | 25%      | 1500x1500  | auto           | 2         | 181            | 174         | 188         |
| metric_clustered | 25%      | 1500x1500  | lapmod         | 2         | 182            | 174         | 191         |

| Regime           | Pattern | Shape     | Solver         | Instances | Median<br>(ms) | Q25<br>(ms) | Q75<br>(ms) |
|------------------|---------|-----------|----------------|-----------|----------------|-------------|-------------|
| metric_clustered | 25%     | 1500x1500 | ramshaw_tarjan | 2         | 335            | 321         | 350         |
| metric_clustered | 25%     | 1500x4500 | jv             | 2         | 87.4           | 85.2        | 89.6        |
| metric_clustered | 25%     | 1500x4500 | auto           | 2         | 90.5           | 89.1        | 92          |
| metric_clustered | 25%     | 1500x4500 | lapmod         | 2         | 102            | 95.3        | 108         |
| metric_clustered | 25%     | 1500x4500 | ramshaw_tarjan | 2         | 138            | 136         | 141         |
| metric_clustered | 25%     | 1500x4500 | sap            | 2         | 164            | 162         | 166         |

## 2.4 Do the solvers agree

Every solver in a cell solves the same instance, so their totals have to agree. A disagreement would be a defect in a solver rather than a property of the regime.

## 348 instances carrying 5 to 12 solvers each; the worst relative disagreement on the optimum is 696.

## 3 The edge-generation loop away from one cloud

The article’s implicit results come from one cloud: eight independent normal covariates with the treated group shifted, one draw per size. This grid sweeps one factor at a time around that base, several seeds deep. Where the dense path can afford the problem, the loop’s answer is compared against it unit by unit.

Table 7: The loop by cell. Rounds is the range over seeds. Graph is the median share of the complete pair count the candidate set ended up holding, in percent. Distances is the median number of distance evaluations over all rounds as a multiple of the complete pair count. Implicit and lazy are median seconds over seeds and speedup is their median ratio.

| Sweep     | Cloud        | Dim | Units  | Caliper  | Seeds | Rounds | Graph<br>% | Distances | Implicit | Lazy    | Speedup |
|-----------|--------------|-----|--------|----------|-------|--------|------------|-----------|----------|---------|---------|
| caliper   | gaussian     | 8   | 10,000 | boundary | 5     | 2-2    | 1.151      | 0.31x     | 0.911 s  | 1.66 s  | 1.79x   |
| caliper   | gaussian     | 8   | 10,000 | mid      | 5     | 2-2    | 1.163      | 0.51x     | 0.991 s  | 1.8 s   | 1.82x   |
| caliper   | gaussian     | 8   | 10,000 | none     | 5     | 2-2    | 1.170      | 1.79x     | 0.862 s  | 1.77 s  | 2.06x   |
| cloud     | clustered    | 8   | 10,000 | none     | 5     | 2-2    | 1.170      | 0.48x     | 0.325 s  | 0.654 s | 2.00x   |
| cloud     | contested    | 8   | 10,000 | none     | 5     | 9-10   | 73.318     | 6.67x     | 133 s    | 426 s   | 3.18x   |
| cloud     | gaussian     | 8   | 10,000 | none     | 5     | 2-2    | 1.170      | 1.79x     | 0.863 s  | 1.77 s  | 2.04x   |
| cloud     | heavy_tailed | 8   | 10,000 | none     | 5     | 2-2    | 1.170      | 1.99x     | 0.894 s  | 0.964 s | 1.08x   |
| cloud     | lattice_ties | 8   | 10,000 | none     | 5     | 2-2    | 1.170      | 1.78x     | 1.09 s   | 1.96 s  | 1.79x   |
| cloud     | shell        | 8   | 10,000 | none     | 5     | 4-7    | 1.183      | 4.96x     | 2.63 s   | 14.8 s  | 5.57x   |
| cloud     | shifted      | 8   | 10,000 | none     | 5     | 7-7    | 3.086      | 5.92x     | 4.81 s   | 54.3 s  | 11.11x  |
| dimension | gaussian     | 2   | 10,000 | none     | 5     | 2-4    | 1.170      | 0.24x     | 0.164 s  | 0.6 s   | 3.81x   |
| dimension | gaussian     | 4   | 10,000 | none     | 5     | 2-2    | 1.170      | 0.69x     | 0.335 s  | 1.45 s  | 4.37x   |
| dimension | gaussian     | 8   | 10,000 | none     | 5     | 2-2    | 1.170      | 1.79x     | 0.864 s  | 1.77 s  | 2.05x   |
| dimension | gaussian     | 16  | 10,000 | none     | 5     | 2-2    | 1.170      | 2.00x     | 2.51 s   | 4.96 s  | 1.97x   |
| dimension | gaussian     | 32  | 10,000 | none     | 5     | 2-2    | 1.170      | 2.00x     | 9.58 s   | 21.3 s  | 2.22x   |
| size      | gaussian     | 8   | 2,000  | none     | 3     | 2-2    | 4.951      | 1.99x     | 0.05 s   | 0.066 s | 1.29x   |
| size      | gaussian     | 8   | 5,000  | none     | 3     | 2-2    | 2.160      | 1.92x     | 0.25 s   | 0.37 s  | 1.48x   |
| size      | gaussian     | 8   | 10,000 | none     | 3     | 2-2    | 1.170      | 1.79x     | 0.863 s  | 1.77 s  | 2.05x   |

| Sweep | Cloud    | Dim | Units  | Caliper | Seeds | Rounds | Graph % | Distances | Implicit | Lazy   | Speedup |
|-------|----------|-----|--------|---------|-------|--------|---------|-----------|----------|--------|---------|
| size  | gaussian | 8   | 20,000 | none    | 3     | 2-2    | 0.630   | 1.61x     | 2.98 s   | 7.7 s  | 2.57x   |
| size  | gaussian | 8   | 50,000 | none    | 3     | 2-3    | 0.288   | 1.94x     | 21.9 s   | 64.4 s | 2.99x   |

Table 8: The certificate by cell. The duality gap is the residual the loop stopped on. Equals the dense solve compares the pairing unit by unit where the dense path could be run, and reads not run where the problem was too large for it.

| Sweep     | Cloud        | Dim | Units  | Caliper  | Worst duality gap | All certified | Equals the dense solve |
|-----------|--------------|-----|--------|----------|-------------------|---------------|------------------------|
| caliper   | gaussian     | 8   | 10,000 | boundary | 0.00e+00          | TRUE          | TRUE                   |
| caliper   | gaussian     | 8   | 10,000 | mid      | 0.00e+00          | TRUE          | TRUE                   |
| caliper   | gaussian     | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| cloud     | clustered    | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| cloud     | contested    | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| cloud     | gaussian     | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| cloud     | heavy_tailed | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| cloud     | lattice_ties | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| cloud     | shell        | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| cloud     | shifted      | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| dimension | gaussian     | 2   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| dimension | gaussian     | 4   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| dimension | gaussian     | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| dimension | gaussian     | 16  | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| dimension | gaussian     | 32  | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| size      | gaussian     | 8   | 2,000  | none     | 0.00e+00          | TRUE          | TRUE                   |
| size      | gaussian     | 8   | 5,000  | none     | 0.00e+00          | TRUE          | TRUE                   |
| size      | gaussian     | 8   | 10,000 | none     | 0.00e+00          | TRUE          | TRUE                   |
| size      | gaussian     | 8   | 20,000 | none     | 0.00e+00          | TRUE          | not run                |
| size      | gaussian     | 8   | 50,000 | none     | 3.64e-12          | TRUE          | not run                |

Table 9: Every implicit run in the grid, one row per seed. Seed width is the number of columns the first round gave each row.

| Sweep     | Cloud    | Dim | Units  | Caliper | Seed | Rounds | Seed width | Graph % | Seconds | Gap      |
|-----------|----------|-----|--------|---------|------|--------|------------|---------|---------|----------|
| dimension | gaussian | 2   | 10,000 | none    | 1    | 2      | 78         | 1.170   | 0.139   | 0.00e+00 |
| dimension | gaussian | 2   | 10,000 | none    | 2    | 3      | 78         | 1.170   | 0.172   | 0.00e+00 |
| dimension | gaussian | 2   | 10,000 | none    | 3    | 3      | 78         | 1.170   | 0.164   | 0.00e+00 |
| dimension | gaussian | 2   | 10,000 | none    | 4    | 2      | 78         | 1.170   | 0.143   | 0.00e+00 |
| dimension | gaussian | 2   | 10,000 | none    | 5    | 4      | 78         | 1.170   | 0.181   | 0.00e+00 |
| dimension | gaussian | 4   | 10,000 | none    | 1    | 2      | 78         | 1.170   | 0.333   | 0.00e+00 |
| dimension | gaussian | 4   | 10,000 | none    | 2    | 2      | 78         | 1.170   | 0.341   | 0.00e+00 |
| dimension | gaussian | 4   | 10,000 | none    | 3    | 2      | 78         | 1.170   | 0.34    | 0.00e+00 |
| dimension | gaussian | 4   | 10,000 | none    | 4    | 2      | 78         | 1.170   | 0.332   | 0.00e+00 |
| dimension | gaussian | 4   | 10,000 | none    | 5    | 2      | 78         | 1.170   | 0.335   | 0.00e+00 |
| dimension | gaussian | 8   | 10,000 | none    | 1    | 2      | 78         | 1.170   | 0.864   | 0.00e+00 |
| dimension | gaussian | 8   | 10,000 | none    | 2    | 2      | 78         | 1.170   | 0.862   | 0.00e+00 |



| Sweep     | Cloud        | Dim | Units  | Caliper | Seed | Rounds | Seed<br>width | Graph<br>% | Seconds | Gap      |
|-----------|--------------|-----|--------|---------|------|--------|---------------|------------|---------|----------|
| dimension | gaussian     | 8   | 10,000 | none    | 3    | 2      | 78            | 1.170      | 0.864   | 0.00e+00 |
| dimension | gaussian     | 8   | 10,000 | none    | 4    | 2      | 78            | 1.170      | 0.866   | 0.00e+00 |
| dimension | gaussian     | 8   | 10,000 | none    | 5    | 2      | 78            | 1.170      | 0.862   | 0.00e+00 |
| dimension | gaussian     | 16  | 10,000 | none    | 1    | 2      | 78            | 1.170      | 2.51    | 0.00e+00 |
| dimension | gaussian     | 16  | 10,000 | none    | 2    | 2      | 78            | 1.170      | 2.51    | 0.00e+00 |
| dimension | gaussian     | 16  | 10,000 | none    | 3    | 2      | 78            | 1.170      | 2.51    | 0.00e+00 |
| dimension | gaussian     | 16  | 10,000 | none    | 4    | 2      | 78            | 1.170      | 2.52    | 0.00e+00 |
| dimension | gaussian     | 16  | 10,000 | none    | 5    | 2      | 78            | 1.170      | 2.52    | 0.00e+00 |
| dimension | gaussian     | 32  | 10,000 | none    | 1    | 2      | 78            | 1.170      | 9.58    | 0.00e+00 |
| dimension | gaussian     | 32  | 10,000 | none    | 2    | 2      | 78            | 1.170      | 9.58    | 0.00e+00 |
| dimension | gaussian     | 32  | 10,000 | none    | 3    | 2      | 78            | 1.170      | 9.58    | 0.00e+00 |
| dimension | gaussian     | 32  | 10,000 | none    | 4    | 2      | 78            | 1.170      | 9.58    | 0.00e+00 |
| dimension | gaussian     | 32  | 10,000 | none    | 5    | 2      | 78            | 1.170      | 9.58    | 0.00e+00 |
| cloud     | gaussian     | 8   | 10,000 | none    | 1    | 2      | 78            | 1.170      | 0.863   | 0.00e+00 |
| cloud     | gaussian     | 8   | 10,000 | none    | 2    | 2      | 78            | 1.170      | 0.864   | 0.00e+00 |
| cloud     | gaussian     | 8   | 10,000 | none    | 3    | 2      | 78            | 1.170      | 0.863   | 0.00e+00 |
| cloud     | gaussian     | 8   | 10,000 | none    | 4    | 2      | 78            | 1.170      | 0.867   | 0.00e+00 |
| cloud     | gaussian     | 8   | 10,000 | none    | 5    | 2      | 78            | 1.170      | 0.861   | 0.00e+00 |
| cloud     | clustered    | 8   | 10,000 | none    | 1    | 2      | 78            | 1.170      | 0.314   | 0.00e+00 |
| cloud     | clustered    | 8   | 10,000 | none    | 2    | 2      | 78            | 1.170      | 0.325   | 0.00e+00 |
| cloud     | clustered    | 8   | 10,000 | none    | 3    | 2      | 78            | 1.170      | 0.333   | 0.00e+00 |
| cloud     | clustered    | 8   | 10,000 | none    | 4    | 2      | 78            | 1.170      | 0.36    | 0.00e+00 |
| cloud     | clustered    | 8   | 10,000 | none    | 5    | 2      | 78            | 1.170      | 0.316   | 0.00e+00 |
| cloud     | shifted      | 8   | 10,000 | none    | 1    | 7      | 78            | 3.062      | 4.8     | 0.00e+00 |
| cloud     | shifted      | 8   | 10,000 | none    | 2    | 7      | 78            | 3.086      | 5.01    | 0.00e+00 |
| cloud     | shifted      | 8   | 10,000 | none    | 3    | 7      | 78            | 3.108      | 4.93    | 0.00e+00 |
| cloud     | shifted      | 8   | 10,000 | none    | 4    | 7      | 78            | 3.099      | 4.71    | 0.00e+00 |
| cloud     | shifted      | 8   | 10,000 | none    | 5    | 7      | 78            | 3.080      | 4.81    | 0.00e+00 |
| cloud     | lattice_ties | 8   | 10,000 | none    | 1    | 2      | 78            | 1.170      | 1.09    | 0.00e+00 |
| cloud     | lattice_ties | 8   | 10,000 | none    | 2    | 2      | 78            | 1.170      | 1.09    | 0.00e+00 |
| cloud     | lattice_ties | 8   | 10,000 | none    | 3    | 2      | 78            | 1.170      | 1.09    | 0.00e+00 |
| cloud     | lattice_ties | 8   | 10,000 | none    | 4    | 2      | 78            | 1.170      | 1.09    | 0.00e+00 |
| cloud     | lattice_ties | 8   | 10,000 | none    | 5    | 2      | 78            | 1.170      | 1.1     | 0.00e+00 |
| cloud     | heavy_tailed | 8   | 10,000 | none    | 1    | 2      | 78            | 1.170      | 0.894   | 0.00e+00 |
| cloud     | heavy_tailed | 8   | 10,000 | none    | 2    | 2      | 78            | 1.170      | 0.89    | 0.00e+00 |
| cloud     | heavy_tailed | 8   | 10,000 | none    | 3    | 2      | 78            | 1.170      | 0.914   | 0.00e+00 |
| cloud     | heavy_tailed | 8   | 10,000 | none    | 4    | 2      | 78            | 1.170      | 0.894   | 0.00e+00 |
| cloud     | heavy_tailed | 8   | 10,000 | none    | 5    | 2      | 78            | 1.170      | 0.893   | 0.00e+00 |
| cloud     | contested    | 8   | 10,000 | none    | 1    | 10     | 78            | 73.287     | 133     | 0.00e+00 |
| cloud     | contested    | 8   | 10,000 | none    | 2    | 9      | 78            | 73.318     | 133     | 0.00e+00 |
| cloud     | contested    | 8   | 10,000 | none    | 3    | 10     | 78            | 73.347     | 134     | 0.00e+00 |
| cloud     | contested    | 8   | 10,000 | none    | 4    | 10     | 78            | 73.355     | 134     | 0.00e+00 |
| cloud     | contested    | 8   | 10,000 | none    | 5    | 9      | 78            | 73.312     | 133     | 0.00e+00 |
| cloud     | shell        | 8   | 10,000 | none    | 1    | 4      | 78            | 1.172      | 1.94    | 0.00e+00 |
| cloud     | shell        | 8   | 10,000 | none    | 2    | 7      | 78            | 1.200      | 3.26    | 0.00e+00 |
| cloud     | shell        | 8   | 10,000 | none    | 3    | 6      | 78            | 2.825      | 3.4     | 0.00e+00 |
| cloud     | shell        | 8   | 10,000 | none    | 4    | 5      | 78            | 1.183      | 2.63    | 0.00e+00 |
| cloud     | shell        | 8   | 10,000 | none    | 5    | 5      | 78            | 1.176      | 2.5     | 0.00e+00 |
| caliper   | gaussian     | 8   | 10,000 | none    | 1    | 2      | 78            | 1.170      | 0.862   | 0.00e+00 |
| caliper   | gaussian     | 8   | 10,000 | none    | 2    | 2      | 78            | 1.170      | 0.862   | 0.00e+00 |
| caliper   | gaussian     | 8   | 10,000 | none    | 3    | 2      | 78            | 1.170      | 0.862   | 0.00e+00 |

| Sweep   | Cloud    | Dim | Units  | Caliper  | Seed | Rounds | Seed width | Graph % | Seconds | Gap      |
|---------|----------|-----|--------|----------|------|--------|------------|---------|---------|----------|
| caliper | gaussian | 8   | 10,000 | none     | 4    | 2      | 78         | 1.170   | 0.866   | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | none     | 5    | 2      | 78         | 1.170   | 0.861   | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | boundary | 1    | 2      | 78         | 1.151   | 0.911   | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | boundary | 2    | 2      | 78         | 1.145   | 0.901   | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | boundary | 3    | 2      | 78         | 1.121   | 0.844   | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | boundary | 4    | 2      | 78         | 1.168   | 1.05    | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | boundary | 5    | 2      | 78         | 1.165   | 1       | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | mid      | 1    | 2      | 78         | 1.166   | 1.04    | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | mid      | 2    | 2      | 78         | 1.125   | 0.852   | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | mid      | 3    | 2      | 78         | 1.142   | 0.883   | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | mid      | 4    | 2      | 78         | 1.167   | 1.02    | 0.00e+00 |
| caliper | gaussian | 8   | 10,000 | mid      | 5    | 2      | 78         | 1.163   | 0.991   | 0.00e+00 |
| size    | gaussian | 8   | 2,000  | none     | 1    | 2      | 66         | 4.951   | 0.05    | 0.00e+00 |
| size    | gaussian | 8   | 2,000  | none     | 2    | 2      | 66         | 4.951   | 0.05    | 0.00e+00 |
| size    | gaussian | 8   | 2,000  | none     | 3    | 2      | 66         | 4.951   | 0.051   | 0.00e+00 |
| size    | gaussian | 8   | 5,000  | none     | 1    | 2      | 72         | 2.160   | 0.248   | 0.00e+00 |
| size    | gaussian | 8   | 5,000  | none     | 2    | 2      | 72         | 2.160   | 0.25    | 0.00e+00 |
| size    | gaussian | 8   | 5,000  | none     | 3    | 2      | 72         | 2.160   | 0.25    | 0.00e+00 |
| size    | gaussian | 8   | 10,000 | none     | 1    | 2      | 78         | 1.170   | 0.866   | 0.00e+00 |
| size    | gaussian | 8   | 10,000 | none     | 2    | 2      | 78         | 1.170   | 0.861   | 0.00e+00 |
| size    | gaussian | 8   | 10,000 | none     | 3    | 2      | 78         | 1.170   | 0.863   | 0.00e+00 |
| size    | gaussian | 8   | 20,000 | none     | 1    | 2      | 84         | 0.630   | 3       | 0.00e+00 |
| size    | gaussian | 8   | 20,000 | none     | 2    | 2      | 84         | 0.630   | 2.98    | 0.00e+00 |
| size    | gaussian | 8   | 20,000 | none     | 3    | 2      | 84         | 0.630   | 2.96    | 0.00e+00 |
| size    | gaussian | 8   | 50,000 | none     | 1    | 2      | 96         | 0.288   | 15.2    | 0.00e+00 |
| size    | gaussian | 8   | 50,000 | none     | 2    | 3      | 96         | 0.288   | 21.9    | 0.00e+00 |
| size    | gaussian | 8   | 50,000 | none     | 3    | 3      | 96         | 0.288   | 21.9    | 3.64e-12 |

## Every cell of the grid finished.

## 4 Peak memory

Peak resident set size is a property of a process, so each cell here is a fresh R process that loads what it needs, does one matching and exits, measured from outside by the platform's `time` utility. The baseline row of a size is a process that loaded the same packages and matched nothing, and it is what the increment column is measured against. `dense_matrix` builds the distance matrix and stops, which separates the representation from the solver that runs on it.

Table 10: Peak resident set size by arm and problem size. Matrix is the bytes the dense cost matrix itself needs at that shape. Package estimate is what `estimate_dense_matrix_mb()` predicts, which is four times the matrix and is what the memory dispatcher warns from.

| Units | Arm      | Peak RSS | Over baseline | R heap peak | Matrix | Package estimate | Seconds | Status |
|-------|----------|----------|---------------|-------------|--------|------------------|---------|--------|
| 2,000 | baseline | 146 MB   | -             | 80 MB       | 7 MB   | 28 MB            | 0.008   | ok     |

| Units  | Arm           | Peak<br>RSS | Over<br>baseline | R heap<br>peak | Matrix    | Package<br>estimate | Seconds | Status |
|--------|---------------|-------------|------------------|----------------|-----------|---------------------|---------|--------|
| 2,000  | baseline__alt | 153<br>MB   | -                | 87 MB          | 7<br>MB   | 28 MB               | 0.008   | ok     |
| 2,000  | dense         | 266<br>MB   | 120 MB           | 153<br>MB      | 7<br>MB   | 28 MB               | 0.167   | ok     |
| 2,000  | dense__matrix | 203<br>MB   | 58 MB            | 137<br>MB      | 7<br>MB   | 28 MB               | 0.069   | ok     |
| 2,000  | implicit      | 157<br>MB   | 12 MB            | 102<br>MB      | 7<br>MB   | 28 MB               | 0.07    | ok     |
| 2,000  | lazy          | 151<br>MB   | 6 MB             | 102<br>MB      | 7<br>MB   | 28 MB               | 0.068   | ok     |
| 2,000  | MatchIt       | 531<br>MB   | 378 MB           | 311<br>MB      | 7<br>MB   | 28 MB               | 2.35    | ok     |
| 2,000  | optmatch      | 483<br>MB   | 331 MB           | 318<br>MB      | 7<br>MB   | 28 MB               | 1.85    | ok     |
| 5,000  | baseline      | 146<br>MB   | -                | 80 MB          | 44<br>MB  | 178 MB              | 0.007   | ok     |
| 5,000  | baseline__alt | 153<br>MB   | -                | 87 MB          | 44<br>MB  | 178 MB              | 0.008   | ok     |
| 5,000  | dense         | 563<br>MB   | 417 MB           | 409<br>MB      | 44<br>MB  | 178 MB              | 0.887   | ok     |
| 5,000  | dense__matrix | 226<br>MB   | 81 MB            | 161<br>MB      | 44<br>MB  | 178 MB              | 0.339   | ok     |
| 5,000  | implicit      | 175<br>MB   | 29 MB            | 104<br>MB      | 44<br>MB  | 178 MB              | 0.315   | ok     |
| 5,000  | lazy          | 154<br>MB   | 8 MB             | 104<br>MB      | 44<br>MB  | 178 MB              | 0.365   | ok     |
| 5,000  | MatchIt       | 2436<br>MB  | 2283<br>MB       | 1474<br>MB     | 44<br>MB  | 178 MB              | 16.6    | ok     |
| 5,000  | optmatch      | 2059<br>MB  | 1906<br>MB       | 1320<br>MB     | 44<br>MB  | 178 MB              | 13.3    | ok     |
| 10,000 | baseline      | 149<br>MB   | -                | 80 MB          | 178<br>MB | 711 MB              | 0.007   | ok     |
| 10,000 | baseline__alt | 153<br>MB   | -                | 87 MB          | 178<br>MB | 711 MB              | 0.008   | ok     |
| 10,000 | dense         | 1429<br>MB  | 1280<br>MB       | 812<br>MB      | 178<br>MB | 711 MB              | 3.13    | ok     |
| 10,000 | dense__matrix | 442<br>MB   | 294 MB           | 374<br>MB      | 178<br>MB | 711 MB              | 1.21    | ok     |
| 10,000 | implicit      | 207<br>MB   | 58 MB            | 108<br>MB      | 178<br>MB | 711 MB              | 1.11    | ok     |
| 10,000 | lazy          | 159<br>MB   | 10 MB            | 108<br>MB      | 178<br>MB | 711 MB              | 1.61    | ok     |
| 10,000 | MatchIt       | 8862<br>MB  | 8709<br>MB       | 6382<br>MB     | 178<br>MB | 711 MB              | 74      | ok     |
| 10,000 | optmatch      | 7344<br>MB  | 7192<br>MB       | 4545<br>MB     | 178<br>MB | 711 MB              | 60.1    | ok     |
| 20,000 | baseline      | 146<br>MB   | -                | 80 MB          | 711<br>MB | 2844 MB             | 0.007   | ok     |
| 20,000 | baseline__alt | 153<br>MB   | -                | 87 MB          | 711<br>MB | 2844 MB             | 0.007   | ok     |
| 20,000 | dense         | 6231<br>MB  | 6085<br>MB       | 3656<br>MB     | 711<br>MB | 2844 MB             | 10.6    | ok     |

| Units  | Arm          | Peak<br>RSS | Over<br>baseline | R heap<br>peak | Matrix | Package<br>estimate | Seconds | Status                                                      |
|--------|--------------|-------------|------------------|----------------|--------|---------------------|---------|-------------------------------------------------------------|
| 20,000 | dense_matrix | 1589 MB     | 1443 MB          | 1145 MB        | 711 MB | 2844 MB             | 4.28    | ok                                                          |
| 20,000 | implicit     | 273 MB      | 127 MB           | 116 MB         | 711 MB | 2844 MB             | 3.93    | ok                                                          |
| 20,000 | lazy         | 166 MB      | 20 MB            | 116 MB         | 711 MB | 2844 MB             | 6.42    | ok                                                          |
| 20,000 | MatchIt      | 8793 MB     | 8640 MB          | 7121 MB        | 711 MB | 2844 MB             | 113     | error: (from optmatch) result would exceed $2^{31}-1$ bytes |
| 20,000 | optmatch     | 27570 MB    | 27417 MB         | 19331 MB       | 711 MB | 2844 MB             | 278     | ok                                                          |

Table 11: The four-fold estimate against the measurement. The estimate is deliberately conservative: it exists to avoid under-warning before a large allocation, not to predict the peak.

| Units  | Matrix | Measured over baseline | Measured / matrix | Package estimate |
|--------|--------|------------------------|-------------------|------------------|
| 2,000  | 7 MB   | 58 MB                  | 8.13              | 28 MB            |
| 5,000  | 44 MB  | 81 MB                  | 1.82              | 178 MB           |
| 10,000 | 178 MB | 294 MB                 | 1.65              | 711 MB           |
| 20,000 | 711 MB | 1443 MB                | 2.03              | 2844 MB          |

## 5 Representation against solver

The implicit mode is a representation and a solver at once: its restricted master is the flow model, and naming a solver under it is an error rather than a setting. Reading its timing as a representation comparison would therefore attribute the flow solver’s behaviour to the representation. The table below is the decomposition: down a column, one solver over two representations; across a row, one representation under two solvers.

Table 12: Wall-clock seconds by representation and solver. Dense and lazy differ only in whether the matrix is materialised; dense and dense-flow differ only in the solver; implicit is the mode as a user meets it.

| Units  | Dense, JV | Lazy, JV | Dense, flow | Implicit |
|--------|-----------|----------|-------------|----------|
| 500    | 21 ms     | 8 ms     | 20 ms       | 11 ms    |
| 2,000  | 156 ms    | 53 ms    | 191 ms      | 55 ms    |
| 5,000  | 814 ms    | 348 ms   | 1.12 s      | 294 ms   |
| 10,000 | 3.07 s    | 1.59 s   | 4.07 s      | 1.1 s    |
| 20,000 | not run   | 6.38 s   | not run     | 3.91 s   |
| 50,000 | not run   | 62.1 s   | not run     | 20.2 s   |

## 6 Scaling, every run

The scaling table in the article reports the median across instances. These are the runs behind it, one row per repetition.

Table 13: Every scaling run. Instance is an independently generated problem of that size; rep is a repetition on that problem. A run that failed or timed out carries its status and no time.

| Units | Package  | Instance | Seed     | Rep | Seconds | Status |
|-------|----------|----------|----------|-----|---------|--------|
| 500   | couplr   | 1        | 20261015 | 1   | 0.101   | ok     |
| 500   | couplr   | 1        | 20261015 | 2   | 0.01    | ok     |
| 500   | couplr   | 1        | 20261015 | 3   | 0.011   | ok     |
| 500   | couplr   | 2        | 21261018 | 1   | 0.011   | ok     |
| 500   | couplr   | 2        | 21261018 | 2   | 0.011   | ok     |
| 500   | couplr   | 2        | 21261018 | 3   | 0.011   | ok     |
| 500   | couplr   | 3        | 22261021 | 1   | 0.011   | ok     |
| 500   | couplr   | 3        | 22261021 | 2   | 0.011   | ok     |
| 500   | couplr   | 3        | 22261021 | 3   | 0.01    | ok     |
| 500   | couplr   | 4        | 23261024 | 1   | 0.011   | ok     |
| 500   | couplr   | 4        | 23261024 | 2   | 0.011   | ok     |
| 500   | couplr   | 4        | 23261024 | 3   | 0.011   | ok     |
| 500   | couplr   | 5        | 24261027 | 1   | 0.011   | ok     |
| 500   | couplr   | 5        | 24261027 | 2   | 0.011   | ok     |
| 500   | couplr   | 5        | 24261027 | 3   | 0.009   | ok     |
| 500   | MatchIt  | 1        | 20261015 | 1   | 0.11    | ok     |
| 500   | MatchIt  | 1        | 20261015 | 2   | 0.112   | ok     |
| 500   | MatchIt  | 1        | 20261015 | 3   | 0.113   | ok     |
| 500   | MatchIt  | 2        | 21261018 | 1   | 0.115   | ok     |
| 500   | MatchIt  | 2        | 21261018 | 2   | 0.113   | ok     |
| 500   | MatchIt  | 2        | 21261018 | 3   | 0.151   | ok     |
| 500   | MatchIt  | 3        | 22261021 | 1   | 0.116   | ok     |
| 500   | MatchIt  | 3        | 22261021 | 2   | 0.117   | ok     |
| 500   | MatchIt  | 3        | 22261021 | 3   | 0.117   | ok     |
| 500   | MatchIt  | 4        | 23261024 | 1   | 0.121   | ok     |
| 500   | MatchIt  | 4        | 23261024 | 2   | 0.124   | ok     |
| 500   | MatchIt  | 4        | 23261024 | 3   | 0.122   | ok     |
| 500   | MatchIt  | 5        | 24261027 | 1   | 0.115   | ok     |
| 500   | MatchIt  | 5        | 24261027 | 2   | 0.117   | ok     |
| 500   | MatchIt  | 5        | 24261027 | 3   | 0.117   | ok     |
| 500   | optmatch | 1        | 20261015 | 1   | 0.097   | ok     |
| 500   | optmatch | 1        | 20261015 | 2   | 0.13    | ok     |
| 500   | optmatch | 1        | 20261015 | 3   | 0.094   | ok     |
| 500   | optmatch | 2        | 21261018 | 1   | 0.097   | ok     |
| 500   | optmatch | 2        | 21261018 | 2   | 0.098   | ok     |
| 500   | optmatch | 2        | 21261018 | 3   | 0.13    | ok     |
| 500   | optmatch | 3        | 22261021 | 1   | 0.098   | ok     |
| 500   | optmatch | 3        | 22261021 | 2   | 0.099   | ok     |
| 500   | optmatch | 3        | 22261021 | 3   | 0.098   | ok     |
| 500   | optmatch | 4        | 23261024 | 1   | 0.107   | ok     |
| 500   | optmatch | 4        | 23261024 | 2   | 0.139   | ok     |
| 500   | optmatch | 4        | 23261024 | 3   | 0.104   | ok     |
| 500   | optmatch | 5        | 24261027 | 1   | 0.099   | ok     |
| 500   | optmatch | 5        | 24261027 | 2   | 0.099   | ok     |
| 500   | optmatch | 5        | 24261027 | 3   | 0.136   | ok     |
| 2,000 | couplr   | 1        | 20262515 | 1   | 0.107   | ok     |
| 2,000 | couplr   | 1        | 20262515 | 2   | 0.139   | ok     |
| 2,000 | couplr   | 1        | 20262515 | 3   | 0.142   | ok     |
| 2,000 | couplr   | 2        | 21262518 | 1   | 0.106   | ok     |

| Units | Package  | Instance | Seed     | Rep | Seconds | Status |
|-------|----------|----------|----------|-----|---------|--------|
| 2,000 | couplr   | 2        | 21262518 | 2   | 0.105   | ok     |
| 2,000 | couplr   | 2        | 21262518 | 3   | 0.109   | ok     |
| 2,000 | couplr   | 3        | 22262521 | 1   | 0.112   | ok     |
| 2,000 | couplr   | 3        | 22262521 | 2   | 0.146   | ok     |
| 2,000 | couplr   | 3        | 22262521 | 3   | 0.114   | ok     |
| 2,000 | couplr   | 4        | 23262524 | 1   | 0.112   | ok     |
| 2,000 | couplr   | 4        | 23262524 | 2   | 0.11    | ok     |
| 2,000 | couplr   | 4        | 23262524 | 3   | 0.11    | ok     |
| 2,000 | couplr   | 5        | 24262527 | 1   | 0.109   | ok     |
| 2,000 | couplr   | 5        | 24262527 | 2   | 0.112   | ok     |
| 2,000 | couplr   | 5        | 24262527 | 3   | 0.11    | ok     |
| 2,000 | MatchIt  | 1        | 20262515 | 1   | 2.076   | ok     |
| 2,000 | MatchIt  | 1        | 20262515 | 2   | 2.105   | ok     |
| 2,000 | MatchIt  | 1        | 20262515 | 3   | 2.103   | ok     |
| 2,000 | MatchIt  | 2        | 21262518 | 1   | 2       | ok     |
| 2,000 | MatchIt  | 2        | 21262518 | 2   | 2.029   | ok     |
| 2,000 | MatchIt  | 2        | 21262518 | 3   | 1.999   | ok     |
| 2,000 | MatchIt  | 3        | 22262521 | 1   | 2.152   | ok     |
| 2,000 | MatchIt  | 3        | 22262521 | 2   | 2.175   | ok     |
| 2,000 | MatchIt  | 3        | 22262521 | 3   | 2.231   | ok     |
| 2,000 | MatchIt  | 4        | 23262524 | 1   | 2.119   | ok     |
| 2,000 | MatchIt  | 4        | 23262524 | 2   | 2.107   | ok     |
| 2,000 | MatchIt  | 4        | 23262524 | 3   | 2.094   | ok     |
| 2,000 | MatchIt  | 5        | 24262527 | 1   | 2.076   | ok     |
| 2,000 | MatchIt  | 5        | 24262527 | 2   | 1.954   | ok     |
| 2,000 | MatchIt  | 5        | 24262527 | 3   | 2.035   | ok     |
| 2,000 | optmatch | 1        | 20262515 | 1   | 1.689   | ok     |
| 2,000 | optmatch | 1        | 20262515 | 2   | 1.716   | ok     |
| 2,000 | optmatch | 1        | 20262515 | 3   | 1.673   | ok     |
| 2,000 | optmatch | 2        | 21262518 | 1   | 1.589   | ok     |
| 2,000 | optmatch | 2        | 21262518 | 2   | 1.583   | ok     |
| 2,000 | optmatch | 2        | 21262518 | 3   | 1.603   | ok     |
| 2,000 | optmatch | 3        | 22262521 | 1   | 1.77    | ok     |
| 2,000 | optmatch | 3        | 22262521 | 2   | 1.733   | ok     |
| 2,000 | optmatch | 3        | 22262521 | 3   | 1.736   | ok     |
| 2,000 | optmatch | 4        | 23262524 | 1   | 1.698   | ok     |
| 2,000 | optmatch | 4        | 23262524 | 2   | 1.684   | ok     |
| 2,000 | optmatch | 4        | 23262524 | 3   | 1.707   | ok     |
| 2,000 | optmatch | 5        | 24262527 | 1   | 1.596   | ok     |
| 2,000 | optmatch | 5        | 24262527 | 2   | 1.654   | ok     |
| 2,000 | optmatch | 5        | 24262527 | 3   | 1.606   | ok     |
| 5,000 | couplr   | 1        | 20265515 | 1   | 0.722   | ok     |
| 5,000 | couplr   | 1        | 20265515 | 2   | 0.71    | ok     |
| 5,000 | couplr   | 2        | 21265518 | 1   | 0.71    | ok     |
| 5,000 | couplr   | 2        | 21265518 | 2   | 0.681   | ok     |
| 5,000 | couplr   | 3        | 22265521 | 1   | 0.771   | ok     |
| 5,000 | couplr   | 3        | 22265521 | 2   | 0.689   | ok     |
| 5,000 | couplr   | 4        | 23265524 | 1   | 0.665   | ok     |
| 5,000 | couplr   | 4        | 23265524 | 2   | 0.649   | ok     |
| 5,000 | couplr   | 5        | 24265527 | 1   | 0.74    | ok     |
| 5,000 | couplr   | 5        | 24265527 | 2   | 0.646   | ok     |
| 5,000 | MatchIt  | 1        | 20265515 | 1   | 16.02   | ok     |

| Units  | Package  | Instance | Seed     | Rep | Seconds | Status                                                              |
|--------|----------|----------|----------|-----|---------|---------------------------------------------------------------------|
| 5,000  | MatchIt  | 1        | 20265515 | 2   | 15.39   | ok                                                                  |
| 5,000  | MatchIt  | 2        | 21265518 | 1   | 15.04   | ok                                                                  |
| 5,000  | MatchIt  | 2        | 21265518 | 2   | 15.06   | ok                                                                  |
| 5,000  | MatchIt  | 3        | 22265521 | 1   | 15.33   | ok                                                                  |
| 5,000  | MatchIt  | 3        | 22265521 | 2   | 15.19   | ok                                                                  |
| 5,000  | MatchIt  | 4        | 23265524 | 1   | 15.01   | ok                                                                  |
| 5,000  | MatchIt  | 4        | 23265524 | 2   | 15.07   | ok                                                                  |
| 5,000  | MatchIt  | 5        | 24265527 | 1   | 14.93   | ok                                                                  |
| 5,000  | MatchIt  | 5        | 24265527 | 2   | 14.76   | ok                                                                  |
| 5,000  | optmatch | 1        | 20265515 | 1   | 13.09   | ok                                                                  |
| 5,000  | optmatch | 1        | 20265515 | 2   | 12.68   | ok                                                                  |
| 5,000  | optmatch | 2        | 21265518 | 1   | 12.27   | ok                                                                  |
| 5,000  | optmatch | 2        | 21265518 | 2   | 12.2    | ok                                                                  |
| 5,000  | optmatch | 3        | 22265521 | 1   | 12.19   | ok                                                                  |
| 5,000  | optmatch | 3        | 22265521 | 2   | 12.27   | ok                                                                  |
| 5,000  | optmatch | 4        | 23265524 | 1   | 12.61   | ok                                                                  |
| 5,000  | optmatch | 4        | 23265524 | 2   | 12.18   | ok                                                                  |
| 5,000  | optmatch | 5        | 24265527 | 1   | 11.84   | ok                                                                  |
| 5,000  | optmatch | 5        | 24265527 | 2   | 11.82   | ok                                                                  |
| 10,000 | couplr   | 1        | 20270515 | 1   | 2.934   | ok                                                                  |
| 10,000 | couplr   | 1        | 20270515 | 2   | 2.858   | ok                                                                  |
| 10,000 | couplr   | 2        | 21270518 | 1   | 2.932   | ok                                                                  |
| 10,000 | couplr   | 2        | 21270518 | 2   | 2.827   | ok                                                                  |
| 10,000 | couplr   | 3        | 22270521 | 1   | 3.127   | ok                                                                  |
| 10,000 | couplr   | 3        | 22270521 | 2   | 3.041   | ok                                                                  |
| 10,000 | MatchIt  | 1        | 20270515 | 1   | 72.36   | ok                                                                  |
| 10,000 | MatchIt  | 1        | 20270515 | 2   | 73.67   | ok                                                                  |
| 10,000 | MatchIt  | 2        | 21270518 | 1   | 72.74   | ok                                                                  |
| 10,000 | MatchIt  | 2        | 21270518 | 2   | 72.92   | ok                                                                  |
| 10,000 | MatchIt  | 3        | 22270521 | 1   | 70.6    | ok                                                                  |
| 10,000 | MatchIt  | 3        | 22270521 | 2   | 72.44   | ok                                                                  |
| 10,000 | optmatch | 1        | 20270515 | 1   | 59.41   | ok                                                                  |
| 10,000 | optmatch | 1        | 20270515 | 2   | 58.93   | ok                                                                  |
| 10,000 | optmatch | 2        | 21270518 | 1   | 59.27   | ok                                                                  |
| 10,000 | optmatch | 2        | 21270518 | 2   | 59.54   | ok                                                                  |
| 10,000 | optmatch | 3        | 22270521 | 1   | 58.28   | ok                                                                  |
| 10,000 | optmatch | 3        | 22270521 | 2   | 57.67   | ok                                                                  |
| 20,000 | couplr   | 1        | 20280515 | 1   | 10.49   | ok                                                                  |
| 20,000 | couplr   | 2        | 21280518 | 1   | 19.25   | ok                                                                  |
| 20,000 | MatchIt  | 1        | 20280515 | 1   | -       | error: (from optmatch) result would exceed 2 <sup>31</sup> -1 bytes |
| 20,000 | MatchIt  | 2        | 21280518 | 1   | -       | error: (from optmatch) result would exceed 2 <sup>31</sup> -1 bytes |
| 20,000 | optmatch | 1        | 20280515 | 1   | 278.8   | ok                                                                  |
| 20,000 | optmatch | 2        | 21280518 | 1   | 291.1   | ok                                                                  |
| 50,000 | couplr   | 1        | 20310515 | 1   | 109.3   | ok                                                                  |
| 50,000 | MatchIt  | 1        | 20310515 | 1   | -       | timeout                                                             |
| 50,000 | optmatch | 1        | 20310515 | 1   | -       | timeout                                                             |

## 7 Software design

The article describes the three layers the package is organised in, the dispatch rules, the memory modes and how the package is verified. This section carries the object model, the C++ organisation and the dependency footprint.

### 7.1 Object model

The object model is S3 throughout. Matching results carry the class vector `c("matching_result", "couplr_result")`, with the design-specific classes (`full_matching_result`, `cem_result`, `subclass_result`) sharing the `couplr_result` parent, which is what lets `balance_diagnostics()`, `join_matched()`, `match_data()` and the `bal.tab()` methods dispatch on it once rather than being rewritten per design. Results are plain lists of tibbles, so a user who wants something the package does not provide can reach in and take it. `lap_solve()` returns a tibble subclassed as `lap_solve_result` with `source`, `target` and `cost` columns, so an optimum is usable in a plot or a join without an extraction step, while `get_total_cost()` and `get_method_used()` recover the metadata carried in attributes.

### 7.2 C++ organisation

All nineteen solvers are written from scratch in C++ and exposed through Rcpp, with `src/core` holding the cost-source types and shared utilities with no Rcpp dependency, `src/solvers` one algorithm per file with a `thin_rcpp.cpp` wrapper each, `src/flow` the flow model, its two solvers and the edge-generation loop, and `src/interface` the conversion between R objects and the internal types. The separation means the solver bodies are ordinary C++ that can be compiled and tested outside R, which is what the C++ test suite does.

Solvers are templated on their cost source rather than written against a concrete matrix. A cost source provides `at(i, j)`, `allowed(i, j)` and `empty()`, and two types satisfy that interface: `CostMatrix`, which owns a dense flat array plus a mask, and `LazyCostMatrix`, which owns the two feature matrices and computes  $C_{ij}$  on demand from the requested metric, applying calipers and the distance ceiling in `allowed()`. A handful of hot loops index the dense mask array directly for speed, which the lazy type cannot support; these are guarded by a `supports_raw_mask` trait and an `if constexpr` branch rather than by duplicating the solver. One solver body therefore serves both cost representations, and the pricing oracle of the edge-generation loop needs `at(i, j)` and `allowed(i, j)` and nothing else, so it consumes the same concept and adds no third representation of the costs.

`LazyCostMatrix` also bakes in the transformations that the dense path applies in a separate preparation pass, namely mapping forbidden entries to a large sentinel and negating costs when maximising. Doing this at construction is what allows the untouched solver body to work on either type.

### 7.3 Dependency footprint

`couplr` declares Rcpp, tibble, dplyr, rlang, purrr and `methods` in `Imports`, and Rcpp alone in `LinkingTo`. The recursive closure of hard dependencies is 17 non-base packages, so a default installation brings 18 packages including `couplr` itself. The comparable figures on the same day were 18 for `optmatch`, 8 for `MatchIt` and 7 for `designmatch`.

Two choices sit behind that number. First, we use no external solver: all nineteen algorithms are package source, so there is no dependency on an LP or MIP library and no system solver to install. That choice also keeps `LinkingTo` to Rcpp alone, which matters because every entry there forces a compile of that package before `couplr` can be built from source; `couplr` declares only what its headers actually include. Second, everything that serves a single feature lives in `Suggests` and is loaded at call time. The animation and image-morphing functions, the plotting helpers, the interoperability layer for `MatchIt`, `optmatch` and `cobalt`, and the parallel back end are all optional in this sense, so a user who only wants to match pays for none of them.